

# THE THREE- AND FOUR-LOOP EQUAL-MASS SUNSET INTEGRALS AND LOCAL CALABI–YAU FOURFOLDS AND FIVEFOLDS

by

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**Abstract.** — We determine the equal-mass two-dimensional three- and four-loop sunset Feynman integrals  $I_{\odot}^{(3)}$  and  $I_{\odot}^{(4)}$  at large Euclidean momentum in terms of the genus-zero Gromov–Witten theory of the local Calabi–Yau fourfold and fivefold  $X_L = \text{Tot}(K_{F_L})$ , where  $F_L \subset (\mathbb{P}^1)^{L+1}$ ,  $L = 3, 4$ , is a smooth hypersurface of multi-degree  $(1, \dots, 1)$ . At two loops the sunset integral is governed by the prepotential of a local Calabi–Yau threefold. For  $L \geq 3$  the moduli spaces of unmarked stable maps have positive virtual dimension and there is no prepotential, therefore the Feynman integral is governed instead by the one-point potential  $\mathcal{F}_{\gamma_L}$  with insertion  $\gamma_L = H^{L-1}/(L-1)!$ , by its gravitational descendants, and by  $\zeta(3)$  boundary terms coming from the Gamma class. At both loop orders the coefficient of  $R^{L-1}$  in the Frobenius expansion in the mirror coordinate  $R$  is  $-\theta_Q \mathcal{F}_{\gamma_L}$  with the Novikov variable  $Q = \exp(R)$ . At three loops the whole Feynman integral is an explicit expression in  $\mathcal{F}_{\gamma_3}$  alone, because genus-zero recursion determines all descendant series from it; at four loops two diagonal series suffice, and the  $\zeta(3)$  term is itself corrected by  $\theta_Q \mathcal{F}_{\gamma_4}$ . At three loops we also derive the analytic normalization independently, from the Picard–Fuchs equation and the polar part of a Mellin transform, prove integrality of the multiple-cover transforms for arbitrary integral insertions together with an explicit quantum- $K$  formula, and compute the exact period connection polynomials. Finally we give endpoint values along the Euclidean axis, bounds on the Taylor radii in the Novikov variable  $Q$ , a computer-assisted proof of the four-loop endpoint nonanalyticity, and a comparison of the normalizations at two, three and four loops.

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## 1. Introduction

The equal-mass two-dimensional sunset integrals are among the simplest Feynman integrals that are not polylogarithmic. For  $L \geq 2$  the  $L$ -loop member, with all  $L + 1$  masses equal to one, has the Schwinger representation

$$I_{\odot}^{(L)}(s) = \int_{\mathbb{R}_+^{L+1}} e^{-\sum_{i=1}^{L+1} x_i + sW(x)} W(x) \prod_{i=1}^{L+1} \frac{dx_i}{x_i}, \quad W(x) = \left( \sum_{i=1}^{L+1} x_i^{-1} \right)^{-1}, \quad (1.1)$$

which converges for  $\operatorname{Re} s < (L+1)^2$  and continues analytically to  $\mathbb{C} \setminus [(L+1)^2, \infty)$ . This paper is about the three- and four-loop integrals  $I_{\odot}^{(3)}$  and  $I_{\odot}^{(4)}$ . We express them exactly, at large Euclidean momentum  $s = -T \rightarrow -\infty$ , through the genus-zero Gromov–Witten invariants of a local Calabi–Yau fourfold and of a local Calabi–Yau fivefold, and we study the enumerative, arithmetic and analytic structure that this identification brings to light.

**From two loops to  $L \geq 3$  loops.** — At two loops the geometry attached to the sunset integral is a family of elliptic curves. Bloch, Kerr and Vanhove [1] related the two-loop integral to the genus-zero prepotential of the local Calabi–Yau threefold  $\operatorname{Tot}(K_{F_2})$  over the del Pezzo surface  $F_2$  of degree six, through a mirror-symmetry and Legendre-transform construction. At  $L$  loops the corresponding geometry is the local Calabi–Yau  $(L+1)$ -fold

$X_L = \operatorname{Tot}(K_{F_L})$ ,  $F_L \subset Y_L = (\mathbb{P}^1)^{L+1}$  a smooth hypersurface of multidegree  $(1, \dots, 1)$ .

Here  $F_L$  is Fano with  $K_{F_L} = \mathcal{O}_{F_L}(-H)$ , where  $H = p_1 + \dots + p_{L+1}$  and  $p_i$  is the pullback of the point class of the  $i$ th factor. The Laurent polynomial

$$\phi(x) = \left( \sum_{i=1}^{L+1} x_i \right) \left( \sum_{i=1}^{L+1} x_i^{-1} \right)$$

is the Landau–Ginzburg superpotential mirror to  $F_L$ , and the exponent in (1.1) factors as  $-W(x)(\phi(x) - s)$ . The sunset integral is therefore a Laplace transform of the oscillatory integral of this mirror (Proposition [2, 8.1]), and it is Iritani’s Gamma-class identity for that oscillatory integral [3] that brings the Gromov–Witten theory of  $X_L$  into play.

Passing from two to three loops is not merely a change of dimension. Three things change, and together they account for what is new here.

The moduli spaces of genus-zero stable maps to  $X_L$  with  $\ell$  marked points have virtual dimension  $\operatorname{vdim} \overline{\mathcal{M}}_{0,\ell}(X_L, \beta) = L - 2 + \ell$ . At two loops the unmarked spaces have virtual dimension zero; their degrees form the prepotential, and one

marked point with the divisor insertion  $H$  gives nothing new, since the divisor equation reduces  $\langle H \rangle_{0,1,\beta}$  to  $\langle H \cdot \beta \rangle_{0,0,\beta}$ . For  $L \geq 3$  the unmarked spaces do not directly define numerical genus-zero invariants without additional insertions. The first numerical observable is the one-point invariant with the insertion

$$\gamma_L = \frac{H^{L-1}}{(L-1)!} = \sum_{|S|=L-1} \prod_{i \in S} p_i \in H^{2(L-1)}(F_L, \mathbb{Z}),$$

whose complex codimension  $L-1$  equals  $\text{vdim } \overline{\mathcal{M}}_{0,1}(X_L, \beta)$ . Its generating series is the one-point potential

$$\mathcal{F}_{\gamma_L}(Q) = \sum_{m \geq 1} Q^m \sum_{H \cdot \beta = m} \int_{[\overline{\mathcal{M}}_{0,1}(X_L, \beta)]^{\text{vir}}} \text{ev}^*(\varpi^* \gamma_L), \quad (1.2)$$

where  $\varpi: X_L \rightarrow F_L$  is the projection and  $Q$  is the Novikov variable. For  $L \geq 3$  the class  $\gamma_L$  is not a divisor, and no divisor equation reduces  $\mathcal{F}_{\gamma_L}$  to unmarked invariants. Accordingly its multiple-cover transform has the one-marking weight  $r^{-2}$  instead of the weight  $r^{-3}$  of the unmarked prepotential used at two loops; Section 5 shows that this is a change of observable, not of the genus-zero cover law.

The coefficients of the lower powers of the mirror coordinate involve the diagonal one-point descendant series

$$\mathcal{D}_a^\Delta(Q) = \sum_{m \geq 1} Q^m \sum_{H \cdot \beta = m} \int_{[\overline{\mathcal{M}}_{0,1}(X_L, \beta)]^{\text{vir}}} \psi^a \text{ev}^*(\varpi^* \delta_a), \quad \delta_a = \frac{H^{L-1-a}}{(L-1-a)!}, \quad (1.3)$$

for  $0 \leq a \leq L-1$ , so that  $\mathcal{D}_0^\Delta = \mathcal{F}_{\gamma_L}$ . At two loops the divisor and dilaton equations reduce both such series to the prepotential. For  $L \geq 3$  they do not, and one of the questions answered below is to what extent the descendants are nevertheless determined by  $\mathcal{F}_{\gamma_L}$ .

The boundary behaviour at large momentum is governed by the Gamma class  $\widehat{\Gamma}_{X_L}|_{F_L} = \exp(\sum_{r \geq 1} \frac{2\zeta(2r+1)}{2r+1} H^{2r+1})$  (Proposition [2, 8.6]). On the surface  $F_2$  one has  $H^3 = 0$ , so no zeta value enters and the two-loop boundary constants vanish (Section 5.7). For  $L = 3, 4$  the term  $\frac{2}{3}\zeta(3)H^3$  survives. It produces the boundary term  $-16\zeta(3) = -\frac{2}{3}\zeta(3) \int_{F_3} H^3$  at three loops and, at four loops, the term  $-80\zeta(3) = -\frac{2}{3}\zeta(3) \int_{F_4} H^4$  multiplying a logarithmic solution that is itself corrected by  $\mathcal{F}_{\gamma_4}$ ; the value  $\zeta(5)$  first enters at five loops.

Two further changes are technical but essential. For  $L = 3$  the hypersurface  $F_3$  has  $b_3(F_3) = 2$  (Remark 3.4) and is therefore not toric, so the toric mirror theorem used at two loops does not apply to it directly; the local theory is computed instead on the ambient space  $Y_L$  with an Euler and an inverse Euler twist [2]. For  $L = 4$  the middle cohomology of  $F_4$  contains an 18-dimensional nonambient

summand, invisible to the differential operator; the one-point evaluation classes are nevertheless ambient (Theorem [2, 14.1]).

Conversely, two hypotheses used throughout for  $L \geq 3$  fail at  $L = 2$ , which is why the two-loop case is not a specialization of the results below. First, the weak Lefschetz isomorphism  $H_2(F_L, \mathbb{Z}) \cong H_2(Y_L, \mathbb{Z})$  requires  $L \geq 3$ . For the del Pezzo surface  $F_2$  the curve lattice has rank four and the ambient one rank three, so an ambient degree does not determine a curve class. Second, at the threshold  $s = (L + 1)^2$  the integrand of the simplex representation of (1.1) behaves like  $\|u - u_*\|^{-2}$  in real dimension  $L$ . This is integrable for  $L \geq 3$ , and for  $L = 3, 4$  the Feynman integral is continuous, but not  $C^1$ , at threshold (Proposition [2, 8.2]); for  $L = 2$  the integral diverges logarithmically at  $s = 9$ . Section 5 treats the two-loop case separately and compares it with three and four loops.

**The Feynman integral in terms of  $\mathcal{F}_{\gamma_L}$ .** — Put  $y = -1/s$ , so that  $y = 1/T > 0$  on the Euclidean axis, and write  $\theta_y = y d/dy$  and  $\theta_Q = Q d/dQ$ . Let  $A(y) = 1 + O(y)$  be the holomorphic period,  $R$  the logarithmic mirror coordinate, normalized by  $\theta_y R = A$  and  $R = \log y + O(y)$ , and  $Q = e^R$ ; put  $\Pi_L = (-1)^L y A$ . With the uniform assembly of [2] (Theorem [2, 9.1]) as input, the three- and four-loop Feynman integrals are expressed through  $\mathcal{F}_{\gamma_L}$  as follows. We use the notation of [2] as follows. For  $\gamma_L = H^{L-1}/(L-1)!$ ,  $\mathcal{F}_{\gamma_L}$  denotes the genus-zero one-point potential associated with the local Calabi–Yau  $X_L = \text{Tot}(K_{F_L})$ . The diagonal descendant series are denoted by  $\mathcal{D}_a^\Delta$ , with

$$\mathcal{D}_0^\Delta = \mathcal{F}_{\gamma_L}.$$

We write  $\theta_Q = Q d/dQ$ ,  $Q = e^R$ , and  $\mathcal{T}$  for the diagonal two-point primary series. All other notation imported from [2] is recalled at its first use below.

At three loops (Theorem 2.1 and Corollary 2.3), with  $\mathcal{F} = \mathcal{F}_{\gamma_3}$  and  $\theta_Q^{-1} \sum_{m \geq 1} c_m Q^m = \sum_{m \geq 1} (c_m/m) Q^m$ ,

$$\begin{aligned} I_\odot^{(3)}(s) = -yA \Big[ & 4R^3 - (\theta_Q \mathcal{F}) R^2 + \frac{(\theta_Q \mathcal{F})^2}{12} R \\ & + 2\theta_Q^{-1} \mathcal{F} + \frac{1}{12} \theta_Q^{-1} ((\theta_Q \mathcal{F})^2) - \frac{1}{6} \mathcal{F} \theta_Q \mathcal{F} - 16\zeta(3) \Big]. \end{aligned} \quad (1.4)$$

The coefficient of  $R^2$  is  $-\theta_Q \mathcal{F}_{\gamma_3}$ , the fourfold analogue of the two-loop theorem of [1]. It is proved in Section 3 independently of the oscillatory Gamma identity on which [2] relies, and Section 6 gives a second, differential proof of the complete formula (1.4), which avoids the signed analytic assembly of [2]. The two lower coefficients, which in the general assembly involve the descendant series  $\mathcal{D}_1^\Delta$  and  $\mathcal{D}_2^\Delta$ , are determined by  $\mathcal{F}_{\gamma_3}$  through the genus-zero topological recursion (Theorem 6.2). Thus at three loops the whole Feynman integral is determined by a single generating series of genus-zero invariants, as at two loops, with  $\mathcal{F}_{\gamma_3}$  in the role of the prepotential. Its multiple-cover expansion  $\mathcal{F}_{\gamma_3} = \sum_{m \geq 1} n_m \text{Li}_2(Q^m)$

has integer coefficients  $n_m = 24, -72, 264, -1488, 9696, \dots$  (Theorem 4.6), so that  $\exp(\theta_Q \mathcal{F}_{\gamma_3}) = \prod_{m \geq 1} (1 - Q^m)^{-mn_m}$  is an integral Euler product.

At four loops (Theorem 2.2 and Corollary 2.4),

$$I_{\odot}^{(4)}(s) = yA \left[ 5R^4 - (\theta_Q \mathcal{F}_{\gamma_4})R^3 + h_2 R^2 + h_1 R + h_0 - 80\zeta(3) \left( R - \frac{\theta_Q \mathcal{F}_{\gamma_4}}{20} \right) \right], \quad (1.5)$$

with  $h_2 = -3\mathcal{F}_{\gamma_4} - \theta_Q \mathcal{D}_1^{\Delta}$ , and with  $h_1$  and  $h_0$  determined by  $\mathcal{F}_{\gamma_4}$  and  $\mathcal{D}_1^{\Delta}$  through the recursion of Theorem 6.5. Thus the four-loop Feynman integral is determined by two diagonal genus-zero series,  $\mathcal{F}_{\gamma_4}$  and  $\mathcal{D}_1^{\Delta}$ , equivalently by  $\mathcal{F}_{\gamma_4}$  and the diagonal two-point primary series  $\mathcal{T}$ . The two-input statement is a sufficiency result. Whether a single diagonal input suffices remains open. This establishes sufficiency of these two inputs, but not their minimality. We do not prove that two independent diagonal inputs are necessary: in particular, it remains open whether the four-loop integral  $I_{\odot}^{(4)}$  can be determined from  $\mathcal{F}_{\gamma_4}$  alone by an additional relation not used in the present reconstruction.

At both loop orders the leading term is  $(L+1)R^L$  and the first instanton coefficient is  $-\theta_Q \mathcal{F}_{\gamma_L}$ :

$$I_{\odot}^{(L)}(s) = \Pi_L \left[ (L+1)R^L - (\theta_Q \mathcal{F}_{\gamma_L})R^{L-1} + \sum_{j=0}^{L-2} c_{L,j}(Q)R^j \right],$$

with  $c_{L,j}$  convergent power series in  $Q$  whose constant terms are the  $\zeta(3)$  terms above. The same shape holds at two loops for the coarsened invariants of Section 5, where  $\gamma_2 = H$  and, by the divisor equation,  $\theta_Q \mathcal{F}_H$  is the second logarithmic derivative of the prepotential. This is the precise sense in which  $\mathcal{F}_{\gamma_L}$  takes over the role of the prepotential for  $L \geq 3$ .

In this work we derive specific results at  $L = 3$  and  $L = 4$  independently of the result of [2].<sup>(1)</sup>

Section 2 states the three- and four-loop identities and their expressions through  $\mathcal{F}_{\gamma_L}$ , with a dictionary between the two loop orders and numerical checks.

Section 3 gives an independent analytic derivation at three loops. In this section, we determine the three Frobenius boundary constants of  $I_{\odot}^{(3)}$  from the inhomogeneous Picard–Fuchs equation and the polar part of a Mellin transform. We show that two vanish and the third is  $-16\zeta(3)$  (Proposition 3.12). A coefficient bridge

<sup>(1)</sup>The all-loop identity is proved in the companion paper [2], whose results are used here without any proofs and refer to the companion paper for details. Namely, we will use the zero-section factorization and the virtual reduction, the twisted mirror theorem and the extraction of the descendants, the operator  $P_L$  with its all-degree certificate, the signed Gamma formula, which rests on Iritani’s oscillatory identity [3], and the uniform assembly (Theorem [2, 9.1]). Theorems 2.1 and 2.2 are its specializations (Corollaries [2, 9.2] and [2, 9.3]). No argument below depends on the internal structure of those proofs.

built on the quantum-Serre identity and a logarithmic B-model recurrence gives  $d_2(m) = -mN_m(\gamma_3)$  for the coefficients of  $Q^m R^2$ , independently of all three constants (Theorems 3.22 and 3.25); a direct count of lines confirms the normalization  $N_1(\gamma_3) = 24$  (Proposition 3.27). The section also identifies the three descendant potentials,  $\mathcal{D}_0^\Delta = -12K_2$ ,  $\mathcal{D}_1^\Delta = -24K_3$  and  $\mathcal{D}_2^\Delta = -24K_4$  (Theorem 3.24), and the exterior domains on which the series converge.

In section 4, we derive an integrality and quantum  $K$ -theory at three loops. We prove that for every integral class  $\eta \in H^4(F_3, \mathbb{Z})$  the  $r^{-2}$  transforms of the one-point invariants are integers (Theorem 4.6). The proof uses the projective-bundle compactification  $\mathbb{P}(\mathcal{O}_{F_3} \oplus K_{F_3})$ , which is semipositive, and the integrality theorem of Ionel and Parker in the algebraic form of Chou [4, 5]. The section also gives an explicit quantum- $K$  formula for the transforms of  $\mathcal{F}_{\gamma_3}$  (Theorem 4.9), the resulting supercongruences, and exact values through degree 19. The parity argument for the symplectic semipositivity of this compactification is specific to complex dimension four; primary integrality for every  $L \geq 3$  is proved in [2] (Corollary [2, 13.10]) with a compact Calabi–Yau cover instead.

Section 5 compares the curve lattices, the normalizations and the multiple-cover conventions at  $L = 2, 3, 4$ , separates the change of observable from the genuine two-loop restrictions, derives the vanishing of the two-loop boundary constants by the Mellin method, records the four-loop specialization with explicit series, and relates the local invariants to logarithmic invariants of the pair  $(F_L, D)$ ,  $D \in |H|$ , through the local–relative correspondence of van Garrel, Graber and Ruddat [6].

In section 6, we give a reconstruction from primary invariants. We apply the genus-zero topological recursion on a compact model of  $X_L$ . At three loops all one-point descendants are determined by the primary evaluation classes, and on the diagonal by  $\mathcal{F}_{\gamma_3}$  alone (Theorem 6.2); this gives (1.4), together with the differential proof mentioned above, which rests on the symmetric-square structure of the Picard–Fuchs operator. At four loops the primary evaluation classes in full multidegree determine every one-point descendant (Theorem 6.4); after restriction to the one-parameter diagonal, the reconstruction closes on two series (Theorem 6.5).

Corollary 2.4 shows that two diagonal inputs suffice at four loops, not that two are needed. The use of a second diagonal input does not contradict the full multidegree reconstruction theorem. The latter reconstructs descendants from the complete collection of primary evaluation classes, whereas after restriction to the one-parameter diagonal that information is no longer retained in a single series  $\mathcal{F}_{\gamma_4}$ . Whether  $\mathcal{F}_{\gamma_4}$  alone determines  $I_\ominus^{(4)}$ , as  $\mathcal{F}_{\gamma_3}$  determines  $I_\ominus^{(3)}$ , remains open.

Section 7 and 8 give the endpoint values of the mirror coordinate and of all descendant potentials along the Euclidean axis. At three loops the endpoint value of  $R$  is  $R_* = -7\zeta(3)/\pi^2$ , minus twice the Mahler measure of  $1 + z_1 + z_2 + z_3$ ; at four loops the endpoint values reduce to two constants. The same section gives

the four-loop threshold expansion. Section 8 computes the exact three-loop period connection polynomials (Theorem 8.1). As a consequence  $K_2, K_3, K_4$  have a one-sided  $C^1$  extension, but an unbounded second derivative, at the endpoint  $Q_* = e^{R_*}$ , and their  $Q$ -Taylor radii lie between  $1/12$  and  $Q_*$ . Sections 9–13 prove the analogous four-loop statement, with radii between  $1/20$  and  $Q_\infty = 0.3366117386\dots$ , by a computer-assisted argument in ball arithmetic; the complete code is in Appendix A.

**The differential operators.** — The Feynman integrals satisfy the inhomogeneous Picard–Fuchs equations

$$P_L\left(\frac{I_\ominus^{(L)}(-1/y)}{y}\right) = (-1)^L(L+1)! \quad (1.6)$$

of [2, Eq. (8.9)], where  $P_L$  is the order- $L$  operator produced by the algorithm of Definition [2, 7.1]. At the two loop orders treated here,

$$P_3 = \theta_y^3 + 64y^2(\theta_y + 1)^3 + y(20\theta_y^3 + 30\theta_y^2 + 18\theta_y + 4),$$

and

$$\begin{aligned} P_4 &= \theta_y^4 + y^3Q_3(\theta_y) + y^2Q_2(\theta_y) + yQ_1(\theta_y), \\ Q_1 &= 35\theta_y^4 + 70\theta_y^3 + 63\theta_y^2 + 28\theta_y + 5, \\ Q_2 &= 259\theta_y^4 + 1036\theta_y^3 + 1580\theta_y^2 + 1088\theta_y + 285, \\ Q_3 &= 225\theta_y^4 + 1350\theta_y^3 + 2925\theta_y^2 + 2700\theta_y + 900. \end{aligned} \quad (1.7)$$

After the substitution  $y = -t$  these are the equal-mass sunset Picard–Fuchs operators recorded in the Feynman-integral literature; they have been derived in the mathematical [7] and physics literature [8–12]. Their holomorphic solution  $A(y) = \sum_{D \geq 0} (-1)^D c_D y^D$ , with  $c_D = \sum_{|\mathbf{d}|=D} \binom{D}{\mathbf{d}}^2$ , is the alternating generating series of the  $(L+1)$ -variable generalized Apéry numbers; for  $L = 3$  these are  $1, 4, 28, 256, 2716, \dots$ . What the companion paper [2] supplies is the uniform construction for all  $L$  together with an all-degree certificate, which is what allows the operator to be used without fitting it to initial coefficients. That  $P_L$  is irreducible, hence the minimal operator annihilating the period, follows from the irreducibility of the symmetric power of the Bessel connection and two Fourier transforms with checked extensions at zero. We refer to Proposition [2, 14.2] of the companion paper [2].

### Acknowledgements and use of AI

This work started as an application of the AI co-mathematician tool [13] developed by Google DeepMind, in which a human works with AI agents to develop a theory. Later, the output of the AI co-mathematician tool was also reviewed by

different AI tools (first Gemini, later also Claude and ChatGPT) and the reviews were used to improve the theorems, sometimes in further collaboration with the AI co-mathematician.

The final paper is human written, using parts of the AI generated proofs, and also proofread by AI.

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## 2. The three- and four-loop identities

We state the three- and four-loop identities, which are the specializations of the all-loop assembly [2, Eq. (9.2)], together with their expressions through the one-point potentials  $\mathcal{F}_{\gamma_L}$ . The notation is that of the introduction, namely  $y = -1/s$ ,  $R$  is the local mirror coordinate,  $A = \theta_y R$ ,  $Q = e^R$ ,  $\Pi_L = (-1)^L y A$ , and  $\mathcal{D}_a^\Delta$  is the diagonal one-point descendant series (1.3), so that  $\mathcal{D}_0^\Delta = \mathcal{F}_{\gamma_L}$ . The  $K_k$  are the normalized coefficients of the ambient hypergeometric series [2, Eq. (5.8)]; for  $L = 3$  they are recalled in (3.47).

**Theorem 2.1 (Three loops;** Corollary [2, 9.2] of [2])

For  $L = 3$  put  $\Pi_3 = -yA$  and  $\delta_0 = H^2/2$ . Then

$$I_\ominus^{(3)}(s) = \Pi_3 [4R^3 + f_2 R^2 + f_1 R + f_0 - 16\zeta(3)],$$

with

$$f_2 = -\theta_Q \mathcal{D}_0^\Delta, \quad f_1 = -2\mathcal{D}_0^\Delta - \theta_Q \mathcal{D}_1^\Delta, \quad f_0 = -\mathcal{D}_1^\Delta - \theta_Q \mathcal{D}_2^\Delta,$$

and  $\mathcal{D}_0^\Delta = -12K_2$ ,  $\mathcal{D}_1^\Delta = -24K_3$ ,  $\mathcal{D}_2^\Delta = -24K_4$ . In particular the coefficient of  $R^2$  is  $-\theta_Q \mathcal{F}_{\gamma_3}$ , where  $\mathcal{F}_{\gamma_3} = \mathcal{D}_0^\Delta$  is the one-point potential (1.2) of the local Calabi–Yau fourfold  $X_3$  with insertion  $\gamma_3 = H^2/2$ .

**Theorem 2.2 (Four loops;** Corollary [2, 9.3] of [2]). — For  $L = 4$  put  $\Pi_4 = yA$ . Then

$$I_\ominus^{(4)}(s) = \Pi_4 \left[ 5R^4 + \sum_{j=0}^3 h_j R^j - 80\zeta(3) \left( R - \frac{1}{20} \theta_Q \mathcal{D}_0^\Delta \right) \right],$$

where

$$\begin{aligned} h_3 &= -\theta_Q \mathcal{D}_0^\Delta, & h_2 &= -3\mathcal{D}_0^\Delta - \theta_Q \mathcal{D}_1^\Delta, \\ h_1 &= -2\mathcal{D}_1^\Delta - \theta_Q \mathcal{D}_2^\Delta, & h_0 &= -\mathcal{D}_2^\Delta - \theta_Q \mathcal{D}_3^\Delta, \end{aligned}$$

and  $(\mathcal{D}_0^\Delta, \mathcal{D}_1^\Delta, \mathcal{D}_2^\Delta, \mathcal{D}_3^\Delta) = (-20K_2, -60K_3, -120K_4, -120K_5)$ . In particular the coefficient of  $R^3$  is  $h_3 = -\theta_Q \mathcal{F}_{\gamma_4}$ , where  $\mathcal{F}_{\gamma_4} = \mathcal{D}_0^\Delta$  is the one-point potential (1.2) of the local Calabi–Yau fivefold  $X_4$  with insertion  $\gamma_4 = H^3/6$ .

The descendant series in Theorems 2.1 and 2.2 are not independent of the primary series  $\mathcal{F}_{\gamma_L}$ , because the genus-zero topological recursion relations, applied in Section 6, express them through primary invariants. On the diagonal this gives the following two statements.

**Corollary 2.3 (Three loops in terms of  $\mathcal{F}_{\gamma_3}$ ).** — Write  $\mathcal{F} = \mathcal{F}_{\gamma_3}$ , and let  $\theta_Q^{-1}$  act on power series without constant term by  $\theta_Q^{-1} \sum_{m \geq 1} c_m Q^m = \sum_{m \geq 1} (c_m/m) Q^m$ . Then

$$\mathcal{D}_2^\Delta = \frac{\mathcal{F}^2}{12}, \quad \mathcal{D}_1^\Delta = -2\theta_Q^{-1}\mathcal{F} - \frac{1}{12}\theta_Q^{-1}((\theta_Q\mathcal{F})^2).$$

Consequently  $f_1 = (\theta_Q\mathcal{F})^2/12$ , and the three-loop Feynman integral is the expression (1.4) in  $\mathcal{F}$  alone. Equivalently, with  $u = \theta_Q\mathcal{F}/12$ ,

$$I_\odot^{(3)}(s) = -yA[4(R-u)^3 + 4u^3 - \mathcal{D}_1^\Delta - 2u\mathcal{F} - 16\zeta(3)].$$

*Proof.* — The identities for  $\mathcal{D}_2^\Delta$  and  $\mathcal{D}_1^\Delta$  are Theorem 6.2, the second one integrated with zero constant term. Substituting them into the coefficients  $f_1$  and  $f_0$  of Theorem 2.1 gives (1.4), and expanding  $(R-u)^3$  gives the last form.  $\square$

**Corollary 2.4 (Four loops in terms of two diagonal series)**

In Theorem 2.2 one has  $h_3 = -\theta_Q\mathcal{F}_{\gamma_4}$  and  $h_2 = -3\mathcal{F}_{\gamma_4} - \theta_Q\mathcal{D}_1^\Delta$ , while  $\mathcal{D}_2^\Delta$  and  $\mathcal{D}_3^\Delta$ , and hence  $h_1$  and  $h_0$ , are the unique power series without constant term that satisfy (6.14) and (6.15), where  $\mathcal{D}_a$  stands for  $\mathcal{D}_a^\Delta$  and  $\mathcal{D}_0^\Delta = \mathcal{F}_{\gamma_4}$ . Thus the four-loop Feynman integral is determined by the pair  $(\mathcal{F}_{\gamma_4}, \mathcal{D}_1^\Delta)$ , or equivalently by  $(\mathcal{F}_{\gamma_4}, \mathcal{T})$ , where  $\mathcal{T}$  is the diagonal two-point primary series of Theorem 6.5.

This should be contrasted with the three-loop result of Corollary 2.3, where the single primary series  $\mathcal{F}_{\gamma_3}$  determines all descendant data entering the Feynman integral. At four loops we prove that two diagonal inputs suffice, but do not prove that two inputs are necessary. In particular, the possible determination of  $I_\odot^{(4)}$  from  $\mathcal{F}_{\gamma_4}$  alone is left open.

*Proof.* — The formulas for  $h_3$  and  $h_2$  are those of Theorem 2.2, and the statement on  $\mathcal{D}_2^\Delta$  and  $\mathcal{D}_3^\Delta$  is Theorem 6.5. The first identity of that theorem,  $(1 - \theta_Q^2\mathcal{F}_{\gamma_4}/20)\mathcal{T} = 2\theta_Q^2\mathcal{D}_1^\Delta + 12\theta_Q\mathcal{F}_{\gamma_4}$ , determines either of  $\mathcal{T}$  and  $\mathcal{D}_1^\Delta$  from the other, because  $1 - \theta_Q^2\mathcal{F}_{\gamma_4}/20$  is an invertible power series and  $\theta_Q^2$  is bijective on power series without constant term. Thus, for fixed  $\mathcal{F}_{\gamma_4}$ , the two pairs  $(\mathcal{F}_{\gamma_4}, \mathcal{D}_1^\Delta)$  and  $(\mathcal{F}_{\gamma_4}, \mathcal{T})$  contain the same information.  $\square$

The four-loop result therefore exhibits the first case in which the diagonal reconstruction does not, with the present argument, collapse to a single primary potential. This is a statement about the information required by our reconstruction theorem, not a minimality result. In fact whether the second input can be eliminated remains an open question. In full multidegree, by contrast, the primary evaluation classes alone determine every one-point descendant (Theorem 6.4).

**Remark 2.5 (Why  $L = 3, 4$  is not the general pattern)**

At  $L = 5$  a  $\zeta(5)$  term first appears, in addition to the  $\zeta(3)$  term multiplying  $\mathcal{T}_2$ , and at  $L = 6$  the product  $\zeta(3)^2$  appears. Formula [2, Eq. (9.2)] fixes both the zeta polynomial and the descendant correction multiplying each of its terms.

The dictionary between the  $L = 3$  and  $L = 4$  loop orders is as follows.

| Quantity                             | $L = 3$       | $L = 4$          |
|--------------------------------------|---------------|------------------|
| $n$                                  | 4             | 5                |
| $\dim F_L, \dim X_L$                 | 3, 4          | 4, 5             |
| One-point local virtual dimension    | 2             | 3                |
| $\int_{F_L} H^L$                     | 24            | 120              |
| Primary insertion $\gamma_L$         | $H^2/2$       | $H^3/6$          |
| Descendants needed                   | $a = 0, 1, 2$ | $a = 0, 1, 2, 3$ |
| Period prefactor                     | $-yA$         | $yA$             |
| Degree-one diagonal primary          | 24            | 60               |
| Degree-one diagonal first descendant | -48           | -120             |
| Degree-two diagonal primary          | -66           | -225             |
| Degree-two transformed primary       | -72           | -240             |
| Possible nonambient even cohomology  | none          | middle degree 4  |

**Remark 2.6 (Numerical verification).** — The Bessel-moment representation [2, Eq. (16.1)] of the Feynman integral was evaluated at forty-digit working precision and compared numerically with the right-hand sides of [2, Eq. (8.8)] and of the assembly [2, Eq. (9.2)]:

| $L$ | $T$ | $I_{\ominus}^{(L)}(-T)$ | agreement |
|-----|-----|-------------------------|-----------|
| 3   | 50  | 5.17515590864132484...  | 35 digits |
| 3   | 100 | 4.08241320270405960...  | 40 digits |
| 4   | 100 | 26.9629554067580390...  | 40 digits |

The table records observed numerical agreement. The agreement uses the descendant series, the mirror map, and the  $\zeta(3)$  term together. The diagonal primaries and their transforms were computed to  $m = 8$  at  $L = 3$  and to  $m = 6$  at  $L = 4$ :

$$n_m(L=3) = 24, -72, 264, -1488, 9696, -69240, 528072, -4238880,$$

$$n_m(L=4) = 60, -240, 1400, -12480, 131220, -1529520.$$

They are all integers as stated in Corollary [2, 13.10]. Whereas, the quantities  $N_m$  are rational number.

### 3. An independent analytic derivation at three loops

The proof of Theorem [2, 9.1] in [2] runs through the oscillatory Gamma-class identity of Iritani’s paper [3, Section 2, equation (10)], which is its analytic input. This section independently determines the  $L = 3$  analytic boundary normalization from the Picard–Fuchs equation and the polar part of a Mellin transform. This determination does not use that identity; the first proof of the full descendant assembly later invokes the companion paper [2]. Section 6 supplies an alternative proof using the primary bridge, recursion, and the symmetric-square equation. The two routes fix the same three Frobenius boundary constants and give the

same expansion. The second route also isolates what does and does not depend on those constants. The coefficient identity  $d_2(m) = -mN_m(\gamma)$  is independent of all three, whereas the explicit boundary term  $-16\zeta(3)\pi(s)$  is not.

Throughout this section  $L = 3$ , so that the number of factors is  $n = L + 1 = 4$ .

**3.1. Three-loop conventions.** — The geometry is  $Y_3 = (\mathbb{P}^1)^4$ , a smooth  $F_3 \in |\mathcal{O}_{Y_3}(1, 1, 1, 1)|$  with  $j : F_3 \hookrightarrow Y_3$ , and  $X_3 = \text{Tot}(K_{F_3}) \xrightarrow{\varpi} F_3$  with zero section  $\iota$ . By Proposition [2, 3.1],  $K_{F_3} = \mathcal{O}_{F_3}(-H)$ ,  $\int_{F_3} H^3 = 24$  and  $H_2(F_3, \mathbb{Z}) \cong \mathbb{Z}^4$ ; we write  $|\beta| = H \cdot \beta$ . Put  $\gamma := H^2/2 = \sum_{i < j} p_i p_j \in H^4(F_3, \mathbb{Z})$ , which is the insertion  $\gamma_3 = \delta_0$  of the introduction, and

$$\begin{aligned} \mathcal{F}_\gamma(Q) &= \sum_{m \geq 1} N_m(\gamma) Q^m, \\ N_m(\gamma) &= \sum_{|\beta|=m} N_\beta(\gamma), \\ N_\beta(\gamma) &= \int_{[\overline{\mathcal{M}}_{0,1}(X_3, \beta)]^{\text{vir}}} \text{ev}^*(\varpi^* \gamma), \end{aligned} \tag{3.1}$$

so  $\mathcal{F}_\gamma = \mathcal{F}_{\gamma_3} = \mathcal{D}_0^\Delta$ ; in this section and the next we keep the shorter notation  $\gamma$  and  $\mathcal{F}_\gamma$ . In the dictionary between this section and [2],

$$\begin{aligned} y &= -1/s, & g(y) &= S(y) = -G(y, y, y, y), \\ R_{\text{body}}(y) &= T(y), & \Pi_3(s) &= -yA(y) = \pi(s), \end{aligned}$$

and the  $K_2(Q)$  used here is the  $K_2(y(Q))$  of [2, Eq. (5.8)]. The symbol  $k$  denotes the number of markings in the statements quoted below; the number of factors is fixed at four. The Schwinger representation is (1.1) with  $L = 3$ , which is [2, Eq. (8.1)] with  $n = 4$ .

The next four statements are the three-loop forms of results proved in [2]. They are recorded here, with their notation, because the arguments of this section are written in it; their proofs are the specializations indicated, and are not repeated.

**Proposition 3.1 (Zero-section factorization).** — *Let  $S$  be an arbitrary  $\mathbb{C}$ -scheme, let  $\varrho_S : \mathcal{C}_S \rightarrow S$  be a flat, proper, finitely presented family of connected nodal curves of arithmetic genus zero, and let  $g : \mathcal{C}_S \rightarrow X_3$  be a morphism whose restriction to every geometric fiber has class  $\beta \neq 0$ . Then  $g$  factors uniquely through the zero section  $\iota : F_3 \hookrightarrow X_3$ . The factorization preserves stability and commutes with arbitrary base change. Consequently*

$$\overline{\mathcal{M}}_{0,k}(F_3, \beta) \xrightarrow{\sim} \overline{\mathcal{M}}_{0,k}(X_3, \beta)$$

*is a canonical isomorphism of stacks, and the latter is a proper Deligne–Mumford stack.*

*Proof.* — This is Theorem [2, 4.4] with  $n = 4$  and  $K_{F_3} = \mathcal{O}_{F_3}(-H)$ ; that statement is functorial over arbitrary bases and preserves stability.  $\square$

For  $k$  markings write  $\mathcal{M}_{F_3} = \overline{\mathcal{M}}_{0,k}(F_3, \beta)$ , let  $\varrho$  and  $f$  be its universal curve and map, and set  $\mathcal{V} = R^1\varrho_* f^* K_{F_3}$ . This is a vector bundle of rank  $m - 1$ , where  $m = H \cdot \beta > 0$ : on every geometric fiber  $f^* K_{F_3}$  has degree  $-m < 0$  and nonpositive degree on every component, so  $H^0(C, f^* K_{F_3}) = 0$  and  $h^1(C, f^* K_{F_3}) = m - 1$  by Lemma [2, 4.1], and cohomology and base change, in the form of Lemma [2, 4.3], makes  $R^1\varrho_* f^* K_{F_3}$  locally free of rank  $m - 1$ , compatibly with arbitrary base change.

**Proposition 3.2 (Virtual reduction).** — *Under the identification of Proposition 3.1,*

$$[\overline{\mathcal{M}}_{0,k}(X_3, \beta)]^{\text{vir}} = c_{\text{top}}(\mathcal{V}) \cap [\mathcal{M}_{F_3}]^{\text{vir}}, \quad (3.2)$$

$$[\overline{\mathcal{M}}_{0,k}(X_3, \beta)]_{\lambda}^{\text{vir}} = e_{\lambda}(\mathcal{V}) \cap [\mathcal{M}_{F_3}]^{\text{vir}}, \quad (3.3)$$

where in (3.3) the fiberwise  $\mathbb{C}^*$ -action on  $X_3$  gives  $\mathcal{V}$  weight one.

*Proof.* — This is Theorem [2, 4.5], whose proof identifies the obstruction-theory morphisms themselves and includes the equivariant formula.  $\square$

For  $k = 1$  define the *proper evaluation cycle* on  $F_3$ ,

$$C_{\beta}^{\lambda} = (\text{ev}_{F_3})_*(e_{\lambda}(\mathcal{V}) \cap [\overline{\mathcal{M}}_{0,1}(F_3, \beta)]^{\text{vir}}) \in H^*(F_3)[\lambda]. \quad (3.4)$$

Its coefficients are honest cohomology classes on the projective variety  $F_3$ ; no localization of the equivariant parameter is involved. Expanding  $e_{\lambda}(\mathcal{V}) = \sum_{k=0}^{m-1} \lambda^{m-1-k} c_k(\mathcal{V})$ , the cycle  $c_k(\mathcal{V}) \cap [\mathcal{M}_{F_3}]^{\text{vir}}$  has dimension  $(m + 1) - k$ , so its pushforward to the threefold  $F_3$  vanishes for  $k \leq m - 3$ , is a multiple of the fundamental class for  $k = m - 2$ , and is a 2-dimensional cycle class for  $k = m - 1$ . Hence

$$C_{\beta}^{\lambda} = C_{\beta}^0 + \lambda c_{\beta} \mathbf{1}, \quad C_{\beta}^0 = (\text{ev}_{F_3})_*(c_{\text{top}}(\mathcal{V}) \cap [\mathcal{M}_{F_3}]^{\text{vir}}) \in H^2(F_3, \mathbb{Q}), \quad c_{\beta} \in \mathbb{Q}, \quad (3.5)$$

homogeneous of total equivariant degree one; its polynomial dependence on  $\lambda$  is what makes the non-equivariant limit  $C_{\beta}^0$  meaningful. Since  $\text{ev}_{X_3} \circ \bar{\iota} = \iota \circ \text{ev}_{F_3}$  for the induced isomorphism  $\bar{\iota}$  of stable-map stacks, and  $\text{ev}_{F_3}$  is proper, the equivariant self-intersection formula  $\iota^* \iota_* = e_{\lambda}(K_{F_3}) \cup (-)$  gives

$$\iota^*(\text{ev}_{X_3})_*[\overline{\mathcal{M}}_{0,1}(X_3, \beta)]_{\lambda}^{\text{vir}} = e_{\lambda}(K_{F_3}) C_{\beta}^{\lambda} = (\lambda - H) C_{\beta}^{\lambda}. \quad (3.6)$$

Let  $j : F_3 \hookrightarrow Y_3$  be the inclusion.

**Lemma 3.3 (State space).** —  $j^* : H^{2k}(Y_3, \mathbb{Q}) \rightarrow H^{2k}(F_3, \mathbb{Q})$  is surjective for every  $k$ , and  $\ker j^* = \text{Ann}(H) \cap H^{2k}(Y_3, \mathbb{Q})$ , where  $\text{Ann}(H) = \{a \in H^*(Y_3, \mathbb{Q}) : aH = 0\}$ . Hence

$$H^{\text{even}}(F_3, \mathbb{Q}) \cong H^*(Y_3, \mathbb{Q}) / \text{Ann}(H),$$

and this presentation is stable under the permutation action of  $S_4$  on the factors of  $Y_3$ . The bilinear form

$$(a, b)_{X_3, \lambda} = \int_{F_3} \frac{ab}{\lambda - H}, \quad \frac{1}{\lambda - H} = \frac{1}{\lambda} + \frac{H}{\lambda^2} + \frac{H^2}{\lambda^3} + \frac{H^3}{\lambda^4},$$

is nondegenerate on  $H^{\text{even}}(F_3, \mathbb{C}(\lambda))$ .

*Proof.* — This is Proposition [2, 3.2] with  $n = 4$ , so  $\dim F_3 = 3$  is odd because all even cohomology is ambient, and the pairing and kernel statements specialize as displayed.  $\square$

**Remark 3.4.** —  $H^{\text{odd}}(F_3, \mathbb{Q}) \neq 0$ : from  $c(T_{F_3}) = 1 + H + H^2 + 2e_3$  one gets  $\int_{F_3} c_3(T_{F_3}) = 8$ , so  $b_3(F_3) = 2 + 2b_2(F_3) - 8 = 2$ . Thus  $H^*(F_3)$  is *not* entirely ambient, and Lemma 3.3 concerns the even part only. All classes occurring below — the evaluation cycles (3.4), the insertion  $\gamma$ , and every coefficient of the  $I$ -function — are even. Odd insertions contribute nothing to one-point genus-zero invariants for degree reasons (the virtual class and  $\psi$  are even), so nothing is lost by working on the even part. Also note that a permutation of the factors of  $Y_3$  need not preserve the particular hypersurface  $F_3$ ; the  $S_4$ -symmetry used below is the ring-level symmetry of the presentation  $H^*(Y_3)/\text{Ann}(H)$ , which is all that the arguments require.

Let  $\mathbb{C}^*$  act on  $X_3 = \text{Tot}(K_{F_3})$  by scaling the fibers, with equivariant parameter  $\lambda$ . Since  $X_3$  is not proper, its equivariant Gromov–Witten theory is defined through the following two conventions, both standard.

**Definition 3.5 (Local correlators and the local  $J$ -function)**

- (i) The state space is  $\mathcal{H}_{X_3} := H^{\text{even}}(F_3, \mathbb{C}(\lambda))$ , identified with the even part of  $H_{\mathbb{C}^*}^*(X_3) \otimes \mathbb{C}(\lambda)$  via  $\varpi^* = (\iota^*)^{-1}$ , with the pairing  $(, )_{X_3, \lambda}$  of Lemma 3.3. This is equivariant integration over  $X_3$  by localization to the fixed locus  $F_3$ , whose normal bundle  $K_{F_3}$  has equivariant Euler class  $\lambda - H$ .
- (ii) For  $\beta \neq 0$ ,  $k \geq 1$ ,  $\phi_i \in \mathcal{H}_{X_3}$  and  $a_i \geq 0$ ,

$$\langle \phi_1 \psi^{a_1}, \dots, \phi_k \psi^{a_k} \rangle_{0, k, \beta}^{X_3, \lambda} := \int_{[\overline{\mathcal{M}}_{0, k}(X_3, \beta)]_{\lambda}^{\text{vir}}} \prod_{i=1}^k \text{ev}_i^*(\varpi^* \phi_i) \psi_i^{a_i} \in \mathbb{C}(\lambda),$$

an honest integral over the proper stack of Proposition 3.1; by (3.3) it equals

$$\int_{[\mathcal{M}_{F_3}]^{\text{vir}}} e_{\lambda}(\mathcal{V}) \prod_i \text{ev}_i^* \phi_i \psi_i^{a_i}.$$

It lies in  $\mathbb{Q}[\lambda]$  for  $\phi_i \in H^{\text{even}}(F_3, \mathbb{Q})$ .

- (iii) The small local  $J$ -function is

$$J_{X_3}^{\lambda}(\mathbf{Q}, z) = e^{\tau/z} \left[ z \mathbf{1} + \sum_{\beta \neq 0} \mathbf{Q}^{\beta} \sum_{\alpha} \phi^{\alpha} \left\langle \frac{\phi_{\alpha}}{z - \psi} \right\rangle_{0, 1, \beta}^{X_3, \lambda} \right], \quad \tau = \sum_{i=1}^4 p_i \log Q_i, \quad (3.7)$$

where  $\{\phi_\alpha\}$  is a basis of  $\mathcal{H}_{X_3}$  and  $\{\phi^\alpha\}$  the dual basis for  $(\cdot, \cdot)_{X_3, \lambda}$ ; the expression is independent of the basis. We write  $\mathcal{J}_{X_3}^\lambda = J_{X_3}^\lambda/z$ , which has leading term  $\mathbf{1}$ ; keeping  $J$  and  $\mathcal{J}$  distinct prevents a one-power error when the primary coefficient is extracted below.

The right-hand side of (3.7) is the usual small  $J$ -function in the divisor coordinate  $\tau \in H^2$ , written with the Novikov variable and  $e^{\tau \cdot \beta}$  identified, as the string and divisor equations permit. We use throughout the convention in which  $J(\tau, z) = z\mathbf{1} + \tau + O(z^{-1})$  is the point of Givental's cone with that leading behaviour; the convention of [14, 15] places  $J(\tau, -z)$  on the cone instead, and the two are related by the substitution  $z \mapsto -z$  throughout, which we do not write out.

Let

$$G(\mathbf{y}) = \sum_{\mathbf{d} \neq 0} \mathbf{y}^{\mathbf{d}} (-1)^{D-1} \frac{D! (D-1)!}{\prod_{i=1}^4 (d_i!)^2}$$

and define the regularized  $I$ -function

$$I^{\mu, \lambda}(\mathbf{y}, z) = e^{\sum_{i=1}^4 p_i \log y_i / z} \sum_{\mathbf{d} \in \mathbb{N}^4} \mathbf{y}^{\mathbf{d}} \frac{\prod_{r=1}^D (\mu + H + rz) \prod_{r=0}^{D-1} (\lambda - H - rz)}{\prod_{i=1}^4 \prod_{r=1}^{d_i} (p_i + rz)^2}, \quad (3.8)$$

$$D = \sum_{i=1}^4 d_i,$$

with values in  $H^*(Y_3, \mathbb{C})[\mu, \lambda]$ ; each coefficient is a polynomial in  $\mu$  and  $\lambda$ . Its specialization

$$\mathcal{I}_{X_3}^\lambda(\mathbf{y}, z) := j^* I^{0, \lambda}(\mathbf{y}, z) = e^{\sum_{i=1}^4 p_i \log y_i / z} \sum_{\mathbf{d} \in \mathbb{N}^4} \mathbf{y}^{\mathbf{d}} \frac{\prod_{r=1}^D (H + rz) \prod_{r=0}^{D-1} (\lambda - H - rz)}{\prod_{i=1}^4 \prod_{r=1}^{d_i} (p_i + rz)^2} \quad (3.9)$$

takes values in  $\mathcal{H}_{X_3}$ ; the further specialization  $\lambda = 0$  is defined, even though the pairing  $(\cdot, \cdot)_{X_3, \lambda}$  has  $(\lambda - H)^{-1}$  in it.

**Proposition 3.6 (Mirror map).** — *In  $\mathcal{H}_{X_3}((z^{-1}))[[\mathbf{Q}]]$ , with  $\mathbf{Q}$  and  $\mathbf{y}$  related by  $Q_i = y_i e^{-G(\mathbf{y})}$ ,*

$$J_{X_3}^\lambda(\mathbf{Q}, z) = z e^{-\lambda G(\mathbf{y})/z} \mathcal{I}_{X_3}^\lambda(\mathbf{y}, z). \quad (3.10)$$

*Proof.* — This is Theorem [2, 5.6] together with Proposition [2, 5.1] and Proposition [2, 3.2]. The normalized local comparison holds against all ambient insertions; since  $L = 3$  is odd these span  $H^{\text{even}}(F_3)$ , so nondegeneracy identifies the whole even one-point series. Here  $g = -G$  and  $Q_i = y_i e^{-G}$ , and removing the unit input  $\lambda G$  gives the factor  $e^{-\lambda G/z}$  in (3.10).  $\square$

### 3.2. Convergence, the differential equation, and Frobenius at infinity.

— Put

$$V(x) = \sum_{i=1}^4 x_i, \quad W(x) = \left( \sum_{i=1}^4 x_i^{-1} \right)^{-1},$$

so that (1.1) at  $L = 3$  becomes

$$I_{\ominus}^{(3)}(s) = \int_{\mathbb{R}_+^4} e^{-V(x)+sW(x)} W(x) \prod_{i=1}^4 \frac{dx_i}{x_i}. \quad (3.11)$$

**Proposition 3.7 (Convergence).** — *The integral (3.11) is absolutely convergent for  $\operatorname{Re} s \leq 16$  and holomorphic for  $\operatorname{Re} s < 16$ . Absolute convergence fails for  $\operatorname{Re} s > 16$ .*

*Proof.* — Since  $|e^{sW}| = e^{aW}$  with  $a = \operatorname{Re} s$ , absolute convergence depends only on  $a$ . Write  $x_i = ru_i$ , where  $r > 0$ ,  $u_i > 0$ , and  $\sum u_i = 1$ ; the Jacobian of  $(r, u_1, u_2, u_3) \mapsto x$  is  $r^3$ , so  $\prod_i dx_i/x_i = (dr/r) d^3u/(u_1u_2u_3u_4)$  on the open simplex  $\Delta_3$ . Both  $V$  and  $W$  are homogeneous of degree one:  $V(x) = r$ ,  $W(x) = rw(u)$  with  $w(u) = (\sum u_i^{-1})^{-1}$ . The arithmetic–harmonic mean inequality gives  $w \leq 1/16$  on  $\Delta_3$ , with equality only at  $u_* = (1/4, \dots, 1/4)$ . Hence for  $a \leq 16$  one has  $1 - aw(u) > 0$  for every  $u \in \Delta_3 \setminus \{u_*\}$ , and on that set the radial integral is  $\int_0^\infty e^{-r(1-aw)} rw \, dr/r = w/(1-aw)$ . By Tonelli (the integrand is nonnegative, and  $\{u_*\}$  is a null set), absolute convergence is equivalent to finiteness of

$$\int_{\Delta_3} \frac{w(u)}{u_1u_2u_3u_4[1-aw(u)]} d^3u.$$

For  $a < 16$  the integrand is bounded by the one for  $a = 16$ , so it suffices to treat  $a = 16$ . Put  $e_3(u) = \sum_j \prod_{k \neq j} u_k$  and  $e_4(u) = u_1u_2u_3u_4$ , so that  $w = e_4/e_3$  and the integrand equals  $1/P(u)$  with  $P = e_3(1 - 16w) = e_3 - 16e_4$ . On the simplex,

$$P(u) = \sum_{i < j} u_k u_l (u_i - u_j)^2, \quad \{k, l\} = \{1, 2, 3, 4\} \setminus \{i, j\}; \quad (3.12)$$

indeed, expanding the right side gives  $\sum_i u_i^2 e_2(u_i) - 12e_4$  (where  $e_2(u_i)$  is the second elementary symmetric function of the three variables other than  $u_i$ ), and  $\sum_i u_i = 1$  gives  $e_3 = \sum_i u_i e_3(u) = \sum_i u_i^2 e_2(u_i) + 4e_4$ . Also  $\sum_{i < j} (u_i - u_j)^2 = 4 \sum_i u_i^2 - 1 = 4\|u - u_*\|^2$  on the simplex. In the neighborhood  $U = \{\max_i u_i - \min_i u_i < 1/8\}$  of  $u_*$  all  $u_i$  are bounded below by  $1/8$  (since  $\max_i u_i \geq 1/4$ ); hence the weights  $u_k u_l$  in (3.12) lie in  $[1/64, 1]$  and  $P \asymp \|u - u_*\|^2$  on  $U$ . Thus  $1/P$  is locally integrable near  $u_*$  in dimension three. On the ordered chamber  $u_1 \geq u_2 \geq u_3 \geq u_4$ , outside  $U$  one has  $u_1 - u_4 \geq 1/8$  and, keeping only the term  $(i, j) = (1, 4)$  of (3.12),  $P \geq u_2 u_3 (u_1 - u_4)^2 \geq u_2 u_3 / 64$ ; enlarging the domain of integration to  $0 \leq u_4 \leq u_3 \leq u_2 \leq 1$  gives

$$\int \frac{du_2 \, du_3 \, du_4}{P} \leq 64 \int_0^1 du_2 \int_0^{u_2} du_3 \int_0^{u_3} \frac{du_4}{u_2 u_3} = 64.$$

The  $S_4$ -symmetry of the integrand completes the estimate. Holomorphy on  $\operatorname{Re} s < 16$  follows by dominated convergence on compact subsets. For  $a > 16$ , the factor  $1 - aw(u)$  is negative on an open neighborhood of  $u_*$ , and the radial integral diverges there, so absolute convergence fails.  $\square$

The period of the three-loop sunset is the  $n = 4$  case of the generalized Apéry numbers [8, 9, 16, 17]

$$A_n = \sum_{d_1+d_2+d_3+d_4=n} \binom{n}{d_1, d_2, d_3, d_4}^2. \quad (3.13)$$

They satisfy the recursion relation [16, 17]

$$n^3 A_n = (20n^3 - 30n^2 + 18n - 4)A_{n-1} - 64(n-1)^3 A_{n-2} \quad (n \geq 1, A_{-1} = 0), \quad (3.14)$$

and  $A_n \leq 16^n$ .

**Convention 3.8 (Branches).** — On the Euclidean base ray  $s = -t$ ,  $t > 16$ , choose

$$y = -s^{-1} = t^{-1} > 0, \quad \text{Log } s = \log t + i\pi, \quad \ell := \text{Log } s - i\pi = \log t,$$

and set

$$\pi(s) = \frac{1}{s} \sum_{n \geq 0} \frac{A_n}{s^n}, \quad (3.15)$$

$$R(s) = i\pi - \text{Log } s + \sigma(s), \quad \sigma(s) = \sum_{n \geq 1} \frac{A_n}{ns^n}, \quad Q = e^{R(s)}. \quad (3.16)$$

Both series converge for  $|s| > 16$ . Then  $R = -\ell + \sigma$ ,  $R(-t) = \log y + \sum_{n \geq 1} (-1)^n A_n y^n / n$ , and  $dR/ds = -\pi(s)$ . The function  $Q(s) = -s^{-1} e^{\sigma(s)}$  is single-valued and analytic on  $|s| > 16$ , with  $Q(s) = -s^{-1} + O(s^{-2})$ ; on the Euclidean ray  $Q = t^{-1} e^{\sigma(-t)} > 0$ . Since  $s^{-1} \mapsto Q$  has derivative  $-1$  at  $s^{-1} = 0$ , it is a local biholomorphism there; hence every function meromorphic at  $s = \infty$  is a convergent Laurent series in  $Q$ , and conversely.

The Picard–Fuchs operator is [8, 18]

$$\mathcal{L}_3 = s^2(s-4)(s-16)\partial_s^3 + 6s(s^2-15s+32)\partial_s^2 + (7s^2-68s+64)\partial_s + (s-4). \quad (3.17)$$

In terms of  $\theta_y = y\partial_y$ , where  $y = -1/s$ , and of the operator

$$P_3 = \theta_y^3 + y p(\theta_y) + 64y^2(\theta_y + 1)^3, \quad p(x) = 20x^3 + 30x^2 + 18x + 4,$$

of the introduction, a direct substitution gives

$$\mathcal{L}_3 u = P_3 \left( \frac{u}{y} \right). \quad (3.18)$$

Thus the equation  $\mathcal{L}_3 I_\odot^{(3)} = -24$  of Lemma 3.9 below is the three-loop case of (1.6). Since  $\theta_y = -s\partial_s$ , one has  $\theta_y s^k = -k s^k$  for every exponent  $k$ , and (3.18) gives

$$\mathcal{L}_3 s^k = (k+1)^3 s^{k+1} - p(k) s^k + 64k^3 s^{k-1}. \quad (3.19)$$

Its singular points are  $s = 0, 4, 16, \infty$ , all regular. By (3.19), the indicial polynomial at infinity is  $(\rho+1)^3$ , and  $\mathcal{L}_3 \pi = 0$ , therefore the coefficient recurrence of  $\mathcal{L}_3$  at infinity for  $s^{-1} \sum c_n s^{-n}$  is exactly (3.14). Thus  $\pi$  is the unique solution of  $\mathcal{L}_3 u = 0$  of the form  $s^{-1} + O(s^{-2})$  without logarithms: at exponent  $-1$  the coefficient of

the  $n$ -th unknown in that recurrence is  $(-n)^3 \neq 0$  for  $n \geq 1$ , so a log-free series solution is determined by its leading coefficient.

**Lemma 3.9 (The inhomogeneous Picard–Fuchs equation)**

*On  $\operatorname{Re} s < 16$  the integral (3.11) satisfies  $\mathcal{L}_3 I_\odot^{(3)}(s) = -24$ .*

That the equal-mass sunset integral satisfies an inhomogeneous Picard–Fuchs equation whose homogeneous part is the operator governing the associated Calabi–Yau periods is standard in the Feynman-integral literature [9, 11, 18, 19], where the statement is proved at three loops and, in the last two, for the sunset family at every loop order. We therefore quote the lemma and record only what fixes the two conventions on which the rest of this section depends.

**Remark 3.10 (Normalization of the source term)**

The right-hand side is a constant, and its value  $-24 = -\int_{F_3} H^3$  is the one compatible with the normalization of the parametric Schwinger representation in (1.1) and the  $\theta_y$ -form (3.18); it is the three-loop case of the source  $H^L y^H$  certified in Theorem [2, 7.2] of [2]. Two checks pin it independently of that certificate. Expanding  $e^{sW}$  under the integral sign for  $|s| < 16$ , which Proposition 3.7 justifies, writes the Taylor coefficients of  $I_\odot^{(3)}$  as Bessel moments,

$$I_\odot^{(3)}(s) = \sum_{k \geq 0} a_k s^k, \quad a_k = \frac{8 c_{4,2k+1}}{4^k k!^2}, \quad c_{4,w} = \int_0^\infty x^w K_0(x)^4 dx,$$

and  $\mathcal{L}_3 I_\odot^{(3)} = -24$  becomes the moment recurrence of [17, 20], explicitly

$$64(k+1)^3 a_{k+1} - (20k^3 + 30k^2 + 18k + 4)a_k + k^3 a_{k-1} = 0 \quad (k \geq 1),$$

whereas its constant coefficient is  $64a_1 - 4a_0 = -24$ . Equivalently, with  $w = 2k+1$ , the moment recurrence is

$$64(w+1)c_{4,w+2} - 4w(5w^2 + 3)c_{4,w} + (w-1)^5 c_{4,w-2} = 0 \quad (w = 3, 5, \dots).$$

The endpoint condition is the initial relation  $16c_{4,3} - 4c_{4,1} = -3$ , which follows from the pole of  $c_{4,2a-1}$  at  $a = 0$ . With  $c_{4,1} = \frac{7}{8}\zeta(3)$  and  $c_{4,3} = \frac{7}{32}\zeta(3) - \frac{3}{16}$  from [17, §4.3], that relation holds; the individual values are not needed anywhere below. Since  $\mathcal{L}_3 I_\odot^{(3)} + 24$  is holomorphic on the connected domain  $\operatorname{Re} s < 16$  and vanishes on  $|s| < 16$ , the equation holds on the whole domain.

This also records a useful distinction between singularities of the operator and singularities of the physical solution. The expansion shows that  $I_\odot^{(3)}$  is holomorphic on all of  $|s| < 16$ . In particular  $s = 4$  is a singular point of (3.17) at which the particular solution  $I_\odot^{(3)}$  is holomorphic; this does not assert that 4 is an apparent singularity of the homogeneous equation, whose exponents there are  $0, \frac{1}{2}, 1$ .

**Frobenius structure at infinity.** — The construction below is the standard Frobenius method at a point of maximal unipotent monodromy, applied to  $\mathcal{L}_3$  at  $s = \infty$ , where the indicial polynomial is  $(\rho+1)^3$ . For the sunset family it is carried

out in [11] and [19]; we record only the normalizations used below, since the later arguments refer to them by name.

Let

$$u(s, \rho) = s^\rho \sum_{n \geq 0} c_n(\rho) s^{-n}, \quad c_0(\rho) = 1,$$

with the  $c_n(\rho) \in \mathbb{Q}(\rho)$  determined by inserting  $u$  into (3.19) and killing the coefficient of  $s^{\rho+1-n}$  for  $n \geq 1$ :

$$(\rho - n + 1)^3 c_n(\rho) = p(\rho - n + 1) c_{n-1}(\rho) - 64(\rho - n + 2)^3 c_{n-2}(\rho) \quad (n \geq 1, c_{-1} = 0). \quad (3.20)$$

Only  $n = 0$  survives, so

$$\mathcal{L}_3 u(s, \rho) = (\rho + 1)^3 s^{\rho+1}. \quad (3.21)$$

The denominators in (3.20) do not vanish for  $\rho$  near  $-1$ , and the nearest finite singularity in the coordinate  $s^{-1}$  is  $1/16$ , so  $\sum_n c_n(\rho) s^{-n}$  converges for  $|s| > 16$ , locally uniformly in  $\rho$  near  $-1$ ; it is therefore holomorphic in  $\rho$  there and may be differentiated termwise. Comparing (3.20) at  $\rho = -1$  with (3.14) gives  $c_n(-1) = A_n$ , and all derivatives  $c_n^{(j)}(-1)$  are rational numbers.

On the Euclidean branch put  $\tilde{u}(s, \rho) = e^{-i\pi(\rho+1)} u(s, \rho) = -e^{\rho\ell} \sum_n c_n(\rho) s^{-n}$ , so that  $\tilde{u}(s, -1) = \pi(s)$ . Define

$$\Phi_j = \frac{1}{j!} \partial_\rho^j \tilde{u} \Big|_{\rho=-1} \quad (j = 0, 1, 2), \quad I_{\text{part}} = -4 \partial_\rho^3 \tilde{u} \Big|_{\rho=-1}. \quad (3.22)$$

By (3.21),  $\mathcal{L}_3 \Phi_j = 0$  for  $j \leq 2$  and  $\mathcal{L}_3 I_{\text{part}} = -24$ . Write

$$r_j(s) = \frac{\sum_n c_n^{(j)}(-1) s^{-n}}{\sum_n A_n s^{-n}}.$$

Then  $r_0 = 1$  and  $r_j(\infty) = 0$  for  $j \geq 1$ . Restricting to a sufficiently small germ at infinity so that  $s\pi(s)$  is nonzero, one has

$$\Phi_j = \pi \sum_{i=0}^j \binom{j}{i} \frac{\ell^{j-i}}{j!} r_i, \quad I_{\text{part}} = -4\pi(\ell^3 + 3r_1\ell^2 + 3r_2\ell + r_3). \quad (3.23)$$

**Lemma 3.11 (Frobenius form of the integral).** — *There are unique convergent power series  $\hat{f}_0, \hat{f}_1, \hat{f}_2 \in \mathbb{C}\{Q\}$  such that, on the Euclidean germ  $s = -t$  with  $t$  sufficiently large,*

$$I_\odot^{(3)}(s) = \pi(s)(4R^3 + \hat{f}_2(Q)R^2 + \hat{f}_1(Q)R + \hat{f}_0(Q)). \quad (3.24)$$

Moreover  $I_\odot^{(3)} = I_{\text{part}} + \alpha_2 \Phi_2 + \alpha_1 \Phi_1 + \alpha_0 \Phi_0$  for unique constants  $\alpha_j \in \mathbb{C}$ , with

$$\hat{f}_2(0) = \alpha_2/2, \quad \hat{f}_1(0) = -\alpha_1, \quad \hat{f}_0(0) = \alpha_0,$$

and the coefficients of  $\hat{f}_r - \hat{f}_r(0)$  lie in  $\mathbb{Q} + \mathbb{Q}\alpha_1 + \mathbb{Q}\alpha_2$ . In particular, if  $\hat{f}_2(0) = \hat{f}_1(0) = 0$  then  $\hat{f}_r - \hat{f}_r(0) \in \mathbb{Q}\mathbb{Q}\{Q\}$  for  $r = 0, 1, 2$ . (The hats are removed in Theorem 3.13, where the constant term is displayed separately.)

*Proof.* — By Lemma 3.9,  $I_{\odot}^{(3)} - I_{\text{part}}$  solves  $\mathcal{L}_3 u = 0$  on the simply connected region  $\{\text{Re } s < 16, |s| > 16\}$ , where  $\Phi_0, \Phi_1, \Phi_2$  form a basis of solutions (Frobenius theory at the regular singular point  $\infty$  with indicial polynomial  $(\rho + 1)^3$ ); this gives the  $\alpha_j$ . Substituting  $\ell = \sigma - R$  into (3.23) expresses each of  $\Phi_j/\pi$  and  $I_{\text{part}}/\pi$  as a polynomial in  $R$  whose coefficients are convergent power series in  $s^{-1}$  with rational coefficients and, for the non-leading coefficients, zero constant term:

$$\begin{aligned} I_{\text{part}}/\pi &= 4R^3 - 12(\sigma + r_1)R^2 + 12(\sigma^2 + 2r_1\sigma + r_2)R - 4(\sigma^3 + 3r_1\sigma^2 + 3r_2\sigma + r_3), \\ \Phi_2/\pi &= \frac{1}{2}R^2 - (\sigma + r_1)R + \frac{1}{2}(\sigma^2 + 2r_1\sigma + r_2), \\ \Phi_1/\pi &= -R + (\sigma + r_1), \quad \Phi_0/\pi = 1. \end{aligned}$$

By Convention 3.8,  $s^{-1} = -Q + O(Q^2)$  is a convergent power series in  $Q$  with rational coefficients and zero constant term, so every coefficient above is a convergent power series in  $Q$ ; collecting powers of  $R$  gives (3.24) and the formulas for  $\hat{f}_r(0)$  and for the field of coefficients. For proving the uniqueness of the  $\hat{f}_r$ , we use that if  $\sum_{j \leq 3} g_j(Q)R^j = 0$  on the germ with  $g_j$  convergent Laurent series, then, since the local monodromy around  $s = \infty$  fixes  $Q$  and sends  $R \mapsto R - 2\pi i$  for the loop traversed counterclockwise in  $s$ , hence  $R \mapsto R - 2\pi i k$  for its iterates  $k \in \mathbb{Z}$  (both sides being analytic continuations along the same loop), one gets  $\sum_j g_j(Q)(R - 2\pi i k)^j = 0$  for  $k = 0, \dots, 3$ , and a Vandermonde argument gives  $g_j = 0$ . Equivalently,  $R$  is transcendental over the field of convergent Laurent series in  $Q$ .  $\square$

The three constants  $\alpha_j$  are connection coefficients between  $s = 0$  and  $s = \infty$ , and cannot be obtained from the differential equation. They are determined by the leading asymptotics of  $I_{\odot}^{(3)}(-t)$  as  $t \rightarrow \infty$ , which we compute from a Mellin transform.

**Proposition 3.12 (Leading asymptotics).** — As  $t \rightarrow +\infty$ ,

$$I_{\odot}^{(3)}(-t) = \frac{4 \log^3 t + 16\zeta(3)}{t} + O\left(\frac{\log^3 t}{t^2}\right). \quad (3.25)$$

Consequently, in Lemma 3.11,  $\hat{f}_2(0) = \hat{f}_1(0) = 0$  and  $\hat{f}_0(0) = -16\zeta(3)$ ; that is,  $\alpha_2 = \alpha_1 = 0$  and  $\alpha_0 = -16\zeta(3)$ .

*Proof.* — *Step 1: a priori form.* By Lemma 3.11, since  $\pi(-t) = -t^{-1}(1 + O(t^{-1}))$ ,  $R(-t) = -\log t + O(t^{-1})$  and  $Q = O(t^{-1})$ , there are constants  $\kappa_0, \dots, \kappa_3$  with

$$I_{\odot}^{(3)}(-t) = \frac{1}{t} \sum_{j=0}^3 \kappa_j \log^j t + O\left(\frac{\log^3 t}{t^2}\right), \quad \kappa_3 = 4, \kappa_2 = -\hat{f}_2(0), \kappa_1 = \hat{f}_1(0), \kappa_0 = -\hat{f}_0(0). \quad (3.26)$$

*Step 2: Mellin transform.* For  $\operatorname{Re} \nu > 0$  put  $M(\nu) = \int_{\mathbb{R}_+^4} e^{-V} W^\nu \prod_i dx_i/x_i$ . As in the proof of Lemma 3.9,

$$M(\nu) = \frac{8 \cdot 4^{1-\nu}}{\Gamma(\nu)} \mathcal{K}(2\nu), \quad \mathcal{K}(w) = \int_0^\infty x^{w-1} K_0(x)^4 dx \quad (\operatorname{Re} w > 0). \quad (3.27)$$

For  $0 < \operatorname{Re} z < 1$ , Fubini and  $\int_0^\infty t^{z-1} e^{-tW} dt = \Gamma(z) W^{-z}$  give

$$\int_0^\infty t^{z-1} I_\odot^{(3)}(-t) dt = \Gamma(z) M(1-z). \quad (3.28)$$

*Step 3: two continuations.* Fix once and for all a  $t_0 > 16$  large enough that (3.26) is valid for  $t \geq t_0$ ; such a  $t_0$  exists because (3.26) was derived from the convergent expansion of Lemma 3.11, which holds on a germ at infinity. Split the left side of (3.28) as  $\int_0^{t_0} + \int_{t_0}^\infty$ . The first integral is holomorphic for  $\operatorname{Re} z > 0$ , since  $I_\odot^{(3)}$  is continuous on the compact interval  $[-t_0, 0]$  by Proposition 3.7 and  $\int_0^{t_0} t^{\operatorname{Re} z-1} dt < \infty$  there. By (3.26), the second equals

$$\begin{aligned} \sum_{j=0}^3 \kappa_j \int_{t_0}^\infty t^{z-2} \log^j t dt + \int_{t_0}^\infty t^{z-1} \left( I_\odot^{(3)}(-t) - \frac{1}{t} \sum_j \kappa_j \log^j t \right) dt \\ = \sum_{j=0}^3 \frac{\kappa_j j!}{(1-z)^{j+1}} + h(z), \end{aligned}$$

with  $h$  holomorphic on  $\operatorname{Re} z < 2$ . Hence the left side of (3.28) continues meromorphically to  $0 < \operatorname{Re} z < 2$  with a single pole at  $z = 1$ , and the polar part there is  $\sum_j \kappa_j j! \nu^{-j-1}$  in the variable  $\nu = 1 - z$ . On the other hand, the right side continues via (3.27). We have near  $x = 0$ ,  $K_0(x) = -(\log(x/2) + \gamma_E) + O(x^2 \log x)$ , so  $K_0(x)^4 = (\log(x/2) + \gamma_E)^4 + O(x^2 \log^4 x)$  and therefore

$$\begin{aligned} \mathcal{K}(w) &= \int_0^1 x^{w-1} (\log(x/2) + \gamma_E)^4 dx + H_0(w) \\ &= \int_0^\infty e^{-wu} (u + c)^4 du + H_0(w), \quad \text{with} \quad c = \log 2 - \gamma_E, \end{aligned}$$

where  $H_0$  is holomorphic on  $\operatorname{Re} w > -2$ , using  $x = e^{-u}$  in the second line. Thus  $\mathcal{K}$  is meromorphic on  $\operatorname{Re} w > -2$  with a single pole at  $w = 0$  and polar part  $\mathcal{K}_{\text{pol}}(w) = \sum_{j=0}^4 \binom{4}{j} j! c^{4-j} w^{-j-1}$ . By the identity theorem, the two continuations of (3.28) agree, so

$$\sum_{j=0}^3 \frac{\kappa_j j!}{\nu^{j+1}} = \text{polar part at } \nu = 0 \text{ of } \Gamma(1-\nu) \frac{8 \cdot 4^{1-\nu}}{\Gamma(\nu)} \mathcal{K}(2\nu). \quad (3.29)$$

*Step 4: the polar part.* Write  $1/\Gamma(\nu) = \nu/\Gamma(1+\nu)$  and use  $\log \Gamma(1+x) = -\gamma_E x + \sum_{k \geq 2} (-1)^k \zeta(k) x^k / k$  to get

$$\frac{\Gamma(1-\nu)}{\Gamma(1+\nu)} = \exp\left(2\gamma_E \nu + \frac{2}{3}\zeta(3)\nu^3 + O(\nu^5)\right), \quad \text{and} \quad 4^{1-\nu} = 4e^{-2\nu \log 2}.$$

Only the polar part of  $\mathcal{K}(2\nu)$  contributes, and  $\mathcal{K}_{\text{pol}}(2\nu) = \int_0^\infty e^{-2\nu u} (u+c)^4 du$ . Hence the right side of (3.29) is the polar part of

$$32\nu \exp(b\nu + \frac{2}{3}\zeta(3)\nu^3) \left( \frac{24}{(2\nu)^5} + \frac{24c}{(2\nu)^4} + \frac{12c^2}{(2\nu)^3} + \frac{4c^3}{(2\nu)^2} + \frac{c^4}{2\nu} \right),$$

with

$$b = 2\gamma_E - 2\log 2 = -2c.$$

Expanding

$$e^{b\nu + \frac{2}{3}\zeta(3)\nu^3} = 1 + b\nu + \frac{b^2}{2}\nu^2 + \left(\frac{b^3}{6} + \frac{2}{3}\zeta(3)\right)\nu^3 + O(\nu^4)$$

and collecting, the polar part equals

$$32 \left[ \frac{3}{4\nu^4} + \frac{3(b+2c)}{4\nu^3} + \frac{3(b+2c)^2}{8\nu^2} + \frac{(b+2c)^3 + 4\zeta(3)}{8\nu} \right] = \frac{24}{\nu^4} + \frac{16\zeta(3)}{\nu},$$

because  $b+2c=0$ , then every term involving  $\gamma_E$  or  $\log 2$  cancels. Comparing with the left side of (3.29),  $\kappa_3 \cdot 3! = 24$ ,  $\kappa_2 = \kappa_1 = 0$  and  $\kappa_0 = 16\zeta(3)$ . This is (3.25), and the values of the  $\hat{f}_r(0)$  follow from (3.26).  $\square$

**Theorem 3.13 (Normalized three-loop expansion ( $n=4$  case))**

*On the Euclidean germ  $s = -t$ ,  $t$  sufficiently large,*

$$I_\odot^{(3)}(s) = \pi(s) (4R^3 + f_2(Q)R^2 + f_1(Q)R + f_0(Q)) - 16\zeta(3)\pi(s), \quad (3.30)$$

where  $f_r(Q) = \sum_{m \geq 1} d_r(m)Q^m \in \mathbb{Q}\mathbb{Q}\{Q\}$ , and the  $f_r$  are uniquely determined by (3.30). In terms of Lemma 3.11,  $f_2 = \hat{f}_2$ ,  $f_1 = \hat{f}_1$  and  $f_0 = \hat{f}_0 + 16\zeta(3)$ . The boundary constant has been displayed separately, so that from here on every  $f_r$  has zero constant term. This is the normalization used in the rest of the paper. Explicitly,  $I_\odot^{(3)} = I_{\text{part}} - 16\zeta(3)\pi$ , so the  $d_r(m)$  are the rational numbers obtained by expanding (3.23). The identity persists under simultaneous analytic continuation of both sides along any path in  $|s| > M$ ,  $M$  large, provided  $\text{Log } s$  and  $R$  are continued along the same path.

*Proof.* — Combine Lemma 3.11 and Proposition 3.12. The last assertion holds because both sides are analytic continuations of the same germ.  $\square$

**Remark 3.14.** — (a) Theorem 3.13 is the  $L=3$  case of [19, Theorem 4.1], whose proof proceeds through the exact convergent residue expansion of [19, Theorem 2.1]. The proof given here uses only the differential equation and the leading Mellin pole; in particular the rationality of all  $d_r(m)$  is obtained from the rationality of the Frobenius recurrence, not from an expansion of Gamma quotients.

The magnitude  $16\zeta(3)$  of the boundary constant agrees with the generating series in [19]; its sign depends on the normalization of the period and of the logarithm, which differ between the two treatments, and the complete convention comparison is now supplied by Section [2, 15.2]. With the period and logarithm fixed here, the coefficient is exactly  $-16\zeta(3)$ .

(b) The first coefficients obtained from (3.23) are

$$\begin{aligned} f_2 &= -24Q + 132Q^2 - 800Q^3 + 6018Q^4 - \frac{242424}{5}Q^5 + \dots, \\ f_1 &= 48Q^2 - 528Q^3 + 4652Q^4 + \dots, \\ f_0 &= 48Q - 138Q^2 + \frac{7144}{9}Q^3 + \dots. \end{aligned}$$

and  $d_2(1), \dots, d_2(4)$  agree with the values tabulated in [19].

(c) The particular solution  $I_{\text{part}}$  is characterized among all solutions of  $\mathcal{L}_3 u = -24$  by  $\hat{f}_2(0) = \hat{f}_1(0) = \hat{f}_0(0) = 0$ ; the Feynman integral differs from it by  $-16\zeta(3)\pi$  only. The coefficient identity of Theorem 3.22 below concerns  $\hat{f}_2 - \hat{f}_2(0)$  and is therefore independent of the values of the boundary constants; only the statement that the boundary term is exactly  $-16\zeta(3)\pi(s)$  uses Proposition 3.12.

**3.3. Diagonal restriction and the primary potential.** — From now on set  $y_i = y$  and  $Q_i = Q$  for all  $i$ , and write  $|\beta| = H \cdot \beta$  so that  $\mathbf{Q}^\beta = Q^{|\beta|}$  and  $\tau = H \log Q$ . The diagonally restricted  $I$ -function is a symmetric expression in  $p_1, \dots, p_4$  read in the  $S_4$ -stable presentation of Lemma 3.3; we use this repeatedly. **Lemma 3.15 (The coordinate  $T$ ).** — *The coefficient of  $z^{-1}$  in  $\mathcal{I}_{X_3}^0(y, y, y, y, z)$  equals  $H T(y)$  with*

$$T(y) = \log y + S(y), \quad S(y) = \sum_{n \geq 1} \frac{(-1)^n A_n}{n} y^n, \quad Q = e^{T(y)} = y e^{S(y)}. \quad (3.31)$$

The holomorphic part  $S$  has radius of convergence exactly  $1/16$ ; the logarithm is read on the branch of Convention 3.8, where  $T(-s^{-1}) = R(s)$ . Moreover  $Q = y - 4y^2 + 22y^3 - 152y^4 + O(y^5)$  is a holomorphic germ with  $Q(0) = 0$ ,  $Q'(0) = 1$ , and its inverse germ is  $y(Q) = Q + 4Q^2 + 10Q^3 + 32Q^4 + O(Q^5)$ .

*Proof.* — Assign degree one to  $z$  and to each of  $H, p_i$ ; the summand of (3.9) is then homogeneous of degree 0, so its  $z^{-1}$  coefficient is its part of cohomological degree one. Every term of a summand with  $D \geq 1$  contains the factor  $-H$  coming from  $r = 0$ ; hence the degree-one part is  $-H$  times the degree-zero part of the rest, i.e.

$$\left[ \frac{H}{z} \right] \frac{\prod_{r=1}^D (H + rz) \prod_{r=0}^{D-1} (-H - rz)}{\prod_{i=1}^4 \prod_{r=1}^{d_i} (p_i + rz)^2} = \frac{(-1)^D}{D} \binom{D}{d_1, d_2, d_3, d_4}^2,$$

using  $D!(D-1)! = (D!)^2/D$ . For the same reason a summand with  $D \geq 1$  has no degree-zero part, so it cannot combine with the degree-one part  $H \log y$  of the exponential prefactor; the degree-one part of  $\mathcal{I}_{X_3}^0$  is therefore  $H \log y +$

$H \sum_{n \geq 1} (-1)^n A_n y^n / n$  by (3.13). Convergence of  $S$  for  $|y| < 1/16$  follows from  $A_n \leq 16^n$ ; the radius is exactly  $1/16$  because, taking  $n = 4k$  and retaining only the term  $d_1 = d_2 = d_3 = d_4 = k$  in (3.13), one gets  $A_{4k} \geq \binom{4k}{k,k,k,k}^2$ , whence  $\limsup A_n^{1/n} \geq 16$  by Stirling's formula for the asymptotic behavior of the gamma function. The identification  $T(-s^{-1}) = R(s)$  is Convention 3.8. Finally  $Q = ye^{S(y)}$  is holomorphic at 0 with  $Q(0) = 0$ ,  $Q'(0) = 1$ , so the inverse germ exists and is unique.  $\square$

**Lemma 3.16 (Well-definedness of  $K_2$ ).** — *The coefficient of  $z^{-2}$  in  $e^{-HT(y)/z} \mathcal{I}_{X_3}^0(y, y, y, y, z)$  is a multiple of  $H^2$  by a holomorphic germ in  $y$  of radius  $\geq 1/16$ . Writing it as  $H^2 K_2(Q)$  with  $y = y(Q)$  defines a unique  $K_2 \in Q\mathbb{Q}\{Q\}$ , and*

$$e^{-HT(y)/z} \mathcal{I}_{X_3}^0(y, y, y, y, z) = 1 + \frac{H^2}{z^2} K_2(Q) + O(z^{-3}). \quad (3.32)$$

*Proof.* — The coefficient of  $z^{-1}$  vanishes by Lemma 3.15. The coefficient of  $z^{-2}$  is a class of cohomological degree two, i.e. an element of  $H^4(Y_3, \mathbb{Q})$  read in  $H^4(F_3, \mathbb{Q})$ ; being obtained from a diagonally restricted expression it is invariant under the permutation action of  $S_4$  on  $p_1, \dots, p_4$ . The invariant subspace of  $H^4(Y_3, \mathbb{Q}) = \bigoplus_{i < j} \mathbb{Q} p_i p_j$  is spanned by  $e_2(p) = \sum_{i < j} p_i p_j = H^2/2$ , so the coefficient is a multiple of  $H^2$ , and the multiple is unique because  $H^2 \neq 0$  in  $H^4(F_3, \mathbb{Q})$  (indeed  $\int_{F_3} H^2 \cdot H = 24$ ). This is the point at which *no division by the nilpotent class  $H$  takes place* because the symmetry, not a division, produces the scalar.

The prefactor is  $e^{-HT/z} \cdot e^{H \log y/z} = e^{-HS(y)/z}$ , so no logarithm survives, and the coefficient is a power series in  $y$ . For its radius, expand the finite products in the  $\mathbf{d}$ -summand of (3.9) to second order in the classes, then the coefficients are  $\binom{D}{\mathbf{d}}^2$  times sums of at most two harmonic numbers  $h_r \leq 1 + \log r$ ,  $r \leq D$ , so the  $y^n$ -coefficient is bounded by a constant times  $A_n(1 + \log n)^2$ , and  $A_n \leq 16^n$  gives radius  $\geq 1/16$ . Composing with the holomorphic inverse germ  $y(Q)$  of Lemma 3.15 gives  $K_2 \in \mathbb{C}\{Q\}$ ; its coefficients are rational, and  $K_2(0) = 0$  because  $\mathcal{I}_{X_3}^0 = 1 + O(y)$  at  $z = \infty$ .  $\square$

**Proposition 3.17.** — *For  $\beta \neq 0$  let  $C_\beta^\lambda$  be the proper evaluation cycle (3.4), and put  $C_+^\lambda(Q) = \sum_{\beta \neq 0} Q^{|\beta|} C_\beta^\lambda$  and  $C_+^0(Q) = \sum_{\beta \neq 0} Q^{|\beta|} C_\beta^0$ . Then*

$$[z^{-2}](e^{-\tau/z} \mathcal{J}_{X_3}^\lambda - \mathbf{1}) = (\lambda - H) C_+^\lambda(Q), \quad (3.33)$$

*$C_+^0(Q) = -H K_2(Q)$ , and the one-point potential (3.1) satisfies*

$$\mathcal{F}_\gamma(Q) = -12 K_2(Q). \quad (3.34)$$

*In particular  $\mathcal{F}_\gamma$  is a convergent power series near  $Q = 0$ , with the same radius as  $K_2$ .*

*Proof.* — Expanding  $1/(z(z - \psi)) = z^{-2} + \psi z^{-3} + \dots$  in (3.7),

$$[z^{-2}](e^{-\tau/z} \mathcal{J}_{X_3}^\lambda - 1) = \sum_{\beta \neq 0} Q^{|\beta|} \sum_{\alpha} \phi^\alpha \langle \phi_\alpha \rangle_{0,1,\beta}^{X_3,\lambda} = \sum_{\beta \neq 0} Q^{|\beta|} \sum_{\alpha} \phi^\alpha \int_{F_3} C_\beta^\lambda \phi_\alpha,$$

by Definition 3.5(ii), (3.3), and the projection formula for  $\text{ev}_{F_3}$ . The element  $u \in \mathcal{H}_{X_3}$  with  $(u, \phi_\alpha)_{X_3,\lambda} = \int_{F_3} C_\beta^\lambda \phi_\alpha$  for all  $\alpha$  is  $u = (\lambda - H)C$ , because  $\int_{F_3} (\lambda - H)C \phi_\alpha / (\lambda - H) = \int_{F_3} C \phi_\alpha$  and the Poincaré pairing is nondegenerate on the even part. This proves (3.33); it is the algebraic form of the equivariant self-intersection formula  $\iota^* \iota_* = e_\lambda(K_{F_3}) \cup (-)$ , as recorded in (3.6).

By (3.5),  $C_\beta^\lambda = C_\beta^0 + \lambda c_\beta \mathbf{1}$  with  $C_\beta^0 \in H^2(F_3, \mathbb{Q})$  and  $c_\beta \in \mathbb{Q}$ , so both sides of (3.33) are polynomial in  $\lambda$  and may be specialized to  $\lambda = 0$ . Using  $\mathcal{J}_{X_3}^0 = \mathcal{I}_{X_3}^0$  (Proposition 3.6 at  $\lambda = 0$ ),  $\tau = HT$ , and (3.32), the specialization reads

$$-H C_+^0(Q) = H^2 K_2(Q), \quad \text{i.e.} \quad H \cup (C_+^0(Q) + H K_2(Q)) = 0.$$

Both  $C_+^0$  and  $H K_2$  lie in  $H^2(F_3)$ , and  $H \cup - : H^2(F_3) \rightarrow H^4(F_3)$  is an isomorphism by Hard Lefschetz; hence

$$C_+^0(Q) = -H K_2(Q). \quad (3.35)$$

This is a cancellation by an isomorphism, not a division by the nilpotent class  $H$ . Pairing with  $\gamma = H^2/2$  and using the projection formula,  $\text{ev}_{X_3}^* \varpi^* \gamma = \text{ev}_{F_3}^* \gamma$ , and  $\int_{F_3} H^3 = 24$ :

$$\mathcal{F}_\gamma(Q) = \int_{F_3} \gamma \cup C_+^0(Q) = -K_2(Q) \int_{F_3} \frac{H^2}{2} \cdot H = -12 K_2(Q). \quad \square$$

**3.4. The logarithmic B-model recurrence.** — This subsection supplies the calculation behind  $\mathcal{P}_y B = 0$  in Lemma 3.21. Its purpose is to avoid treating the diagonally restricted logarithmic series as if its differential equation followed automatically from that of  $A$ . All computations take place over the dual numbers  $\mathbb{D} = \mathbb{C}[\varepsilon]/(\varepsilon^2)$ ; every equality below is an equality in  $\mathbb{D}$ , i.e. modulo  $\varepsilon^2$ . (We write  $\varepsilon$ , not  $\rho$ , to keep this deformation parameter distinct from the Frobenius exponent of Section 3.2, although the two play analogous roles.)

Throughout,  $y^\varepsilon$  means  $1 + \varepsilon \log y$ , and  $S_{-1} = S_{-2} = 0$ . For  $\mathbf{d} \in \mathbb{N}^4$  put

$$C_{\mathbf{d}}(\varepsilon) = \frac{\Gamma(|\mathbf{d}| + 1 + \varepsilon)^2}{\prod_{i=1}^4 \Gamma(d_i + 1 + \varepsilon/4)^2}, \quad S_n(\varepsilon) = \sum_{|\mathbf{d}|=n} C_{\mathbf{d}}(\varepsilon).$$

From  $\frac{d}{d\varepsilon} \log C_{\mathbf{d}} = 2\psi(|\mathbf{d}| + 1 + \varepsilon) - \frac{1}{2} \sum_i \psi(d_i + 1 + \varepsilon/4)$  and  $\psi(k+1) = h_k - \gamma_E$ ,

$$S_n(0) = A_n, \quad S'_n(0) = \sum_{|\mathbf{d}|=n} \frac{(n!)^2}{\prod_{i=1}^4 (d_i!)^2} \left( 2h_n - \frac{1}{2} \sum_{i=1}^4 h_{d_i} \right), \quad (3.36)$$

the Euler constants cancelling because  $2 - 4 \cdot \frac{1}{2} = 0$ , in the same way as in the proof of Lemma 3.20. Hence, by Lemma 3.20,

$$y^\varepsilon \sum_{n \geq 0} (-1)^n S_n(\varepsilon) y^n = A(y) + \varepsilon B(y). \quad (3.37)$$

Let  $\theta_y = y \partial_y$  and put

$$F(y, \varepsilon) = \sum_{k \geq 0} \frac{y^k}{\Gamma(k + 1 + \varepsilon/4)^2}, \quad h(y, \varepsilon) = F(y, \varepsilon)^4.$$

Since  $\Gamma(k + 1 + \frac{\varepsilon}{4}) = (k + \frac{\varepsilon}{4}) \Gamma(k + \frac{\varepsilon}{4})$ ,

$$(\theta_y + \frac{\varepsilon}{4})^2 F = \sum_{k \geq 0} \frac{y^k}{\Gamma(k + \varepsilon/4)^2} = yF + \frac{1}{\Gamma(\varepsilon/4)^2} = yF,$$

the correction term vanishing because  $\Gamma(\varepsilon/4)^{-2} = \varepsilon^2/16 + O(\varepsilon^3) = 0$  in  $\mathbb{D}$ . (This is the one step where working modulo  $\varepsilon^2$  is essential and not merely convenient.) Hence  $\tilde{F} = y^{\varepsilon/4} F$  satisfies  $\theta_y^2 \tilde{F} = y \tilde{F}$ . With  $v = \tilde{F}^4$ , repeated differentiation gives

$$\begin{aligned} \theta_y v &= 4 \tilde{F}^3 \theta_y \tilde{F}, \\ \theta_y^2 v &= 4 y v + 12 \tilde{F}^2 (\theta_y \tilde{F})^2, \\ \theta_y^3 v &= 4 y v + 10 y \theta_y v + 24 \tilde{F} (\theta_y \tilde{F})^3, \\ \theta_y^4 v &= 4 y v + 14 y \theta_y v + 16 y \theta_y^2 v - 24 y^2 v + 24 (\theta_y \tilde{F})^4, \\ \theta_y^5 v &= 4 y v + 18 y \theta_y v + 30 y \theta_y^2 v + 16 y \theta_y^3 v - 48 y^2 v - 24 y^2 \theta_y v + 96 y \tilde{F} (\theta_y \tilde{F})^3. \end{aligned}$$

Eliminating the last cubic term with the third line yields

$$[\theta_y^5 - y p(\theta_y) + 64 y^2 (\theta_y + 1)] v = 0, \quad p(x) = 20x^3 + 30x^2 + 18x + 4. \quad (3.38)$$

Since  $v = y^\varepsilon h$  and  $\theta_y(y^\varepsilon h) = y^\varepsilon (\theta_y + \varepsilon) h$ ,

$$[(\theta_y + \varepsilon)^5 - y p(\theta_y + \varepsilon) + 64 y^2 (\theta_y + \varepsilon + 1)] h = 0. \quad (3.39)$$

The diagonal Gamma series is obtained from  $h$  by the coefficientwise multiplier

$$C^+(y, \varepsilon) = \Gamma(\theta_y + 1 + \varepsilon)^2 h(y, \varepsilon) = \sum_{n \geq 0} S_n(\varepsilon) y^n;$$

here  $\Gamma(\theta_y + 1 + \varepsilon)^2$  is the formal operator multiplying the coefficient of  $y^n$  by the invertible element  $\Gamma(n + 1 + \varepsilon)^2 \in \mathbb{D}$ , not an analytic function of  $\theta_y$ . Writing  $\Phi = \Gamma(\theta_y + 1 + \varepsilon)^2$ , one has  $\Phi \circ y \circ \Phi^{-1} = y(\theta_y + 1 + \varepsilon)^2$  and  $\Phi \circ y^2 \circ \Phi^{-1} = y^2(\theta_y + 2 + \varepsilon)^2(\theta_y + 1 + \varepsilon)^2$ , while  $\Phi$  commutes with  $f(\theta_y)$ . Conjugating (3.39)

therefore gives

$$\begin{aligned} & \left[ (\theta_y + \varepsilon)^5 - y(\theta_y + \varepsilon + 1)^2 p(\theta_y + \varepsilon) \right. \\ & \quad \left. + 64y^2(\theta_y + \varepsilon + 1)^3(\theta_y + \varepsilon + 2)^2 \right] C^+(y, \varepsilon) = 0. \end{aligned} \quad (3.40)$$

For  $n \geq 1$  the coefficient of  $y^n$  reads

$$(n + \varepsilon)^5 S_n(\varepsilon) - (n + \varepsilon)^2 p(n - 1 + \varepsilon) S_{n-1}(\varepsilon) + 64(n - 1 + \varepsilon)^3 (n + \varepsilon)^2 S_{n-2}(\varepsilon) = 0,$$

while for  $n = 0$  it reads  $\varepsilon^5 S_0 = 0$ , which holds in  $\mathbb{D}$ . The factor  $(n + \varepsilon)^2$  is invertible in  $\mathbb{D}$  for  $n \geq 1$ , so

$$(n + \varepsilon)^3 S_n(\varepsilon) - p(n - 1 + \varepsilon) S_{n-1}(\varepsilon) + 64(n - 1 + \varepsilon)^3 S_{n-2}(\varepsilon) = 0. \quad (3.41)$$

Since  $p(-x-1) = -p(x)$  one has  $p(n-1+\varepsilon) = 20(n+\varepsilon)^3 - 30(n+\varepsilon)^2 + 18(n+\varepsilon) - 4$ , so (3.41) is the deformation of (3.14) used in Lemma 3.21; at  $n = 0$  the third-order relation reads  $\varepsilon^3 S_0 = 0$ , again valid in  $\mathbb{D}$ , so no division by a nilpotent element occurs anywhere. Substituting  $y \mapsto -y$  turns (3.41) into the statement

$$[(\theta_y + \varepsilon)^3 + y p(\theta_y + \varepsilon) + 64y^2(\theta_y + \varepsilon + 1)^3] \sum_{n \geq 0} (-1)^n S_n(\varepsilon) y^n = 0,$$

and multiplication by  $y^\varepsilon$  conjugates every  $\theta_y + \varepsilon$  to  $\theta_y$ . By (3.37) the constant term in  $\varepsilon$  is  $\mathcal{P}_y A = 0$  and the coefficient of  $\varepsilon$  is  $\mathcal{P}_y B = 0$ . This proves the logarithmic equation coefficientwise, hence in the ring of convergent germs by the bounds  $A_n \leq 16^n$  and  $h_n = O(\log n)$  used in the proof of Theorem 3.22.

**3.5. Quantum-Serre coefficients and the coefficient bridge.** — Introduce the sign-twisted series

$$I^\sharp(y, z) = e^{H \log y/z} \sum_{D \geq 0} (-1)^D y^D \sum_{|\mathbf{d}|=D} \frac{\prod_{m=1}^D (H + mz)^2}{\prod_{i=1}^4 \prod_{m=1}^{d_i} (p_i + mz)^2}, \quad (3.42)$$

and write  $I^\sharp = A(y)\mathbf{1} + B(y)H/z + O(z^{-2})$ ,  $\tau^\sharp = B/A$ .

**Remark 3.18.** — Up to the sign  $(-1)^D$ , (3.42) is the hypergeometric  $I$ -function of a smooth member  $W \in |-K_{F_3}|$ , that is, of a  $(1, 1, 1, 1)$ -complete intersection of two divisors in  $Y_3$ , which is a  $K3$  surface [12, 18]. Lemma 3.21 are statements about the formal series (3.42) alone, and no mirror theorem for  $W$  is invoked.

**Lemma 3.19 (A coefficient identity).** — *Coefficientwise, with the canonical ambient lift of (3.8) used before restriction to  $F_3$ ,*

$$z\theta_y \mathcal{I}_{X_3}^0 = H I^\sharp, \quad A = \theta_y T, \quad \tau^\sharp - T = \theta_Q K_2.$$

*Proof.* — On the degree- $D$  summand,  $z\theta_y$  multiplies by  $H + Dz$ . For  $D \geq 1$ ,

$$\prod_{m=0}^{D-1} (-H - mz) = (-1)^D H \prod_{m=1}^{D-1} (H + mz),$$

and  $(H + Dz) \prod_{m=1}^{D-1} (H + mz) = \prod_{m=1}^D (H + mz)$ , so the numerator of the  $\mathbf{d}$ -summand of  $z\theta_y \mathcal{I}_{X_3}^0$  becomes  $(-1)^D H \prod_{m=1}^D (H + mz)^2$ , which is  $H$  times the corresponding numerator in (3.42); the  $D = 0$  terms agree as well, both being  $He^{H \log y/z}$ . This proves the first identity. Applying  $z\theta_y$  to (3.32) in the form  $\mathcal{I}_{X_3}^0 = 1 + \frac{HT}{z} + \frac{H^2}{z^2} \left( \frac{T^2}{2} + K_2 \right) + O(z^{-3})$  gives  $z\theta_y \mathcal{I}_{X_3}^0 = H\theta_y T + \frac{H^2}{z} \theta_y \left( \frac{T^2}{2} + K_2 \right) + O(z^{-2})$ . Comparing coefficients of  $H$  and of  $H^2/z$  with  $HI^\sharp = HA + \frac{H^2}{z} B + O(z^{-2})$  yields  $A = \theta_y T$  and  $B = \theta_y \left( \frac{T^2}{2} + K_2 \right) = T\theta_y T + \theta_y K_2$ . Since  $T = \log Q$  we have  $\theta_y = (\theta_y T)\theta_Q = A\theta_Q$ , so  $\theta_y K_2 = A\theta_Q K_2$  and  $B = A(T + \theta_Q K_2)$ , i.e.  $\tau^\sharp = B/A = T + \theta_Q K_2$ . (The comparison of coefficients of  $H$  and  $H^2$  is legitimate for the same  $S_4$ -symmetry reason as in Lemma 3.16.)  $\square$

Set  $a_n = (-1)^n A_n$ , so that  $A(y) = \sum_{n \geq 0} a_n y^n$  by (3.13).

**Lemma 3.20 (Expansion of  $B$ ).** —  $B(y) = A(y) \log y + \tilde{B}(y)$  with  $\tilde{B}$  a holomorphic germ of radius  $\geq 1/16$ :

$$\tilde{B}(y) = \sum_{n \geq 1} b_n y^n, \quad b_n = (-1)^n \sum_{|\mathbf{d}|=n} \frac{(n!)^2}{\prod_{i=1}^4 (d_i!)^2} \left( 2h_n - \frac{1}{2} \sum_{i=1}^4 h_{d_i} \right), \quad h_n = \sum_{j=1}^n \frac{1}{j}. \quad (3.43)$$

*Proof.* — Expanding the  $\mathbf{d}$ -summand of (3.42) to first order in the classes,

$$\frac{\prod_{m=1}^D (H + mz)^2}{\prod_i \prod_{m=1}^{d_i} (p_i + mz)^2} = \binom{D}{\mathbf{d}}^2 \left[ 1 + \frac{1}{z} \left( 2h_D H - 2 \sum_i h_{d_i} p_i \right) + O(z^{-2}) \right],$$

which gives  $A$  at order  $z^0$ . For the  $H/z$  coefficient, the sum over  $\{\mathbf{d} : |\mathbf{d}| = D\}$  is  $S_4$ -symmetric, so  $\sum_{\mathbf{d}} \binom{D}{\mathbf{d}}^2 \sum_i h_{d_i} p_i = \frac{H}{4} \sum_{\mathbf{d}} \binom{D}{\mathbf{d}}^2 \sum_i h_{d_i}$ ; this is the origin of the factor  $\frac{1}{2} = \frac{2}{4}$  in (3.43). Adding the contribution  $A \log y$  of the exponential prefactor gives the expansion; the radius follows from  $h_n \leq 1 + \log n$  and  $A_n \leq 16^n$ .  $\square$

**Lemma 3.21 (The scalar B-model operator).** — Both  $A(y)$  and  $B(y)$  are annihilated by

$$\mathcal{P}_y = \theta_y^3 + y p(\theta_y) + 64y^2(\theta_y + 1)^3, \quad p(x) = 20x^3 + 30x^2 + 18x + 4. \quad (3.44)$$

Under  $y = -s^{-1}$  and the gauge  $u(s) = s^{-1}v(-s^{-1})$  one has  $(\mathcal{P}_y v)(-s^{-1}) = -\mathcal{L}_3 u(s)$ . Consequently

$$\pi(s) = \frac{A(-s^{-1})}{s}, \quad \pi(s) \tau^\sharp(-s^{-1}) = \frac{B(-s^{-1})}{s}$$

solve  $\mathcal{L}_3 u = 0$ .

*Proof.* — Reading off the coefficient of  $y^n$  in  $\mathcal{P}_y A = 0$  gives  $n^3 a_n + p(n-1)a_{n-1} + 64(n-1)^3 a_{n-2} = 0$ , which for  $a_n = (-1)^n A_n$  is (3.14), since  $p(n-1) = 20n^3 - 30n^2 + 18n - 4$ . For  $B$ , Subsection 3.4 derives — rather than assumes — the

deformed recurrence. Over the dual numbers  $\mathbb{D} = \mathbb{C}[\varepsilon]/(\varepsilon^2)$  the sums

$$S_n(\varepsilon) = \sum_{|\mathbf{d}|=n} \frac{\Gamma(n+1+\varepsilon)^2}{\prod_{i=1}^4 \Gamma(d_i+1+\varepsilon/4)^2}$$

satisfy  $S_n(0) = A_n$ ,  $S'_n(0) =$  the unsigned harmonic sum of (3.43) (Euler's constant cancels because  $2 - 4 \cdot \frac{1}{2} = 0$ ; see (3.36)), and

$$\mathcal{P}_y \left( y^\varepsilon \sum_{n \geq 0} (-1)^n S_n(\varepsilon) y^n \right) = 0 \quad \text{in } \mathbb{D}.$$

The constant term in  $\varepsilon$  is  $\mathcal{P}_y A = 0$  and the coefficient of  $\varepsilon$  is  $\mathcal{P}_y B = 0$ , by (3.43).

For the conjugation,  $\mathcal{P}_y$  is the operator  $P_3$  of (3.18), and  $v(-s^{-1}) = s u(s) = -u(s)/y$  under  $y = -s^{-1}$ ; hence  $(\mathcal{P}_y v)(-s^{-1}) = -P_3(u/y) = -\mathcal{L}_3 u$  by (3.18). Finally  $A(-s^{-1}) = \sum_n A_n s^{-n}$ , so  $A(-s^{-1})/s = \pi(s)$  by (3.15), and  $B(-s^{-1})/s = \pi \tau^\sharp(-s^{-1})$  since  $\tau^\sharp = B/A$ .  $\square$

**Theorem 3.22 (Coefficient bridge).** — *As convergent analytic germs at  $Q = 0$ ,*

$$f_2(Q) = 12(\tau^\sharp(y(Q)) - T(y(Q))) = 12\theta_Q K_2(Q) = -\theta_Q \mathcal{F}_\gamma(Q). \quad (3.45)$$

*Proof.* — *Convergence.* Neither  $\tau^\sharp = B/A$  nor  $T$  is a holomorphic germ at  $Q = 0$ , since both contain  $\log y$ ; their difference is. Indeed  $\tau^\sharp - T = \tilde{B}/A - S$  by Lemmas 3.15 and 3.20, and  $A, \tilde{B}, S$  are holomorphic germs in  $y$  of radius  $\geq 1/16$  with  $A(0) = 1$ ; composing with the holomorphic inverse germ  $y(Q)$  shows that every term of (3.45) is a convergent germ at  $Q = 0$ .

*The scalar equation in the coordinate  $R$ .* On the germ,  $Q$  and  $s$  determine each other (Convention 3.8), and  $\partial_R$  acts on functions of  $Q$  as  $\theta_Q$ , since  $Q = e^R$ . Since  $R'(s) = -\pi(s)$ , the chain rule reads  $\partial_s = -\pi \partial_R$  on functions of  $R$ . Writing  $\mathcal{L}_3 = A_3 \partial_s^3 + A_2 \partial_s^2 + A_1 \partial_s + A_0$  (the coefficients read off (3.17); the letter  $A$  avoids the  $a_n$  of Lemma 3.21) and  $\pi', \pi''$  for  $s$ -derivatives, one finds for any  $\mathcal{H}$

$$\begin{aligned} \mathcal{L}_3(\pi \mathcal{H}) &= \pi(L_3 \partial_R^3 + L_2 \partial_R^2 + L_1 \partial_R) \mathcal{H}, \\ L_3 &= -A_3 \pi^3, \quad L_2 = 6A_3 \pi \pi' + A_2 \pi^2, \\ L_1 &= -A_3 \left( \frac{3\pi'^2}{\pi} + 4\pi'' \right) - 3A_2 \pi' - A_1 \pi. \end{aligned}$$

with no zeroth-order term because  $\mathcal{L}_3 \pi = 0$ . The  $L_i$  are analytic on a punctured disc at  $Q = 0$  and meromorphic at  $Q = 0$ ; explicitly  $L_3 = -s^2(s-4)(s-16)\pi^3 \sim -s \sim Q^{-1}$ , so they are *not* power series in  $Q$ . All the statements below are equalities in the field of meromorphic germs, and the operator  $L(Q, \theta_Q) = L_3 \theta_Q^3 + L_2 \theta_Q^2 + L_1 \theta_Q$  is used only through its kernel.

*The  $R^2$ -component.* Substitute (3.30) into  $\mathcal{L}_3 I_\ominus^{(3)} = -24$  (Lemma 3.9) and divide by  $\pi$ ; the source  $-24/\pi$  is a function of  $Q$  alone, with no powers of  $R$ . Since

$\partial_R(gR^k) = (\theta_Q g)R^k + kgR^{k-1}$ , the coefficient of  $R^2$  is

$$L(Q, \theta_Q)f_2 + 12L_1 = 0, \quad (3.46)$$

the term  $12L_1$  coming from  $L_1\partial_R(4R^3) = 12L_1R^2$  and the constancy of the leading coefficient 4 (Lemma 3.11). Comparison of coefficients of powers of  $R$  is legitimate by the transcendence of  $R$  over the field of convergent Laurent series in  $Q$ , proved in Lemma 3.11.

*The A-model solution of the same equation.* By Lemmas 3.19 and 3.21,  $\pi\tau^\sharp = \pi(R + \theta_Q K_2)$  solves  $\mathcal{L}_3 u = 0$ . Dividing by  $\pi$  and collecting the coefficient of  $R^0$  gives  $L_1 + L(Q, \theta_Q)\theta_Q K_2 = 0$ ; multiplying by 12 shows that  $12\theta_Q K_2$  also satisfies (3.46).

*Uniqueness.* Put  $\Delta = f_2 - 12\theta_Q K_2$ . Then  $L(Q, \theta_Q)\Delta = 0$ , i.e.  $\pi\Delta$  solves  $\mathcal{L}_3 u = 0$ ; and  $\pi\Delta$  is log-free, being  $\pi$  times a convergent power series in  $Q$ . The coefficient recurrence (3.20) at  $\rho = -1$  has nonzero coefficient  $(-n)^3$  for every subsequent unknown,  $n \geq 1$ ; therefore the space of log-free solutions of this form is one-dimensional and spanned by  $\pi$ ; hence  $\Delta$  is constant. Both  $f_2$  and  $\theta_Q K_2$  vanish at  $Q = 0$  — for  $f_2$  this is Proposition 3.12, for  $K_2$  it is Lemma 3.16 — so  $\Delta = 0$ . The last equality in (3.45) is Proposition 3.17.  $\square$

**Remark 3.23.** — The identity also explains the absence of a quadratic term after passing to the logarithmic period-ratio coordinate  $\tau^\sharp$ . Since  $\tau^\sharp = R + \theta_Q K_2$  and  $f_2 = 12\theta_Q K_2$ , the  $R^2$  term of  $4R^3 + f_2 R^2 + \dots$  cancels when the expansion is rewritten in  $\tau^\sharp$ . This matches the distinction between  $R^{(L)}$  and the Kähler parameter noted in [19].

**3.6. The three-loop theorem and all three numerical descendant potentials.** — For  $a = 0, 1, 2$  define

$$\mathcal{D}_a(Q) = \sum_{\beta > 0} Q^{|\beta|} \int_{[\overline{\mathcal{M}}_{0,1}(X_3, \beta)]^{\text{vir}}} \psi^a \text{ev}_{X_3}^* \varpi^* \left( \frac{H^{2-a}}{(2-a)!} \right).$$

Thus  $\mathcal{D}_0 = \mathcal{F}_\gamma$ . These insertions have total complex codimension two, equal to the local one-marked virtual dimension. To define the two further coefficients without losing information, retain  $I^{0,0}(y, y, y, y, z)$  from (3.8) in  $\mathcal{R} = \mathbb{Q}[p_1, \dots, p_4]/(p_1^2, \dots, p_4^2)$ , before applying  $j^*$ , and set

$$e^{-HT(y)/z} I^{0,0}(y, y, y, y, z) = 1 + \frac{H^2}{z^2} K_2(Q) + \frac{H^3}{z^3} K_3(Q) + \frac{H^4}{z^4} K_4(Q), \quad Q = e^{T(y)}. \quad (3.47)$$

Its  $K_2$  agrees with the definition above. On  $F_3$  one has  $H^4 = 0$ ; therefore  $K_4$  must be read from the ambient series, divided as in the companion paper [2] (see Theorem [2, 5.9]), and only then used for the numerical descendant.

**Theorem 3.24 (Complete descendant specialization)**

*As convergent germs at  $Q = 0$ ,*

$$\mathcal{D}_0 = -12K_2, \quad \mathcal{D}_1 = -24K_3, \quad \mathcal{D}_2 = -24K_4, \quad (3.48)$$

and the coefficients of the three-loop expansion satisfy

$$f_2 = -\theta_Q \mathcal{D}_0, \quad f_1 = -2\mathcal{D}_0 - \theta_Q \mathcal{D}_1, \quad f_0 = -\mathcal{D}_1 - \theta_Q \mathcal{D}_2. \quad (3.49)$$

Thus if  $N_{a,m} = [Q^m] \mathcal{D}_a$ , then

$$d_2(m) = -mN_{0,m}, \quad d_1(m) = -2N_{0,m} - mN_{1,m}, \quad d_0(m) = -N_{1,m} - mN_{2,m}.$$

The full physical expression is

$$I_{\odot}^{(3)}(s) = \pi(s) [4R^3 - (\theta_Q \mathcal{D}_0)R^2 - (2\mathcal{D}_0 + \theta_Q \mathcal{D}_1)R - \mathcal{D}_1 - \theta_Q \mathcal{D}_2 - 16\zeta(3)].$$

*Proof.* — The multipliers in (3.48) are  $-4!/(2-a)!$  from Theorem [2, 5.9], and the assembly with its coefficients is Corollary [2, 9.2]. Since  $\Pi_3 = -yA = \pi$  no further sign conversion is required. This proof uses the general analytic assembly; the alternative in Section 6 avoids it. In particular, the identity for  $f_2$  alone, which is Theorem 3.25, does not use it.  $\square$

The theorem also identifies  $d_1$  and  $d_0$  with descendants. It requires no conjecture on middle cohomology. In this dimension all even one-point evaluation classes are ambient, as in Lemma 3.3.

**Theorem 3.25.** — *On the Euclidean germ  $s = -t$  for  $t$  sufficiently large, and under simultaneous analytic continuation of both sides along any path in  $|s| > M$  with  $M$  large,*

$$I_{\odot}^{(3)}(s) = \pi(s) \left[ 4R(s)^3 + \sum_{m \geq 1} Q^m (d_0(m) + d_1(m)R(s) - mN_m(\gamma)R(s)^2) \right] - 16\zeta(3)\pi(s), \quad (3.50)$$

where  $Q = e^{R(s)}$ ,  $d_0(m), d_1(m) \in \mathbb{Q}$ , and  $N_m(\gamma)$  is the genus-zero invariant (3.1). In particular

$$d_2(m) = -m N_m(\gamma) \quad (m \geq 1). \quad (3.51)$$

*Proof.* — Combine Theorem 3.13 with Theorem 3.22 and compare coefficients of  $Q^m$  in  $f_2 = -\theta_Q \mathcal{F}_\gamma$ . Properness of the moduli stacks and the virtual definition of the  $N_m(\gamma)$  are Proposition 3.1 and Proposition 3.2.  $\square$

**Remark 3.26 (Independence of the boundary constants)**

Only the explicit term  $-16\zeta(3)\pi(s)$  in (3.50) uses Proposition 3.12 in full. Identity (3.51) uses it only through  $\hat{f}_2(0) = 0$ ; and even that could be replaced by any determination of the single constant  $\alpha_2$ , since Theorem 3.22 shows unconditionally that  $\hat{f}_2 - \hat{f}_2(0) = 12\theta_Q K_2$ .

Expanding (3.9) through cohomological degree two and converting from  $y$  to  $Q$  gives

$$K_2(Q) = -2Q + \frac{11}{2}Q^2 - \frac{200}{9}Q^3 + \frac{1003}{8}Q^4 - \frac{20202}{25}Q^5 + O(Q^6),$$

whence  $\mathcal{F}_\gamma = -12K_2$  and  $f_2 = -\theta_Q \mathcal{F}_\gamma$  give, in particular,  $d_2(1), \dots, d_2(4) = -24, 132, -800, 6018$ , in agreement with the independent Feynman-side computation of Remark 3.14(b).

The following elementary check confirms the normalization of the whole chain  $K_2 \rightsquigarrow \mathcal{F}_\gamma \rightsquigarrow f_2$  at  $m = 1$  without using mirror symmetry at all.

**Proposition 3.27 (Classical verification at  $m = 1$ )**

$$N_1(\gamma) = 24.$$

*Proof.* — A class  $\beta$  with  $|\beta| = 1$  satisfies  $p_i \cdot \beta = \delta_{ij}$  for exactly one  $j$ ; call it  $\beta_j$ . A stable map of class  $\beta_j$  has irreducible image, so its image is a curve meeting the  $j$ -th ruling once and contracted by the other three projections, that is  $L_b = \mathbb{P}^1 \times \{b\}$  with  $b \in (\mathbb{P}^1)^3$  (coordinates in the factors other than  $j$ ). Writing the defining multilinear equation of  $F_3$  as  $A(b)x_j^{(0)} + B(b)x_j^{(1)}$ , one has  $L_b \subset F_3$  if and only if  $A(b) = B(b) = 0$ . Thus the lines of class  $\beta_j$  are parametrized by the complete intersection  $C_j = \{A = B = 0\}$ , of class  $h^2$  in  $(\mathbb{P}^1)^3$ , where  $h$  is the sum of the three remaining point classes.

At a point of  $C_j$ , dependence of  $dA$  and  $dB$  would give  $[x_j^{(0)} : x_j^{(1)}]$  for which the base differential of  $Ax_j^{(0)} + Bx_j^{(1)}$  vanishes. Its fiber differential already vanishes there; this would be a singular point of  $F_3$ . Hence  $dA, dB$  are independent, so  $C_j$  is a smooth regular codimension-two zero locus. The ambient one-marked degree- $\beta_j$  moduli is  $\mathbb{P}^1 \times (\mathbb{P}^1)^3$ ; the section cutting out the hypersurface moduli cuts out  $\mathbb{P}^1 \times C_j$  over arbitrary bases.

Since  $L = L_b \subset F_3 \subset Y_3$  and  $N_{L/Y_3} = \mathcal{O}_L^{\oplus 3}$ , the normal sequence of  $L \subset F_3 \subset Y_3$  reads  $0 \rightarrow N_{L/F_3} \rightarrow \mathcal{O}_L^{\oplus 3} \rightarrow \mathcal{O}_L(H \cdot \beta_j) = \mathcal{O}_L(1) \rightarrow 0$ . A surjection  $\mathcal{O}_L^{\oplus 3} \twoheadrightarrow \mathcal{O}_L(1)$  is given by three sections of  $\mathcal{O}_L(1)$  without common zero, hence spanning  $H^0(\mathcal{O}_L(1))$ ; after a change of basis the map is  $(f_1, f_2, f_3) \mapsto f_1 s_0 + f_2 s_1$  for a basis  $s_0, s_1$ , so

$$N_{L/F_3} \cong \mathcal{O}_L \oplus \mathcal{O}_L(-1).$$

Therefore  $H^1(L, N_{L/F_3}) = 0$ , and from  $0 \rightarrow T_L \rightarrow T_{F_3}|_L \rightarrow N_{L/F_3} \rightarrow 0$  with  $H^1(\mathcal{O}_L(2)) = 0$  we get  $H^1(L, T_{F_3}|_L) = 0$ . Consequently  $\overline{\mathcal{M}}_{0,1}(F_3, \beta_j)$  is unobstructed of dimension  $1 + 1 = 2 = \text{vdim}$ , its virtual class is its fundamental class, and it has no boundary, because a contracted subtree with  $v \geq 1$  components, one attachment, and at most one marking has at most  $2v$  special points in total, whereas stability requires at least  $3v$ . Thus it is the universal line  $\mathbb{P}^1 \times C_j$ , with  $\text{ev}(t, b) = (t, b)$ . Since  $\text{rk } \mathcal{V} = |\beta_j| - 1 = 0$ , Proposition 3.2 identifies  $N_{\beta_j}(\gamma)$  with  $\int_{\mathbb{P}^1 \times C_j} \text{ev}^* \gamma$ .

Split  $\gamma = \sum_{i < k} p_i p_k$  into the part containing  $p_j$  and the rest. The second part is pulled back from the one-dimensional  $C_j$  in degree 4, hence vanishes. The first part is  $p_j \cdot h$ , and  $\text{ev}^* p_j$  is the point class of the  $\mathbb{P}^1$  factor, so

$$\int_{\mathbb{P}^1 \times C_j} \text{ev}^* \gamma = \deg(h|_{C_j}) = \int_{(\mathbb{P}^1)^3} h \cdot h^2 = \int_{(\mathbb{P}^1)^3} h^3 = 3! = 6.$$

Summing over the four classes gives  $N_1(\gamma) = 4 \cdot 6 = 24$ .  $\square$

This matches the coefficient of  $Q$  in  $\mathcal{F}_\gamma = -12K_2$  read off above, and hence  $d_2(1) = -1 \cdot 24 = -24$ , which is the coefficient produced by the Frobenius recurrence in Remark 3.14(b). In particular the sign conventions of Convention 3.8, the normalization  $\gamma = H^2/2$ , and the factor  $-12$  relating  $\mathcal{F}_\gamma$  to  $K_2$  are all consistent.

**3.7. The Gamma-class boundary number.** — We use the convention  $\widehat{\Gamma}(E) = \prod_j \Gamma(1 + \delta_j)$  for the Chern roots  $\delta_j$  of  $E$ , with no factors of  $2\pi i$ . In  $K^0(F_3)$ ,

$$[T_{X_3}|_{F_3}] = [T_{F_3}] + [K_{F_3}] = [T_{Y_3}|_{F_3}] - [\mathcal{O}_{F_3}(H)] + [\mathcal{O}_{F_3}(-H)],$$

using  $T_{Y_3}|_{F_3} = T_{F_3} + N_{F_3/Y_3}$  and  $N_{F_3/Y_3} = \mathcal{O}_{F_3}(H)$ ,  $K_{F_3} = \mathcal{O}_{F_3}(-H)$ . Since the Chern roots of  $T_{Y_3}$  are  $2p_1, \dots, 2p_4$ ,

$$\widehat{\Gamma}_{X_3}|_{F_3} = \frac{\Gamma(1-H)}{\Gamma(1+H)} \prod_{i=1}^4 \Gamma(1+2p_i). \quad (3.52)$$

**Proposition 3.28 (Numerical Gamma identity).** —  $\log(\widehat{\Gamma}_{X_3}|_{F_3}) = \frac{2}{3}\zeta(3)H^3$  exactly, and

$$\int_{F_3} [\widehat{\Gamma}_{X_3}|_{F_3}]_6 = \frac{2\zeta(3)}{3} \int_{F_3} H^3 = 16\zeta(3).$$

*Proof.* — From  $\log \Gamma(1+x) = -\gamma_E x + \sum_{k \geq 2} (-1)^k \zeta(k) x^k/k$  and  $p_i^2 = 0$  we get  $\sum_i \log \Gamma(1+2p_i) = -2\gamma_E H$ , while

$$\log \Gamma(1-H) - \log \Gamma(1+H) = 2\gamma_E H + \sum_{k \geq 2} \frac{\zeta(k)}{k} (1 - (-1)^k) H^k = 2\gamma_E H + \frac{2\zeta(3)}{3} H^3,$$

because the even- $k$  terms cancel and  $H^k = 0$  on  $F_3$  for  $k \geq 4$ . The  $\gamma_E$ -terms cancel between the two contributions, so  $\log \widehat{\Gamma}_{X_3}|_{F_3}$  is purely of degree six; exponentiating, its degree-six part is itself. Now  $\int_{F_3} H^3 = 24$ .  $\square$

This is the absolute value of the boundary coefficient in (3.30). With the normalization  $R' = -\pi$  used in this work, it is the negative of the coefficient of the Feynman integral. The Mellin computation and the direct Gamma calculation are independent, and Proposition 3.28 does *not* identify the integral lattice underlying the Feynman Frobenius basis with the Gamma-class integral structure; no such global normalization/continuation theorem is claimed here.

**3.8. Analytic domain of the three-loop identity.** — Theorem 3.25 is an identity of germs at  $s = \infty$ , together with its analytic continuation along paths; the two sides of (3.50) are solutions of  $\mathcal{L}_3 u = -24$  and continue along any path in  $|s| > 16$ , where  $\mathcal{L}_3$  has no singular point. The following explicit statements concern the *series* on the right of (3.50), which need not converge on all of  $|s| > 16$ .

By Convention 3.8,  $Q(s) = -s^{-1}e^{\sigma(s)}$  is single-valued and analytic on  $|s| > 16$ , whereas  $R = \log Q$  is determined only up to  $2\pi i\mathbb{Z}$  and is single-valued on the universal cover of  $\{|s| > 16\}$ . From  $A_n \leq 16^n$ ,

$$|Q(s)| \leq \frac{1}{|s|} \exp\left(\sum_{n \geq 1} \frac{16^n}{n|s|^n}\right) = \frac{1}{|s| - 16} \quad (|s| > 16).$$

Let  $\rho > 0$  be the common radius of convergence of  $f_2$  and  $\mathcal{F}_\gamma$  (Proposition 3.17 and Theorem 3.22; the two series differ by the operation  $\theta_Q$  and a sign, which do not change the radius). For  $M > 16 + \rho^{-1}$  and  $|s| > M$  one has  $|Q(s)| < \rho$ , and the identity

$$f_2(Q(s)) = -(\theta_Q \mathcal{F}_\gamma)(Q(s)) \quad (3.53)$$

holds as an equality of *single-valued* analytic functions on  $\{|s| > M\}$ , since only  $Q$  and not  $R$  enters. Likewise, if  $\rho' > 0$  is a common radius of convergence of  $f_0, f_1, f_2$  (which exists by Lemma 3.11) and  $M' > 16 + \rho'^{-1}$ , then the full formula (3.50) holds on the universal cover of  $\{|s| > M'\}$  with all series convergent. No continuation through the singular points  $s = 4, 16$  of  $\mathcal{L}_3$ , and no control of the associated monodromy, is claimed by these exterior-domain estimates.

Corollary [2, 10.5] of the companion paper [2], makes these bounds explicit, namely all the three-loop descendant series are holomorphic for  $|Q| < 1/12$ . Thus  $|s| > 28$  is a sufficient exterior domain for the convergent series identities above, using  $|Q(s)| \leq 1/(|s| - 16)$ . The analogous four-loop estimate is  $|s| > 45$ , using  $1/20$  and  $25$ . These sufficient domains do not assert optimality. They concern convergence of the descendant Taylor series in the mirror coordinate  $Q$ . Separately, Proposition [2, 8.2] of the companion paper [2] proves that the Taylor series of the physical Feynman integral in  $s$  has exact radius 16 at three loops and 25 at four loops; in both cases the Feynman integral is continuous but not  $C^1$  at the physical threshold. Thus the  $Q$ -Taylor and  $s$ -Taylor disks are distinct analytic statements.

**Additional real-axis continuation from the companion paper [2].** By Theorem [2, 10.1] and Corollary [2, 10.2],  $A(y) > 0$  and the real mirror coordinate  $Q(y) = ye^{S(y)}$  is strictly increasing for  $y > 0$ . Its range is  $(0, Q_\infty)$ , where

$$Q_\infty = \exp\left(-\int_{(S^1)^4} \log |w_1 + w_2 + w_3 + w_4|^2 d\nu\right), \quad 1/16 \leq Q_\infty \leq 1$$

for normalized Haar measure  $\nu$ . The three descendant germs have the specified real analytic continuation along this interval, and the complete formula of Theorem 3.24 holds for all real  $s < 0$  with these continuations. This extends the exterior-domain

statement without asserting Taylor convergence on the entire interval. No positive physical singular point is crossed on this negative-real path.

#### 4. Three-loop integrality for arbitrary integral insertions, and an explicit quantum- $K$ formula

Corollary [2, 13.10] states primary integrality for  $\gamma_L = \delta_0 = H^{L-1}/(L-1)!$  in every  $L \geq 3$ . Here we give the simpler fourfold projective-bundle argument for every integral primary class  $\eta \in H^4(F_3, \mathbb{Z})$  and specialize the quantum- $K$  formula. The dimension restriction concerns this compactification's semipositivity argument, not the scope of primary integrality on the compact Calabi–Yau cover of the companion paper [2]. The notation is that of Section 3.1.

For  $\eta \in H^4(F_3, \mathbb{Z})$  define classwise numbers  $b_\beta(\eta)$  by

$$N_\beta(\eta) = \sum_{\substack{k \geq 1, \alpha \\ \beta = k\alpha}} \frac{b_\alpha(\eta)}{k^2}, \quad (4.1)$$

and aggregate numbers  $n_m(\gamma)$  by

$$\mathcal{F}_\gamma(Q) = \sum_{m \geq 1} n_m(\gamma) \text{Li}_2(Q^m), \quad \text{Li}_2(q) = \sum_{k \geq 1} \frac{q^k}{k^2}. \quad (4.2)$$

Here  $N_\beta(\eta)$  denotes (3.1) with  $\gamma$  replaced by  $\eta$ . The classwise sums range over integral roots in  $H_2(F_3, \mathbb{Z})$ ; we assign zero to  $N_\alpha$  and  $b_\alpha$  on noneffective classes of positive  $H$ -degree. This convention makes all root sums unambiguous. Both transforms are triangular with unit diagonal, hence uniquely solvable over  $\mathbb{Q}$ ; no convergence is needed for this formal step. Comparing coefficients,

$$N_m(\gamma) = \sum_{k|m} \frac{n_{m/k}(\gamma)}{k^2}, \quad (4.3)$$

$$n_m(\gamma) = \sum_{k|m} \frac{\mu(k)}{k^2} N_{m/k}(\gamma), \quad (4.4)$$

the second by multiplying (4.3) by  $m^2$ , putting  $d = m/k$  to get  $m^2 N_m = \sum_{d|m} d^2 n_d$ , and applying ordinary Möbius inversion to  $G(m) = m^2 N_m$ ,  $F(d) = d^2 n_d$ .

**Theorem 4.1 (Genus-zero Gopakumar–Vafa integrality, one insertion; quoted)**

Let  $Z'$  be a smooth projective complex fourfold which is semi-positive, and let  $\Gamma \in H^4(Z', \mathbb{Z})$ . For an effective class  $A$  with  $K_{Z'} \cdot A = 0$  define  $B_A(\Gamma)$  by

$$\text{GW}_{0,A}^{Z'}(\Gamma) = \sum_{\substack{k \geq 1, A' \\ A = kA'}} \frac{B_{A'}(\Gamma)}{k^2}, \quad \text{GW}_{0,A}^{Z'}(\Gamma) = \int_{[\mathcal{M}_{0,1}(Z', A)]^{\text{vir}}} \text{ev}^* \Gamma.$$

Then  $B_A(\Gamma) \in \mathbb{Z}$ .

**Remark 4.2.** — This is the one-insertion, fourfold,  $K$ -degree-zero case of [5, Thm. 1.1], which restates [4, Thm. 9.2]. For a semi-positive variety of dimension  $\geq 3$  and integral even insertions  $\Gamma_1, \dots, \Gamma_n$ , the genus-zero Gopakumar–Vafa type invariants defined through the Klemm–Pandharipande multiple-cover weight  $k^{-(3-n)}$  in the  $K$ -degree-zero sector [21] are integers; for  $n = 1$  the weight is  $k^{-2}$ . Chou’s proof is algebraic as it compares the invariants with integral quantum  $K$ -theoretic Euler characteristics through virtual Kawasaki–Riemann–Roch, semi-positivity being what bounds the virtual dimensions of the nontrivial Kawasaki strata; the one-insertion computation there [5, Prop. 3.4] concerns insertions of degree  $2(\dim Z' - K_{Z'} \cdot A - 2)$ , which for  $\dim Z' = 4$  and  $K_{Z'} \cdot A = 0$  is exactly  $H^4(Z')$ , the degree used below. The semi-positivity hypothesis is satisfied by  $Z$  in either of the usual senses, namely the algebraic one, that  $c_1(TZ) \cdot A \geq 0$  for every effective curve class, holds because  $-K_Z$  is nef (Lemma 4.3); and the symplectic one of [4], which in real dimension eight forbids classes  $A$  with  $\omega(A) > 0$  and  $c_1(A) = -1$ , holds because  $c_1(TZ) = 2\xi$  takes only even values on integral classes.

We now prove integrality rather than infer it from a finite table. Put

$$Z = \mathbb{P}_{\text{lines}}(\mathcal{O}_{F_3} \oplus K_{F_3}), \quad \varpi_Z : Z \rightarrow F_3, \quad \xi = c_1(\mathcal{O}_Z(1)),$$

let  $D_0, D_\infty$  be the sections corresponding to the summands  $\mathcal{O}_{F_3}$  and  $K_{F_3}$ , and let  $i : F_3 = D_0 \hookrightarrow Z$ .

**Lemma 4.3 (Compactification geometry).** — *The variety  $Z$  is a smooth projective fourfold, being a  $\mathbb{P}^1$ -bundle over the smooth projective threefold  $F_3$ . Moreover*

$$Z \setminus D_\infty \cong X_3, \quad [D_\infty] = \xi, \quad i^*\xi = 0, \quad \mathcal{O}_Z(D_\infty)|_{D_\infty} \cong K_{F_3}^{-1}, \quad c_1(TZ) = 2\xi,$$

and  $\xi$ , hence  $-K_Z$ , is nef.

**Lemma 4.4 (No maps at infinity).** — *Every connected genus-zero stable map to  $Z$  of nonzero class  $i_*\beta$  has image in  $Z \setminus D_\infty = X_3$ , and*

$$\overline{\mathcal{M}}_{0,1}(X_3, \beta) \cong \overline{\mathcal{M}}_{0,1}(Z, i_*\beta) \tag{4.5}$$

as stacks, with identical obstruction theories and virtual classes.

**Lemma 4.5 (Curve classes).** —  *$H_2(Z, \mathbb{Z}) = i_*H_2(F_3, \mathbb{Z}) \oplus \mathbb{Z}[\ell]$  with  $\xi \cdot \ell = 1$ . If  $kA = i_*\beta$  in  $H_2(Z, \mathbb{Z})$  with  $k \geq 1$ , then  $A = i_*\alpha$  with  $k\alpha = \beta$ ; and  $i_*\alpha$  is effective if and only if  $\alpha$  is.*

**Theorem 4.6 (BPS integrality).** — *For every  $\eta \in H^4(F_3, \mathbb{Z})$  and every nonzero effective  $\beta$  one has  $b_\beta(\eta) \in \mathbb{Z}$ . Consequently  $n_m(\gamma) \in \mathbb{Z}$  for every  $m \geq 1$ .*

Combining (3.51), (4.3) and Theorem 4.6,

$$d_2(m) = -m N_m(\gamma) = -m \sum_{k|m} \frac{n_{m/k}(\gamma)}{k^2}, \quad n_m(\gamma) \in \mathbb{Z}, \tag{4.6}$$

and, since  $\theta_Q \text{Li}_2(Q^m) = -m \log(1 - Q^m)$ , the formal Euler product

$$\exp(f_2(Q)) = \prod_{m \geq 1} (1 - Q^m)^{m n_m(\gamma)} \in 1 + Q \mathbb{Z}[[Q]]. \quad (4.7)$$

Putting  $A_m = m^2 N_m(\gamma) = -m d_2(m)$  gives  $A_m = \sum_{d|m} d^2 n_d(\gamma) \in \mathbb{Z}$  and, for  $p$  prime,  $a \geq 1$  and  $p \nmid e$ ,

$$A_{p^a e} - A_{p^{a-1} e} \equiv 0 \pmod{p^{2a}}, \quad (4.8)$$

since subtracting the two divisor sums leaves exactly the divisors of  $p^a e$  of  $p$ -adic valuation  $a$ , each contributing  $p^{2a} d'^2 n_{p^a d'}$ .

Table 1 records the exact values through  $m = 19$ . The Gromov–Witten column is obtained from (3.34), the sunset column can also be obtained independently from (3.20), and the BPS column is the Möbius transform (4.4) of the Gromov–Witten column. All nineteen rows agree by exact rational arithmetic. The BPS column is therefore not an independent enumerative computation, and no finite part of the table is used as a proof of the all-degree statements.

Table 1: Sunset coefficients, Gromov–Witten aggregates, and BPS transforms.

| $m$ | $d_2(m)$                 | $N_m(\gamma) = -d_2(m)/m$ | $n_m(\gamma)$      |
|-----|--------------------------|---------------------------|--------------------|
| 1   | −24                      | 24                        | 24                 |
| 2   | 132                      | −66                       | −72                |
| 3   | −800                     | 800/3                     | 264                |
| 4   | 6018                     | −3009/2                   | −1488              |
| 5   | −242424/5                | 242424/25                 | 9696               |
| 6   | 415088                   | −207544/3                 | −69240             |
| 7   | −25875552/7              | 25875552/49               | 528072             |
| 8   | 33914049                 | −33914049/8               | −4238880           |
| 9   | −954894248/3             | 954894248/27              | 35366424           |
| 10  | 15208750932/5            | −7604375466/25            | −304177440         |
| 11  | −324462499968/11         | 324462499968/121          | 2681508264         |
| 12  | 289525458104             | −72381364526/3            | −24127104048       |
| 13  | −37323117924480/13       | 37323117924480/169        | 220846851624       |
| 14  | 201031033602384/7        | −100515516801192/49       | −2051337209592     |
| 15  | −289441832285280         | 19296122152352            | 19296122151264     |
| 16  | 5872668577670337/2       | −5872668577670337/32      | −183520891992384   |
| 17  | −509335737033576096/17   | 509335737033576096/289    | 1762407394579848   |
| 18  | 921823061292056348/3     | −460911530646028174/27    | −17070797440168296 |
| 19  | −60149764030617173760/19 | 60149764030617173760/361  | 166619844960158376 |

#### 4.1. Proofs of the supporting statements. —

*Compactification and curve-by-curve nefness.* —

*Proof.* — A line in  $\mathcal{O}_x \oplus K_{F_3, x}$  with nonzero first coordinate is spanned by a unique  $(1, v)$ , which gives  $Z \setminus D_\infty \cong X_3$ ; the projection of the tautological subbundle to  $\mathcal{O}_{F_3}$  is a section of  $\mathcal{O}_Z(1)$  vanishing simply along  $D_\infty$ , which gives  $[D_\infty] = \xi$ ,

$i^*\xi = 0$  and the normal bundle  $N_{D_\infty/Z} \cong K_{F_3}^{-1}$ . The relative Euler sequence gives  $c_1(T_{Z/F_3}) = 2\xi + \varpi_Z^*c_1(K_{F_3})$ ; adding  $\varpi_Z^*c_1(TF_3) = -\varpi_Z^*c_1(K_{F_3})$  gives  $c_1(TZ) = 2\xi$ . For nefness let  $C \subset Z$  be an integral curve. If  $C \not\subset D_\infty$  then  $\xi \cdot C = D_\infty \cdot C \geq 0$ . If  $C \subset D_\infty$  then  $\xi \cdot C = \deg(N_{D_\infty/Z}|_C) = -K_{F_3} \cdot \varpi_{Z*}C = H \cdot \varpi_{Z*}C > 0$ , as  $H$  is ample on  $F_3$ .  $\square$

*Excluding the infinity section in families. —*

*Proof.* — Since  $i^*\xi = 0$ , the total intersection is  $\xi \cdot i_*\beta = 0$ . By the proof of Lemma 4.3 each nonconstant component has nonnegative  $\xi$ -degree, strictly positive if contained in  $D_\infty$ ; as the total is zero, no nonconstant component lies in  $D_\infty$  and each has  $\xi$ -degree 0, hence (all local intersection multiplicities being nonnegative) meets  $D_\infty$  nowhere. A contracted component mapping into  $D_\infty$  would be joined through a chain of contracted components to a nonconstant one, which would then meet  $D_\infty$ ; contradiction.

For the stack statement,  $X_3 \hookrightarrow Z$  is an open immersion, so  $\overline{\mathcal{M}}_{0,1}(X_3, \beta)$  is the open substack of  $\overline{\mathcal{M}}_{0,1}(Z, i_*\beta)$  on which the universal image avoids  $D_\infty$ . In a family, the locus where the image meets  $D_\infty$  is closed, being the image under the proper map  $\varrho$  of the closed substack  $\mu^{-1}(D_\infty)$  of the universal curve. The preceding paragraph says this closed substack has no geometric points, hence is empty. This gives an isomorphism of stacks including nilpotent structure, not merely a bijection on geometric points. Finally  $T_Z|_{X_3} = T_{X_3}$ , so the relative tangent-obstruction complexes and their maps to the cotangent complex agree, and so do the virtual classes; properness is that of  $\overline{\mathcal{M}}_{0,1}(Z, i_*\beta)$ .  $\square$

*Integral covering roots and effectivity. —*

*Proof.* — The splitting is the projective-bundle formula together with the section  $i$ , and  $H_2(F_3, \mathbb{Z}) \cong \mathbb{Z}^4$  by Proposition [2, 3.1]. If  $kA = i_*\beta$ , intersecting with  $\xi$  gives  $k\xi \cdot A = 0$ , so the fiber coefficient of  $A$  vanishes and  $A = i_*\alpha$ ; applying  $\varpi_{Z*}$  gives  $k\alpha = \beta$ , and  $H_2(F_3, \mathbb{Z})$  is torsion-free. If  $\alpha$  is effective so is  $i_*\alpha$ . Conversely, an effective cycle representing  $i_*\alpha$  has total  $D_\infty$ -intersection 0, so by the proof of Lemma 4.4 each of its components avoids  $D_\infty$ , and its image under  $\varpi_Z$  is an effective cycle of class  $\alpha$ .  $\square$

*Fourfold semipositivity and arbitrary integral insertions. —*

*Proof.* — By Lemma 4.3 and Remark 4.2,  $Z$  is a smooth projective semi-positive fourfold, and  $c_1(TZ) \cdot i_*\beta = 2\xi \cdot i_*\beta = 0$ , so  $i_*\beta$  lies in the  $K_Z$ -degree-zero sector. The class  $\varpi_Z^*\eta \in H^4(Z, \mathbb{Z})$  is integral, and

$$\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,1}(Z, i_*\beta) = (4 - 3) + c_1(TZ) \cdot i_*\beta + 1 = 2,$$

so  $\mathrm{GW}_{0, i_*\beta}^Z(\varpi_Z^*\eta)$  is a well-defined number and Theorem 4.1 applies. By Lemma 4.4 and  $\varpi_Z \circ i = \mathrm{id}_{F_3}$ ,  $\mathrm{GW}_{0, i_*\beta}^Z(\varpi_Z^*\eta) = N_\beta(\eta)$ ; by Lemma 4.5 the  $k$ -th roots of  $i_*\beta$  in  $H_2(Z, \mathbb{Z})$  are exactly the images of the  $k$ -th roots of  $\beta$  in  $H_2(F_3, \mathbb{Z})$ , with the same

effectivity. Hence the two multiple-cover transforms (4.1) and that of Theorem 4.1 coincide, and by uniqueness of the triangular transform  $b_\beta(\eta) = B_{i_*\beta}(\varpi_Z^*\eta) \in \mathbb{Z}$ .

For fixed  $m$ , the nef classes  $p_i$  give nonnegative integer coordinates for effective  $\beta$ , summing to  $m$ ; the curve-lattice identification of Proposition [2, 3.1] therefore gives only finitely many such classes. Summing (4.1) over them and using  $|k\alpha| = k|\alpha|$  gives  $N_m(\gamma) = \sum_{k|m} k^{-2} \sum_{|\alpha|=m/k} b_\alpha(\gamma)$ ; comparing with (4.3) and using uniqueness again,  $n_m(\gamma) = \sum_{|\beta|=m} b_\beta(\gamma) \in \mathbb{Z}$ .  $\square$

This section incorporates the quantum- $K$  strengthening for the same  $F_3$ ,  $X_3 = \text{Tot}(K_{F_3})$ , and insertion  $\gamma = H^2/2$ . It is independent of the analytic-enumerative bridge, but uses the same compactification and the quantum- $K$  theorem underlying Chou's integrality proof. It is therefore an explicit refinement of that integrality argument, not a second independent proof of Chou's result.

**4.2. Compactification and identification of invariants.** — Use the lines convention and notation of Section 4:

$$Z = \mathbb{P}_{\text{lines}}(\mathcal{O}_{F_3} \oplus K_{F_3}), \quad \varpi_Z : Z \longrightarrow F_3, \quad i : F_3 = D_0 \hookrightarrow Z.$$

Lemmas 4.3–4.5 establish, respectively,  $-K_Z = 2\xi$  nef, the isomorphism of stable-map stacks

$$\overline{\mathcal{M}}_{0,1}(X_3, \beta) \simeq \overline{\mathcal{M}}_{0,1}(Z, i_*\beta),$$

with their standard obstruction theories, and the equality of effective integral root sets. The stack isomorphism is induced by an open immersion and holds over arbitrary bases. It does not pass through a coarse moduli space or discard nilpotents. In particular

$$N_\beta(\gamma) = \text{GW}_{0,i_*\beta}^Z(\varpi_Z^*\gamma), \quad b_\beta(\gamma) = B_{i_*\beta}(\varpi_Z^*\gamma). \quad (4.9)$$

The second equality follows from the first by uniqueness of the classwise multiple-cover transform, with the zero convention for noneffective roots. All these statements were proved without a mirror theorem.

**4.3. An integral lift with exact Chern character.** — Set  $L_i = \varpi_Z^*\mathcal{O}_{F_3}(p_i)$  and define

$$\mathcal{E}_\gamma = \sum_{1 \leq i < j \leq 4} (1 - [L_i^{-1}]) (1 - [L_j^{-1}]) \in K_{\text{alg}}^0(Z). \quad (4.10)$$

Here  $K_{\text{alg}}^0$  is the Grothendieck ring of algebraic vector bundles; its unit is the trivial line bundle. The lift is an integral virtual class, not a vector bundle of positive rank.

**Lemma 4.7.** — *In rational even cohomology,  $\text{ch}(\mathcal{E}_\gamma) = \varpi_Z^*\gamma$  exactly.*

For  $\alpha \neq 0$  effective define the true quantum- $K$  invariant

$$\kappa_\alpha = \chi(\overline{\mathcal{M}}_{0,1}(Z, i_*\alpha), \text{ev}_1^* \mathcal{E}_\gamma \otimes \mathcal{O}^{\text{vir}}) \in \mathbb{Z}, \quad (4.11)$$

where  $\mathcal{O}^{\text{vir}}$  is the virtual structure sheaf for the standard stable-map obstruction theory [22]. Put  $\kappa_0 = 0$  and put  $\kappa_\alpha = 0$  for noneffective classes of positive  $H$ -degree. The Euler characteristic is taken on the proper Deligne–Mumford stack. In characteristic zero its coherent cohomology groups are finite-dimensional and vanish in sufficiently high degree. The tame coarse-space morphism has exact pushforward, and the coarse space is proper. Taking their alternating sum, extended additively to integral virtual coherent classes, gives an integer.

**Remark 4.8 (True versus fake quantum  $K$ -theory)**

Equation (4.11) is not obtained by integrating just  $\text{ch}(\text{ev}_1^* \mathcal{E}_\gamma) \text{Td}(T^{\text{vir}})$  against the virtual class of the stack. That integral is the identity-sector, or fake, invariant. The true Euler characteristic includes the other inertia-sector contributions in virtual Kawasaki–Riemann–Roch; these account for the multiple-cover correction in Chou’s result. Thus the construction does not silently replace a stack by a scheme when applying Riemann–Roch.

**4.4. The explicit formula. —**

**Theorem 4.9 (Quantum- $K$  formula).** — *For every nonzero effective  $\beta \in H_2(F_3, \mathbb{Z})$ ,*

$$b_\beta(\gamma) = \sum_{\substack{k \geq 1, \\ \alpha \in H_2(F_3, \mathbb{Z}) \\ k\alpha = \beta}} \mu(k) k \kappa_\alpha. \quad (4.12)$$

*In particular  $b_\beta(\gamma)$  is integral. Equivalently,*

$$\kappa_\beta = \sum_{k\alpha = \beta} k b_\alpha(\gamma). \quad (4.13)$$

Put  $\kappa_m = \sum_{|\alpha|=m} \kappa_\alpha$ . Finiteness of each sum and Theorem 4.9 give

$$n_m(\gamma) = \sum_{k|m} \mu(k) k \kappa_{m/k}, \quad \kappa_m = \sum_{k|m} k n_{m/k}(\gamma).$$

For example the already established BPS values imply  $\kappa_1 = 24$ ,  $\kappa_2 = -24$ ,  $\kappa_3 = 336$ , and  $\kappa_4 = -1536$ . These values are consequences of the quantum- $K$  formula, not independent quantum- $K$  localization computations. The identification of the quantum- $K$  BPS transforms with the BPS coefficients extracted from the sunset expansion is provided by the coefficient bridge Theorem 3.22, in particular

$$f_2(Q) = -\theta_Q \mathcal{F}_\gamma(Q).$$

Thus Theorem 4.9 supplies a quantum- $K$  realization of the BPS transforms already identified with the sunset coefficients by the coefficient bridge.

**4.5. Proofs for the quantum- $K$  specialization. —**

*The exact integral Chern-character lift. —*

*Proof.* — The relation  $p_i^2 = 0$  on  $(\mathbb{P}^1)^4$  remains true after restriction and pullback. Hence

$$\mathrm{ch}(\mathcal{E}_\gamma) = \sum_{i < j} \varpi_Z^*(p_i p_j) = \varpi_Z^* \gamma.$$

Thus  $\mathcal{E}_\gamma$  is an integral  $K$ -theory lift of the degree-four cohomology class  $\varpi_Z^* \gamma$  appearing in the one-insertion specialization of Chou’s formula [5, Prop. 3.4].  $\square$

*The explicit fourfold specialization and inversion. —*

*Proof.* — We apply [5, Prop. 3.4], with target  $Z$ , class  $A = i_* \beta$ , cohomology insertion  $\varpi_Z^* \gamma$ , and its exact  $K$ -lift (4.10). The hypotheses are checked as follows. The target is a smooth projective fourfold, with  $c_1(TZ) = 2\xi$ , which is nef, and  $c_1(TZ) \cdot A = 0$ . (It also satisfies symplectic semipositivity since its  $c_1$  takes even values on integral classes, as explained in Remark 4.2.) The insertion has real cohomological degree  $\deg_{\mathbb{R}}(\varpi_Z^* \gamma) = 4$ . For the fourfold  $Z$  and  $c_1(TZ) \cdot A = 0$ , one has  $\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,1}(Z, A) = (\dim_{\mathbb{C}} Z - 3) + c_1(TZ) \cdot A + 1 = 2$ . Since  $\deg_{\mathbb{R}}(\varpi_Z^* \gamma) = 4$ , the insertion has complex codimension 2, so the resulting invariant is numerical. Lemma 4.7 supplies an integral  $K$ -theory lift of Chou’s cohomological insertion, with no rational denominators or unwanted higher-degree components. His  $K$ -degree-zero formula is precisely

$$B_A(\varpi_Z^* \gamma) = \sum_{kA'=A} \mu(k)k \chi(\overline{\mathcal{M}}_{0,1}(Z, A'), \mathrm{ev}_1^* \mathcal{E}_\gamma \otimes \mathcal{O}^{\mathrm{vir}}).$$

The lattice and effectivity statement of Lemma 4.5 identifies  $A' = i_* \alpha$  in every term; (4.9) then proves (4.12). The sum is finite because  $k$  divides  $H \cdot \beta > 0$ . Its terms and coefficients are integral. Finally, the arithmetic functions  $k \mapsto k$  and  $k \mapsto \mu(k)k$  are Dirichlet-convolution inverses, which gives (4.13).  $\square$

## 5. Two, three and four loops, and the local–relative comparison

This section collects the comparisons with the adjacent loop orders. The first part is the two-loop  $L = 2$  comparison which is not covered by the hypotheses used in this paper, and the point of the comparison is to separate the cover exponent from the genuine lower-loop restrictions on curve lattices and on threshold convergence. The second part records the  $L = 3$  and  $L = 4$  normalizations against the published coefficient tables. The third records the Euler factors relating the local invariants to relative ones.

**5.1. The two-loop threefold with zero or one primary marking. —** The  $r^{-3}$  weight familiar from the two-loop/CY3 prepotential and the  $r^{-2}$  weight used

here are compatible conventions for different observables, not a change of the genus-zero cover law with target dimension. The primary convention with  $\ell$  marked insertions has weight  $r^{\ell-3}$ ; see [21, equation (2)] and Theorem [2, 13.11]. In this section  $r$  is the cover degree, not the loop number or the number of insertions.

Let  $X_2$  be a smooth Calabi–Yau threefold with proper positive-degree stable-map moduli, and let  $H$  be an integral divisor. This includes the local two-loop setting when its positive-degree local invariants are formed using the actual curve lattice of the base surface. Only the positive-degree part of the prepotential is considered. Define

$$U_\beta = \int_{[\overline{\mathcal{M}}_{0,0}(X_2, \beta)]^{\text{vir}}} 1, \quad U_\beta = \sum_{r\alpha=\beta} \frac{\nu_\alpha}{r^3}.$$

The unmarked virtual dimension is zero. The one-divisor invariant, in contrast, is defined on the one-marked stack, of virtual dimension one. The primary divisor equation gives the exact comparison

$$\begin{aligned} P_{H,\beta} &:= \langle H \rangle_{0,1,\beta} = (H \cdot \beta) U_\beta \\ &= \sum_{r\alpha=\beta} \frac{(H \cdot \alpha) \nu_\alpha}{r^2}. \end{aligned} \tag{5.1}$$

In the last equality  $H \cdot \beta = r(H \cdot \alpha)$  for the indicated covering root. Thus the *same threefold* already has the one-point  $r^{-2}$  transform, with

$$b_\alpha(H) = (H \cdot \alpha) \nu_\alpha.$$

This is not an identification of the inserted integers with the raw unmarked numbers  $\nu_\alpha$ . The divisor identity itself does not prove integrality of the unmarked transform; it compares the two normalizations whenever those quantities are defined.

Equivalently, define the logarithmic Novikov derivative by  $\theta_H \mathbf{Q}^\beta = (H \cdot \beta) \mathbf{Q}^\beta$ . As positive-degree formal potentials one has

$$\Phi_0^+(\mathbf{Q}) = \sum_{\alpha>0} \nu_\alpha \text{Li}_3(\mathbf{Q}^\alpha), \tag{5.2}$$

$$\theta_H \Phi_0^+(\mathbf{Q}) = \sum_{\alpha>0} (H \cdot \alpha) \nu_\alpha \text{Li}_2(\mathbf{Q}^\alpha). \tag{5.3}$$

The class variable is the actual integral curve class. A later one-variable specialization may aggregate classes of fixed  $H$ -degree, but only after their lattice and the finiteness of that aggregation have been specified. Under the one-variable specialization  $\mathbf{Q}^\beta \mapsto Q^{H \cdot \beta}$  the operator  $\theta_H$  becomes  $\theta_Q = Q d/dQ$ , which is the form used in (5.8).

**5.2. Why the higher-loop insertion gives the same power.** — For the local  $(L + 1)$ -fold of this paper, genus-zero virtual dimension is

$$\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,\ell}(X_L, \beta) = L - 2 + \ell.$$

At  $L = 2$ , the unmarked space has virtual dimension zero and supplies an uninserted numerical invariant. At  $L > 2$ , its virtual dimension is positive; integrating the unit alone does not supply the numerical curve count in question. We use one marking and

$$\delta_0 = \frac{H^{L-1}}{(L-1)!}, \quad \mathrm{codim}_{\mathbb{C}}(\delta_0) = L - 1 = \mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,1}(X_L, \beta).$$

The insertion's higher degree does not turn it into several independently marked insertions. A power such as  $H^2$  evaluated at one marking is a different observable from two divisor insertions at two markings. Thus  $\ell = 1$ , and the primary weight is  $r^{-2}$  for every  $L \geq 3$ .

In the isolated-rational-curve model on a threefold, the unmarked Aspinwall–Morrison obstruction integral contributes  $r^{-3}$ , and a marked incidence condition has  $r$  preimages under a degree- $r$  cover. This explains the extra factor of  $r$  geometrically; equation (5.1) makes the comparison exact without a heuristic count of automorphisms. Moving-curve models in higher dimensions also have incidence constraints on the image family. Such constraints account for the codimension of an insertion, not for additional marked points. No ideal normal-bundle model for every curve in this particular local target is assumed in the primary integrality proof.

**5.3. What the comparison does not identify.** — The two-loop sunset Feynman integral is related to the local prepotential by the mirror-symmetry and Legendre-transform construction of Bloch–Kerr–Vanhove [1, introduction, equations (1.5)–(1.6)]. The Feynman integral itself is not being identified with  $\Phi_0^+$ , nor does eq. (5.3) say that higher-loop Feynman integrals are obtained by simply differentiating the two-loop Feynman integral. The comparison concerns primary potentials. It supplies no multiple-cover law for the descendants  $\psi^a \delta_a$ ,  $a > 0$ , appearing in the full assembly [2, Eq. (9.2)].

There are also genuine reasons not to substitute  $L = 2$  mechanically into every theorem of this paper. The weak-Lefschetz proof of  $H_2(F_L, \mathbb{Z}) \cong H_2((\mathbb{P}^1)^{L+1}, \mathbb{Z})$  requires  $2 < L$ . For a surface at  $L = 2$  it does not give that isomorphism; its actual curve lattice must be used instead. Independently, at the physical threshold  $s = (L+1)^2$  the local simplex density is proportional to  $\|u - u_*\|^{-2}$  in real dimension  $L$ . At  $L = 2$  its radial integral is proportional to  $\int_0^\varepsilon dr/r$  and diverges logarithmically. For  $L \geq 3$  it is integrable, as used in Proposition [2, 8.1]. These lattice and endpoint restrictions are not the origin of the primary exponent change.

**5.4. Curve classes and normalizations at two, three and four loops.** — For the purposes of this section set

$$Y_L = (\mathbb{P}^1)^{L+1}, \quad F_L \in |\mathcal{O}_{Y_L}(1, \dots, 1)| \text{ smooth}, \quad X_L = \text{Tot}(K_{F_L}), \quad \gamma_L = \frac{H^{L-1}}{(L-1)!}.$$

Adjunction gives  $K_{F_L} = -H$  and  $\int_{F_L} H^L = (L+1)!$ . Moreover  $H^k/k! = e_k(p_1, \dots, p_{L+1})$ , so  $\gamma_L$  is integral. Its codimension is  $L-1 = \text{vdim } \overline{\mathcal{M}}_{0,1}(X_L, \beta)$ . Here and below all curve classes have positive  $H$ -degree.

**5.5. The surface case and its curve classes.** — The surface  $F_2$  is a del Pezzo surface of degree six, hence isomorphic to  $\text{Bl}_3 \mathbb{P}^2$  with the three points noncollinear. The restricted ambient divisor classes span a rank-three sublattice, whereas  $H_2(F_2, \mathbb{Z})$  has rank four. Consequently an ambient degree  $\mathbf{d} = (p_1 \cdot \beta, p_2 \cdot \beta, p_3 \cdot \beta)$  need not specify  $\beta$ . We use the coarsened invariants

$$N_{\mathbf{d}}(H) = \sum_{j_* \beta = \mathbf{d}} N_{\beta}(H), \quad N_m(H) = \sum_{H \cdot \beta = m} N_{\beta}(H).$$

The sums are finite because the effective cone of this del Pezzo surface is rational polyhedral, generated by its six  $(-1)$ -curves, and  $H$  is ample. The zero locus on the ambient stable-map stack is the disjoint union of the stacks indexed by all these lifts. The virtual comparison must be applied to this union. Convexity of  $\mathcal{O}(H)$ , concavity of  $\mathcal{O}(-H)$  in positive degree, the arbitrary-base factorization, and the compatibility of obstruction theories used in Sections [2, 4] and [2, 5] apply to each summand. The two-class modification and extraction then compute the ambient numerical pairings of this coarsened theory; no identification of the full curve lattice with the ambient lattice is used.

This is the sextic del Pezzo geometry of [1, Theorem 6.1 and §6.5]. It is toric, whereas  $F_3$  has  $b_3 = 2$  by Remark 3.4, and hence is not a smooth projective toric variety. Thus the toric-target mirror theorem applies directly to the surface, while at three loops we apply it to  $Y_3$  and use the two twists to obtain the local hypersurface theory.

**5.6. A coefficient calculation with fixed normalizations.** — In this section use the same period sign at each loop order:

$$A_L(y) = \sum_{m \geq 0} (-1)^m c_m^{(L)} y^m, \quad c_m^{(L)} = \sum_{|\mathbf{d}|=m} \binom{m}{\mathbf{d}}^2, \quad \pi_L(s) = -y A_L(y), \quad y = -s^{-1}. \quad (5.4)$$

Set  $g_L = \sum_{m \geq 1} (-1)^m c_m^{(L)} y^m / m$ ,  $T_L = \log y + g_L$ ,  $Q = y e^{g_L}$ , and  $\theta_y = y d/dy$ . Thus  $\theta_y T_L = A_L$  and  $dT_L/ds = -\pi_L$ . The coefficient denoted  $K_2^{(L)}$  is the scalar multiplying  $H^2 z^{-2}$  after multiplication of the local  $I$ -function by  $e^{-HT_L/z}$ . The coefficient of  $z^{-1}$  after this normalization is zero. For  $L = 2, 3, 4$ , numerical primary extraction gives

$$\mathcal{F}_{\gamma_L} = -L(L+1)K_2^{(L)}. \quad (5.5)$$

For  $L \geq 3$  this is Theorem [2, 5.9]; at  $L = 2$  use the coarsened virtual comparison just described. Equivalently, pair the proper primary evaluation cycle with  $H$ , before

removing the equivariant parameter. The factor is  $-\int_{F_2} H^2 = -6$ , and all insertions in this pairing are ambient. This does not require ambient surjectivity in degree two.

Here is an exact finite-sum method for the displayed coefficients. Put  $h_j = \sum_{r=1}^j 1/r$ ,  $h_0 = 0$ , and

$$b_L(y) = \sum_{m \geq 1} (-1)^m y^m \sum_{|\mathbf{d}|=m} \binom{m}{\mathbf{d}}^2 \left( 2h_m - \frac{2}{L+1} \sum_{i=1}^n h_{d_i} \right).$$

The coefficient identity of Lemma 3.19, with  $L+1$  factors, gives

$$\theta_Q K_2^{(L)} = \frac{b_L}{A_L} - g_L, \quad \theta_y K_2^{(L)} = b_L - A_L g_L, \quad K_2^{(L)}(0) = 0. \quad (5.6)$$

These formulas follow by expanding the finite Euler products to first order in the divisor classes and using permutation symmetry; in the average each  $p_i$  is replaced by  $H/(L+1)$ . Integrating the second formula coefficientwise and substituting the unique inverse of  $Q = ye^{g_L}$  determines  $K_2^{(L)}$  over  $\mathbb{Q}$ . The multinomial bound  $c_m^{(L)} \leq (L+1)^{2m}$  and the harmonic-number bound give convergent germs. For  $L=2$  this yields

$$K_2^{(2)} = -Q + \frac{7}{4}Q^2 - \frac{28}{9}Q^3 + \frac{135}{16}Q^4 - \frac{626}{25}Q^5 + \frac{751}{9}Q^6 + O(Q^7).$$

The two-loop operator, acting on  $G(y) = I_{\ominus}^{(2)}(-1/y)/y$ , is

$$P_2 = \theta_y^2 + y(10\theta_y^2 + 10\theta_y + 3) + 9y^2(\theta_y + 1)^2, \quad P_2 G = 6. \quad (5.7)$$

If instead  $I_{\ominus}^{(2)}(s) = s^{-1}v(-s^{-1})$ , then  $v = -G$  and  $P_2 v = -6$ . For a direct normalization of its Frobenius solutions, set  $a_0(\rho) = 1$ ,  $a_{-1}(\rho) = 0$  and, for  $m \geq 1$ , impose

$$(m+\rho)^2 a_m(\rho) + [10(m-1+\rho)^2 + 10(m-1+\rho) + 3]a_{m-1}(\rho) + 9(m-1+\rho)^2 a_{m-2}(\rho) = 0.$$

Then  $U(\rho, y) = y^\rho \sum_{m \geq 0} a_m(\rho) y^m$  satisfies  $P_2 U = \rho^2 y^\rho$ . Its first two jets are  $U(0, y) = A_2$  and  $\partial_\rho U(0, y) = A_2 \log y + b_2$ ; these coefficient identities follow by substitution in the displayed recurrence, or the Bessel symmetric-power calculation with three factors. The boundary calculation below selects  $G = 3\partial_\rho^2 U(0, y)$ . Collecting powers of  $R = T_2$  therefore gives

$$I_{\ominus}^{(2)}(s) = \pi_2(s)(-3R^2 + f_1^{(2)}(Q)R + f_0^{(2)}(Q)), \quad f_1^{(2)} = -6 \left( \frac{b_2}{A_2} - g_2 \right) = \theta_Q \mathcal{F}_H. \quad (5.8)$$

In particular this is an identity of germs, independently of any finite coefficient check.

The divisor equation gives  $N_m(H) = mN_m^{\text{loc}}$ . Define the ordinary threefold multiple-cover coefficients by  $N_m^{\text{loc}} = \sum_{k|m} n_{m/k}/k^3$ . Exact rational expansion gives

| $m$ | $[Q^m]f_1^{(2)}$ | $N_m(H)$   | $n_m$ |
|-----|------------------|------------|-------|
| 1   | 6                | 6          | 6     |
| 2   | -21              | -21/2      | -6    |
| 3   | 56               | 56/3       | 6     |
| 4   | -405/2           | -405/8     | -12   |
| 5   | 3756/5           | 3756/25    | 30    |
| 6   | -3004            | -1502/3    | -84   |
| 7   | 86442/7          | 86442/49   | 252   |
| 8   | -209301/4        | -209301/32 | -816  |

The next three values are 2790, -9930, 36498. In degree one the six  $(-1)$ -curves each have local normal bundle  $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and contribute one. The table is a finite computation, rather than a proof of an all-degree integrality assertion.

**5.7. Two-loop boundary normalization.** — The Schwinger integral converges for  $\operatorname{Re} s < 9$ , but diverges at  $s = 9$ . Near the symmetric point of the two-dimensional simplex its positive integrand is comparable to  $|u - u_*|^{-2}$ . The asymptotic normalization at negative infinity is

$$I_{\ominus}^{(2)}(-t) = \frac{3 \log^2 t}{t} + O(t^{-2}(1 + \log t)^2). \quad (5.9)$$

For completeness this follows from the Mellin method of Section 3 without numerical boundary fitting. With  $M_2(\nu) = \int e^{-V} W^\nu \prod_{i=1}^3 dx_i/x_i$  and  $\mathcal{K}_3(w) = \int_0^\infty x^{w-1} K_0(x)^3 dx$ ,

$$M_2(\nu) = \frac{16 \cdot 4^{-\nu}}{\Gamma(\nu)} \mathcal{K}_3(2\nu), \quad \int_0^\infty t^{z-1} I_{\ominus}^{(2)}(-t) dt = \Gamma(z) M_2(1-z).$$

For  $c = \log 2 - \gamma_E$ , the polar part of  $\mathcal{K}_3(w)$  at zero is  $\sum_{j=0}^3 \binom{3}{j} j! c^{3-j} w^{-j-1}$ . Consequently the polar part of  $\Gamma(1-\nu) M_2(\nu)$  is  $6\nu^{-3}$ : the linear exponential cancels, and

$$\frac{\Gamma(1-\nu)}{\Gamma(1+\nu)} e^{-2\gamma_E \nu} = 1 + \frac{2}{3} \zeta(3) \nu^3 + O(\nu^5).$$

There are no  $\nu^{-2}$  or  $\nu^{-1}$  terms. The convergent regular-singular expansion and the cutoff argument of Section 3 then give (5.9). Thus both homogeneous connection constants relative to  $3\partial_\rho^2 U(0, y)$  vanish.

The Gamma-class computation is consistent with this analytic result:  $c(TF_2) = 1 + H + H^2$  and  $c(TX_2)|_{F_2} = (1 + H + H^2)(1 - H) = 1$ . In general for a del Pezzo surface  $S$ ,

$$\int_S c_2(T_{\operatorname{Tot}(K_S)}|_S) = \chi_{\operatorname{top}}(S) - K_S^2 = 12 - 2K_S^2.$$

This is an equality of characteristic numbers. Its vanishing at degree six does not assert the same boundary normalization for other local surfaces.

**5.8. Four loops.** — For  $L = 4$ ,  $\int_{F_4} H^4 = 120$  and  $\gamma_4 = H^3/6$ . The general results in Theorems [2, 5.9], [2, 7.2] and [2, 8.7] and Corollary [2, 9.3] apply since  $F_4$  is smooth and ample in  $Y_4$ ,  $L \geq 3$ , and the insertion is ambient. In particular full even-cohomology surjectivity is unnecessary; indeed  $b_4(F_4) = 28$ , whereas the ambient image in degree four has dimension 10, by Proposition [2, 3.2]. The operator acting on  $I_{\odot}^{(4)}(-1/y)/y$  is the operator  $P_4 = \theta_y^4 + \sum_{j=1}^3 y^j Q_j(\theta_y)$  of the introduction, in which  $Q_3(\theta_y) = 225(\theta_y + 1)^2(\theta_y + 2)^2$ , and

$$P_4 \left( \frac{I_{\odot}^{(4)}(-1/y)}{y} \right) = 120.$$

Its finite nonzero singularities in the  $s$  coordinate are 1, 9, 25. Formula (5.6) gives

$$K_2^{(4)} = -3Q + \frac{45}{4}Q^2 - \frac{211}{3}Q^3 + \frac{10029}{16}Q^4 - \frac{164028}{25}Q^5 + O(Q^6),$$

$$\mathcal{F}_{\gamma_4} = -20K_2^{(4)} = 60Q - 225Q^2 + \frac{4220}{3}Q^3 - \frac{50145}{4}Q^4 + \frac{656112}{5}Q^5 - 1529195Q^6 + O(Q^7).$$

With the negative period  $\pi_4$  of (5.4),

$$I_{\odot}^{(4)}(s) = \pi_4(s) \left[ -5R^4 + \sum_{j=0}^3 f_j^{(4)}(Q)R^j + 80\zeta(3) \left( R - \frac{1}{20}\theta_Q \mathcal{F}_{\gamma_4} \right) \right], \quad (5.10)$$

where the  $f_j^{(4)}$  have rational coefficients and zero constant terms, and

$$f_3^{(4)} = \theta_Q \mathcal{F}_{\gamma_4} = 60Q - 450Q^2 + 4220Q^3 - 50145Q^4 + 656112Q^5 + O(Q^6).$$

This is Corollary [2, 9.3] with  $\pi_4 = -\Pi_4$ ; the rational coefficients here are the negatives of its  $h_j$ . Thus with a period asymptotic to  $1/s$ , the leading rational coefficient is  $(-1)^{L+1}(L+1)$  and the next one satisfies

$$f_{L-1}^{(L)} = (-1)^L \theta_Q \mathcal{F}_{\gamma_L}. \quad (5.11)$$

For  $L \geq 3$  this is Theorem [2, 9.1]; the two-loop case is (5.8).

In particular the four-loop boundary contribution is nonzero:

$$I_{\odot}^{(4)}(-1/y) = y(5 \log^4 y - 80\zeta(3) \log y) + O(y^2(1 + |\log y|)^4).$$

Although the degree-four part of  $\widehat{\Gamma}_{X_4}|_{F_4}$  vanishes, its degree-three part is  $\frac{2}{3}\zeta(3)H^3$ . Multiplication by the degree-one part  $-H \log y$  of  $y^{-H}$  contributes  $-\frac{2}{3}\zeta(3) \int_{F_4} H^4 \log y = -80\zeta(3) \log y$ . This also explains why the boundary term in (5.10) contains a logarithmic homogeneous solution, rather than just a constant multiple of the holomorphic period.

The dilogarithmic transform, defined by  $N_m(\gamma_4) = \sum_{k|m} b_{m/k}/k^2$ , starts

$$b_m = 60, -240, 1400, -12480, 131220, -1529520, 19192260, -254513280, \dots$$

These displayed values follow by rational Möbius inversion of (5.6). They are numerical checks only. The all-degree arithmetic statement for the distinguished insertion is Corollary [2, 13.10], with its explicitly stated algebraic quantum- $K$  input (Theorem [2, 13.8]); finite integrality checks do not replace that input.

**5.9. Euler factors and the local–relative comparison.** — For an anticanonical divisor class  $D$  and positive integer  $k = D \cdot \beta$ , the finite-product identity is

$$\prod_{a=0}^{k-1} (-D - az) = (-1)^k D \prod_{a=1}^{k-1} (D + az).$$

It is an equality in the cohomology-valued coefficient ring. Multiplication by the nilpotent class  $D$  cannot be inverted there. Moreover replacing  $(k-1)!$  by  $k!$  multiplies a scalar coefficient by the integer  $k$ ; it does not insert the cohomology class  $D$ . The degree-zero term is specified separately by the normalization of the generating function.

There is also a precise numerical correspondence. Choose a smooth  $D \in |H|$  on  $F_L$ ; it exists by Bertini’s theorem, since  $|H|$  is the restriction of the very ample system  $|\mathcal{O}(1, \dots, 1)|$  and is therefore basepoint free. The target  $F_L$  is smooth projective,  $D$  is smooth and ample,  $\mathcal{O}_{F_L}(-D) = K_{F_L}$ , and  $k = D \cdot \beta > 0$ . Therefore [6, Theorem 1.1 and Corollary 1.2] apply to  $\text{Tot}(\mathcal{O}_{F_L}(-D)) = X_L$ . With one ordinary marking carrying the primary insertion  $\gamma_L$  and one marking of maximal contact  $k$ , they give

$$R_\beta(\gamma_L) = (-1)^{k+1} k N_\beta(\gamma_L).$$

Here  $R_\beta(\gamma_L)$  is the genus-zero logarithmic invariant of the pair  $(F_L, D)$  in class  $\beta$ , with one interior marking carrying  $\gamma_L$  and one marking of maximal contact order  $k = D \cdot \beta$  with  $D$ , carrying no insertion; the ordinary primary insertion pulls back under the forgetful morphism, as required by the cited corollary. In particular, summing at degree  $m$  in the three-loop case gives the numerical consequence

$$d_2(m) = (-1)^m \sum_{H \cdot \beta = m} R_\beta(\gamma).$$

This uses the virtual-cycle correspondence and the established coefficient bridge; no division by  $D$  in cohomology is involved.

## 6. Reconstruction from primary evaluation classes

Throughout this section we write  $\mathcal{D}_a$  for the diagonal series  $\mathcal{D}_a^\Delta$ , as in Section 3, so that  $\mathcal{D}_0 = \mathcal{F}_{\gamma_L}$ , and  $\theta_Q^{-1} \sum_{m>0} c_m Q^m = \sum_{m>0} (c_m/m) Q^m$ . Here  $L = 3$  or  $L = 4$ ,  $n = L+1$ ,  $F_L \subset (\mathbb{P}^1)^{L+1}$  is the smooth  $(1, \dots, 1)$  hypersurface and  $X_L = \text{Tot}(K_{F_L})$ . For a positive curve class  $\beta$  write  $|\beta| = m_\beta = H \cdot \beta$  (both notations are used) and  $\beta_i = p_i \cdot \beta$ , the integral curve-lattice coordinates of Proposition [2, 3.1]. The *primary evaluation divisor* is the proper one-point evaluation class

$$D_\beta = C_{0,\beta}^0 = (\text{ev}_{F_L})_*(c_{m_\beta-1}(\mathcal{V}) \cap [\overline{\mathcal{M}}_{0,1}(F_L, \beta)]^{\text{vir}}) \in H^2(F_L, \mathbb{Q})$$

of [2, Eq. (4.2)], written  $D_\beta = \sum_i d_{\beta,i} p_i$ ; thus  $\int_{F_L} D_\beta b$  is the one-point primary invariant of  $X_L$  in class  $\beta$  with insertion  $\varpi^* b$ , for  $b \in H^{2L-2}(F_L, \mathbb{Q})$ . Section [2, 17] of the companion paper [2] gives its full multidegree extraction, Theorem [2, 17.1],

and its integral cover transform

$$B_\beta = \sum_{r|\beta} \frac{\mu(r)}{r^2} D_{\beta/r}, \quad D_\beta = \sum_{r|\beta} \frac{1}{r^2} B_{\beta/r},$$

where  $r \mid \beta$  means that  $\beta/r$  is an integral class. The classwise one-point descendants are, as in [2, Eq. (4.1)],

$$N_{a,\beta} = \int_{[\overline{\mathcal{M}}_{0,1}(X_L, \beta)]^{\text{vir}}} \psi^a \text{ev}^* \varpi^* \frac{H^{L-1-a}}{(L-1-a)!}, \quad 0 \leq a \leq L-1,$$

so that  $\mathcal{D}_a = \sum_{\beta > 0} N_{a,\beta} Q^{|\beta|}$ . All sums  $\alpha + \gamma = \beta$  below are ordered sums of positive effective classes.

By Theorem [2, 13.3] of the companion paper [2] there are a smooth connected projective  $(L+1)$ -fold  $\tilde{Z}$  with an actual isomorphism  $K_{\tilde{Z}} \cong \mathcal{O}_{\tilde{Z}}$ , a retraction  $q: \tilde{Z} \rightarrow F_L$ , and an embedding  $j_+: F_L \hookrightarrow \tilde{Z}$  onto a divisor  $F_{L,+}$  with normal bundle  $K_{F_L} = \mathcal{O}_{F_L}(-H)$ , such that for every positive  $\beta$  and every number of markings the stable-map stack of  $\tilde{Z}$  in class  $j_{+*}\beta$  is identified, together with its standard obstruction theory and virtual class, with that of  $X_L$  in class  $\beta$ . We fix the embedding  $j_+$  throughout this section. A bracket

$$\langle \tau_{a_1}(\mu_1), \dots, \tau_{a_k}(\mu_k) \rangle_\beta = \langle \tau_{a_1}(q^*\mu_1), \dots, \tau_{a_k}(q^*\mu_k) \rangle_{0,k,j_{+*}\beta}^{\tilde{Z}},$$

with classes  $\mu_l$  on  $F_L$ , denotes a genus-zero descendant invariant of  $\tilde{Z}$ ; brackets with insertions given directly by classes on  $\tilde{Z}$  are understood in the same way. By the marked virtual comparison such a bracket equals the local invariant of  $X_L$  in class  $\beta$  with insertions  $\varpi^*\mu_l$ . A divisor  $E$  on  $F_L$  enters through  $q^*E$ , and  $q^*E \cdot j_{+*}\beta = E \cdot \beta$ . The divisor, string, dilaton, and genus-zero topological recursion relations, and the Lee–Pandharipande relation recalled below, hold on the smooth projective target  $\tilde{Z}$ ; we apply them there and transport the results.

Let  $\{\mathbf{b}_k\}$  be a homogeneous basis of  $H^*(\tilde{Z}, \mathbb{Q})$  and  $\{\mathbf{b}^k\}$  the Poincaré-dual basis, so that  $\sum_k \mathbf{b}_k \otimes \mathbf{b}^k$  is the class of the diagonal. Since the evaluation maps of the sector factor through  $j_+$ ,  $\text{ev}_*[\overline{\mathcal{M}}_{0,1}(\tilde{Z}, j_{+*}\alpha)]^{\text{vir}} = j_{+*}D_\alpha$ , and therefore

$$\sum_k \langle \mathbf{b}_k \rangle_\alpha \mathbf{b}^k = j_{+*}D_\alpha, \quad j_+^* j_{+*}D_\alpha = c_1(K_{F_L})D_\alpha = -HD_\alpha. \quad (6.1)$$

Only classes of even degree contribute, and on even cohomology the roles of  $\{\mathbf{b}_k\}$  and  $\{\mathbf{b}^k\}$  may be exchanged. The second identity is the self-intersection formula for the smooth divisor  $F_{L,+}$  of the smooth variety  $\tilde{Z}$ , with  $N_{F_{L,+}/\tilde{Z}} \cong K_{F_L}$ . Consequently an insertion  $j_{+*}D_\alpha$  in a bracket of class  $\gamma$  contributes  $-HD_\alpha$ , that is,  $\langle \dots, j_{+*}D_\alpha \rangle_\gamma = -\langle \dots, HD_\alpha \rangle_\gamma$ . Thus these arguments do not replace the full diagonal by an inverse ambient intersection matrix. In particular they do not assume that the ambient classes exhaust the middle cohomology of the four-dimensional

$F_4$  at four loops. The two-mark cotangent relation used below is the virtual relation of Lee–Pandharipande [23, Corollary 1]; their more restrictive reconstruction theorem is not needed.

**6.1. All cohomological descendants from full primaries.** — Here  $n = 4$ ,  $N_{0,\beta} = 3 \sum_i d_{\beta,i}$ , and the relevant compact-target pairing is

$$C(\alpha, \gamma) = \int_{\tilde{Z}} j_{+*} D_\alpha j_{+*} D_\gamma = - \int_{F_3} H D_\alpha D_\gamma = -2 \sum_{i \neq j} d_{\alpha,i} d_{\gamma,j}. \quad (6.2)$$

The minus sign is the normal bundle  $K_{F_3}$ .

**Theorem 6.1 (Fourfold recursion).** —

$$N_{1,\beta} = -\frac{2}{m_\beta} N_{0,\beta} + \frac{1}{2m_\beta} \sum_{\alpha+\gamma=\beta} |\alpha||\gamma| C(\alpha, \gamma), \quad N_{2,\beta} = -\frac{1}{2} \sum_{\alpha+\gamma=\beta} C(\alpha, \gamma). \quad (6.3)$$

*Proof.* — All brackets are those of the compact target, with  $L = 3$ , and  $m = m_\beta$ . Applying the descendant divisor equation twice gives

$$\langle \tau_1(H), H, H \rangle_\beta = m^2 N_{1,\beta} + 4m N_{0,\beta}, \quad (6.4)$$

$$\langle \tau_2(1), H, H \rangle_\beta = m^2 N_{2,\beta} + 2m N_{1,\beta} + 2N_{0,\beta}; \quad (6.5)$$

for instance  $\langle \tau_1(H), H \rangle_\beta = m N_{1,\beta} + \langle H^2 \rangle_\beta = m N_{1,\beta} + 2N_{0,\beta}$ . On the other hand, genus-zero topological recursion expresses  $\psi_1$  on  $\overline{\mathcal{M}}_{0,3}(\tilde{Z}, j_{+*}\beta)$  as the boundary divisor separating the first marking from the other two. A degree-zero component carrying the first marking and the node alone is unstable and does not occur. The stable degree-zero component carrying the markings 2, 3 and the node contributes, after contraction of the diagonal,  $\langle H, H^2 \rangle_\beta = 2m N_{0,\beta}$  in the first case and  $\langle \tau_1(1), H^2 \rangle_\beta = -\langle H^2 \rangle_\beta = -2N_{0,\beta}$  in the second, the latter by the dilaton equation with one other marking. For a positive splitting  $\alpha + \gamma = \beta$  with the first marking on  $\alpha$ , the divisor equation gives  $\langle \mathbf{b}^k, H, H \rangle_\gamma = m_\gamma^2 \langle \mathbf{b}^k \rangle_\gamma$ , the divisor and dilaton equations give  $\langle H, \mathbf{b}_k \rangle_\alpha = m_\alpha \langle \mathbf{b}_k \rangle_\alpha$  and  $\langle \tau_1(1), \mathbf{b}_k \rangle_\alpha = -\langle \mathbf{b}_k \rangle_\alpha$ , and (6.1) turns  $\sum_k \langle \mathbf{b}_k \rangle_\alpha \langle \mathbf{b}^k \rangle_\gamma$  into  $\int_{\tilde{Z}} j_{+*} D_\alpha j_{+*} D_\gamma = C(\alpha, \gamma)$ . Hence

$$\langle \tau_1(H), H, H \rangle_\beta = 2m N_{0,\beta} + \sum_{\alpha+\gamma=\beta} m_\alpha m_\gamma^2 C(\alpha, \gamma), \quad (6.6)$$

$$\langle \tau_2(1), H, H \rangle_\beta = -2N_{0,\beta} - \sum_{\alpha+\gamma=\beta} m_\gamma^2 C(\alpha, \gamma). \quad (6.7)$$

Since  $C$  is symmetric, symmetrizing the ordered sums gives  $\sum m_\alpha m_\gamma^2 C = \frac{m}{2} \sum m_\alpha m_\gamma C$  and  $\sum m_\gamma^2 C + \sum m_\alpha m_\gamma C = \frac{m^2}{2} \sum C$ . Comparing (6.4) with (6.6) gives the first identity of (6.3); substituting it into (6.5) and comparing with (6.7) gives the second. The full compact-target diagonal is contracted before restriction, so no primitive cohomology is discarded, and the minus sign in (6.2) is the normal bundle  $K_{F_3}$ .  $\square$

By Section [2, 17] of the companion paper [2], on a coordinate ray  $D_{re_i} = (H - 2p_i)/r^2$  has zero self-pairing, and one recovers  $(N_0, N_1, N_2) = (6/r^2, -12/r^3, 0)$ .

**Theorem 6.2 (One diagonal primary series suffices)**

*With zero constant terms,*

$$\mathcal{D}_2 = \frac{\mathcal{D}_0^2}{12}, \quad \theta_Q \mathcal{D}_1 = -2\mathcal{D}_0 - \frac{(\theta_Q \mathcal{D}_0)^2}{12}. \quad (6.8)$$

If  $\mathcal{D}_0 = \sum_{m \geq 1} n_m \text{Li}_2(Q^m)$ , then

$$\mathcal{D}_1 = -2 \sum_{m \geq 1} \frac{n_m}{m} \text{Li}_3(Q^m) - \frac{1}{12} \theta_Q^{-1} \left( \sum_{m \geq 1} m n_m \text{Li}_1(Q^m) \right)^2,$$

$$\mathcal{D}_2 = \frac{1}{12} \left( \sum_{m \geq 1} n_m \text{Li}_2(Q^m) \right)^2.$$

*Proof.* — By the dimension-uniform primary extraction, Theorem [2, 17.1] of the companion paper [2],  $d_{\beta,i}$  is the coefficient of  $\mathbf{x}^\beta$  in an explicit series in which  $g$  is symmetric in  $x_1, \dots, x_4$  and the factor  $f_i$  depends on  $i$  only through the  $i$ th exponent. Hence  $d_{\sigma\beta, \sigma(i)} = d_{\beta,i}$  for every permutation  $\sigma$  of the coordinates, where  $\sigma\beta$  permutes the multidegree. Summing over the permutation-invariant set of classes with  $|\beta| = m$  shows that the coefficient of  $p_i$  in  $\sum_{|\beta|=m} D_\beta$  is independent of  $i$ , so  $\sum_{|\beta|=m} D_\beta = c_m H$ . Pairing with  $H^2/2$  and using  $\int_{F_3} H^3 = 24$  gives  $12c_m = [Q^m] \mathcal{D}_0$ , that is,  $\sum_{|\beta|=m} D_\beta = [Q^m] \mathcal{D}_0 H/12$ . No deformation argument and no automorphism of a fixed hypersurface is used. Using  $\int_F H^3 = 24$ , the aggregate of (6.2) in bidegree  $(k, l)$  is  $-[Q^k] \mathcal{D}_0 [Q^l] \mathcal{D}_0/6$ . Summing (6.3) proves (6.8). Formal integration with zero constants proves the displayed integrated formulas. The  $n_m$  are the primary integers of Corollary [2, 13.10] of the companion paper [2].  $\square$

**6.2. Simplified assembly and a differential proof.** — The fourfold identities give

$$f_2 = -\theta_Q \mathcal{D}_0, \quad f_1 = (\theta_Q \mathcal{D}_0)^2/12,$$

$$f_0 = 2\theta_Q^{-1} \mathcal{D}_0 + \frac{1}{12} \theta_Q^{-1} ((\theta_Q \mathcal{D}_0)^2) - \frac{1}{6} \mathcal{D}_0 \theta_Q \mathcal{D}_0, \quad K_4 = -\frac{1}{2} K_2^2.$$

These supplement the assembly proof above with a derivation from the primary series and the scalar differential equation. Set  $u = \theta_Q \mathcal{D}_0/12$ ,  $\tau = W_1/A = R - u$ , and  $B = I_\ominus^{(3)}/(-yA)$ . The identity for  $\tau$  follows from the primary mirror comparison. Put  $\Delta = (1 + 4y)(1 + 16y)$ . The monic third-order Picard–Fuchs operator is the symmetric square of

$$\partial_y^2 + \left( \frac{1}{y} + \frac{\Delta'}{2\Delta} \right) \partial_y + \frac{2(1 + 8y)}{y\Delta}.$$

Writing  $A = h_0^2$ ,  $W_1 = h_0 h_1$  gives  $\tau' = 1/(y\sqrt{\Delta}A)$  and  $R' = A/y$  by the normalized Wronskian. Conjugating the physical equation  $P_3(AB) = 24$  therefore gives

$$\frac{d^3 B}{d\tau^3} = 24\sqrt{\Delta}A^2 = 24\frac{dR}{d\tau}, \quad \frac{d^2 B}{d\tau^2} = 24R.$$

The last integration constant is zero by the normalized logarithmic germ. For a direct comparison with the enumerative formula set

$$B_{\text{can}} = 4\tau^3 + 4u^3 - \mathcal{D}_1 - 2\mathcal{D}_0 u - 16\zeta(3).$$

Since  $d\mathcal{D}_0/dR = 12u$  and  $d\mathcal{D}_1/dR = -2\mathcal{D}_0 - 12u^2$ , differentiation gives  $dB_{\text{can}}/d\tau = 12R^2 - 24uR + 2\mathcal{D}_0$  and  $d^2 B_{\text{can}}/d\tau^2 = 24R$ . Both  $B$  and  $B_{\text{can}}$  have germ  $4\log^3 y - 16\zeta(3) + O(y(1 + |\log y|)^3)$ , so their possible affine difference in  $\tau$  vanishes. Positivity of  $A$  and  $\Delta$  extends this identity along every  $y > 0$ . This proof avoids the signed analytic assembly of the companion paper [2], while retaining its geometric primary mirror comparison. Finally the primary Lambert expansion gives the formal product

$$e^{12(R-\tau)} = \prod_{m \geq 1} (1 - Q^m)^{-mn_m}.$$

Thus  $e^\tau$  and  $Q = e^R$  are distinct coordinates.

**6.3. The full diagonal and a triangular primary algorithm.** — For  $n = 5$  set

$$P_\beta(\eta) = \int_F D_\beta \eta, \quad T_\beta(\mu, \nu) = \langle \mu, \nu \rangle_\beta, \quad A_\beta(\nu) = \langle \tau_1(\nu) \rangle_\beta,$$

where  $\eta \in H^6(F_4, \mathbb{Q})$  has complex codimension three and  $\mu, \nu \in H^4(F_4, \mathbb{Q})$  have complex codimension two; the brackets are those of the compact target, with  $L = 4$ , and  $\beta_i = p_i \cdot \beta$ . Contraction of the *full* compact-target diagonal against the primary evaluation class gives  $j_{+*}D_\alpha$ ; restriction then gives  $-HD_\alpha$ . This is why all subsequent expressions close on the ambient tensor without inverting an ambient pairing.

**Theorem 6.3 (Triangular two-point reconstruction)**

$$T_\beta(p_i p_j, \nu) = \beta_i \beta_j A_\beta(\nu) + \beta_i P_\beta(p_j \nu) + \beta_j P_\beta(p_i \nu) + \sum_{\alpha + \gamma = \beta} \alpha_i \alpha_j T_\gamma(HD_\alpha, \nu). \quad (6.9)$$

For any  $i$  with  $\beta_i > 0$ ,

$$A_\beta(\nu) = -\frac{2}{\beta_i} P_\beta(p_i \nu) - \frac{1}{\beta_i^2} \sum_{\alpha + \gamma = \beta} \alpha_i^2 T_\gamma(HD_\alpha, \nu). \quad (6.10)$$

Thus the classes  $D_\alpha$  with  $|\alpha| \leq |\beta|$  determine  $A_\beta(\nu)$  for every  $\nu$ , and  $T_\beta(\mu, \nu)$  for every  $\nu$  and every ambient  $\mu$ .

*Proof.* — For divisor classes  $E, E'$  on  $F_4$  write  $e_\beta = E \cdot \beta$  and  $e'_\beta = E' \cdot \beta$ , and let  $\nu \in H^4(F_4, \mathbb{Q})$ . Applying the descendant divisor equation twice,

$$\langle \tau_1(\nu), E, E' \rangle_\beta = e_\beta e'_\beta A_\beta(\nu) + e'_\beta P_\beta(E\nu) + e_\beta P_\beta(E'\nu).$$

Genus-zero topological recursion for  $\psi_1$  on  $\overline{\mathcal{M}}_{0,3}(\tilde{Z}, j_{+*}\beta)$  separates the first marking from the two divisor markings. A degree-zero component carrying the first marking and the node alone is unstable. The stable degree-zero component carrying the two divisor markings and the node contributes  $\langle \nu, EE' \rangle_\beta = T_\beta(EE', \nu)$ . A positive splitting with the first marking on  $\gamma$  and the divisor markings on  $\alpha$  contributes, by the divisor equation and (6.1),  $e_\alpha e'_\alpha \langle \nu, j_{+*}D_\alpha \rangle_\gamma = -e_\alpha e'_\alpha T_\gamma(HD_\alpha, \nu)$ . Comparing the two evaluations gives, for arbitrary divisors  $E, E'$ ,

$$T_\beta(EE', \nu) = e_\beta e'_\beta A_\beta(\nu) + e'_\beta P_\beta(E\nu) + e_\beta P_\beta(E'\nu) + \sum_{\alpha+\gamma=\beta} e_\alpha e'_\alpha T_\gamma(HD_\alpha, \nu). \quad (6.11)$$

Taking  $E = p_i$  and  $E' = p_j$  gives (6.9). Taking  $E = E' = p_i$  and using  $p_i^2 = 0$  gives (6.10). Every tensor on the right of (6.10) has smaller positive degree, and the classes  $p_i p_j$  span the ambient part of  $H^4(F_4, \mathbb{Q})$ ; induction on  $|\beta|$  gives the final assertion. Compatibility between pivot choices follows from the virtual identities of the actual theory.  $\square$

**6.4. Explicit inverse for the numerical one-point descendants.** — Fix a divisor class  $E$  on  $F_4$  with  $e_\beta = E \cdot \beta \neq 0$  for the classes considered; one may always take  $E = H$ , since  $H$  is ample on  $F_4$  and hence  $H \cdot \beta > 0$  for every nonzero effective class  $\beta$ . For another divisor  $\lambda$  define ordinary two-point descendants

$$J_\beta(\lambda; \nu) = \langle \tau_1(\lambda), \nu \rangle_\beta, \quad K_\beta(\nu) = \langle \tau_2(1), \nu \rangle_\beta.$$

These are ordinary genus-zero descendant Gromov–Witten invariants; they should not be confused with the Pandharipande–Zinger [24] multiple-cover transforms of primary genus-zero invariants on Calabi–Yau fivefolds. What is reconstructed below are the dimensionally relevant numerical one-point descendants  $N_{a,\beta}$ ,  $0 \leq a \leq 3$ , of the Calabi–Yau fivefold sector, together with the auxiliary two-point invariants  $T_\beta$ ,  $J_\beta$  and  $K_\beta$ ; no assertion is made about other descendant insertions.

**Theorem 6.4 (All-degree inverse).** —

$$\begin{aligned} e_\beta^2 A_\beta(\nu) &= T_\beta(E^2, \nu) - 2e_\beta P_\beta(E\nu) - \sum_{\alpha+\gamma=\beta} e_\alpha^2 T_\gamma(HD_\alpha, \nu), \\ J_\beta(\lambda; \nu) &= -(\lambda \cdot \beta) A_\beta(\nu) - P_\beta(\lambda\nu) - \sum_{\alpha+\gamma=\beta} (\lambda \cdot \alpha) T_\gamma(HD_\alpha, \nu), \\ K_\beta(\nu) &= A_\beta(\nu) + \sum_{\alpha+\gamma=\beta} T_\gamma(HD_\alpha, \nu). \end{aligned} \quad (6.12)$$

Writing  $Q_{2,\beta}(\lambda) = \langle \tau_2(\lambda) \rangle_\beta$  and  $Q_{3,\beta} = \langle \tau_3(1) \rangle_\beta$ ,

$$\begin{aligned} e_\beta^2 Q_{2,\beta}(\lambda) &= J_\beta(\lambda; E^2) - \sum_{\alpha+\gamma=\beta} e_\gamma^2 J_\alpha(\lambda; HD_\gamma) - 2e_\beta A_\beta(E\lambda) - P_\beta(E^2\lambda), \\ e_\beta^2 Q_{3,\beta} &= K_\beta(E^2) - \sum_{\alpha+\gamma=\beta} e_\gamma^2 K_\alpha(HD_\gamma) - 2e_\beta Q_{2,\beta}(E) - A_\beta(E^2). \end{aligned} \quad (6.13)$$

The answer is independent of  $E$ , and

$$(N_0, N_1, N_2, N_3)_\beta = (P_\beta(H^3/6), A_\beta(H^2/2), Q_{2,\beta}(H), Q_{3,\beta}).$$

The full primary divisors  $D_\beta$ , or equivalently  $B_\beta$ , alone determine every displayed invariant.

*Proof.* — The first identity is (6.11) with  $E' = E$ . On  $\overline{\mathcal{M}}_{0,2}(\tilde{Z}, j_{+*}\beta)$ ,  $\beta \neq 0$ , the Lee–Pandharipande relation reads  $\psi_1 + \psi_2 = [D_{1|2}]$ , where  $D_{1|2}$  parametrizes domains split by a node with the first marking on a component of class  $j_{+*}\alpha$  and the second on one of class  $j_{+*}\gamma$ ,  $\alpha + \gamma = \beta$ ; both classes are positive, because a degree-zero component with one marking and one node is unstable. Capping this relation with  $\text{ev}_1^* \lambda \text{ev}_2^* \nu$ , respectively with  $\psi_1 \text{ev}_2^* \nu$ , and splitting the boundary along the diagonal gives

$$\begin{aligned} J_\beta(\lambda; \nu) + \langle \lambda, \tau_1(\nu) \rangle_\beta &= \sum_{\alpha+\gamma=\beta} \sum_k \langle \lambda, \mathbf{b}_k \rangle_\alpha \langle \mathbf{b}^k, \nu \rangle_\gamma, \\ K_\beta(\nu) + \langle \tau_1(1), \tau_1(\nu) \rangle_\beta &= \sum_{\alpha+\gamma=\beta} \sum_k \langle \tau_1(1), \mathbf{b}_k \rangle_\alpha \langle \mathbf{b}^k, \nu \rangle_\gamma. \end{aligned}$$

By the divisor equation  $\langle \lambda, \tau_1(\nu) \rangle_\beta = (\lambda \cdot \beta) A_\beta(\nu) + P_\beta(\lambda\nu)$  and  $\langle \lambda, \mathbf{b}_k \rangle_\alpha = (\lambda \cdot \alpha) \langle \mathbf{b}_k \rangle_\alpha$ ; by the dilaton equation with one other marking  $\langle \tau_1(1), \tau_1(\nu) \rangle_\beta = -A_\beta(\nu)$  and  $\langle \tau_1(1), \mathbf{b}_k \rangle_\alpha = -\langle \mathbf{b}_k \rangle_\alpha$ . Together with  $\sum_k \langle \mathbf{b}_k \rangle_\alpha \langle \mathbf{b}^k, \nu \rangle_\gamma = \langle j_{+*} D_\alpha, \nu \rangle_\gamma = -T_\gamma(HD_\alpha, \nu)$  from (6.1), this gives the formulas for  $J_\beta$  and  $K_\beta$  in (6.12). For (6.13), the divisor equation applied twice gives

$$\begin{aligned} \langle \tau_2(\lambda), E, E \rangle_\beta &= e_\beta^2 Q_{2,\beta}(\lambda) + 2e_\beta A_\beta(E\lambda) + P_\beta(E^2\lambda), \\ \langle \tau_3(1), E, E \rangle_\beta &= e_\beta^2 Q_{3,\beta} + 2e_\beta Q_{2,\beta}(E) + A_\beta(E^2), \end{aligned}$$

while topological recursion for  $\psi_1$  on  $\overline{\mathcal{M}}_{0,3}(\tilde{Z}, j_{+*}\beta)$ , separating the first marking from the two  $E$  markings, gives

$$\begin{aligned} \langle \tau_2(\lambda), E, E \rangle_\beta &= J_\beta(\lambda; E^2) - \sum_{\alpha+\gamma=\beta} e_\gamma^2 J_\alpha(\lambda; HD_\gamma), \\ \langle \tau_3(1), E, E \rangle_\beta &= K_\beta(E^2) - \sum_{\alpha+\gamma=\beta} e_\gamma^2 K_\alpha(HD_\gamma). \end{aligned}$$

Here the first terms come from the stable degree-zero component carrying the two  $E$  markings and the node; a degree-zero component carrying the first marking and the node alone is unstable; and in a positive splitting the first marking lies on  $\alpha$ , the

divisor equation gives  $\langle \mathbf{b}^k, E, E \rangle_\gamma = e_\gamma^2 \langle \mathbf{b}^k \rangle_\gamma$ , and (6.1) turns the contraction into an insertion  $j_{+*} D_\gamma$ , that is, into  $-H D_\gamma$ . Comparing the two evaluations proves (6.13). Theorem 6.3 supplies the tensor from  $D$ ; the cover inversion supplies  $D$  from  $B$ . All equations hold for the actual invariants, hence are independent of the auxiliary divisor.  $\square$

### 6.5. Two-input diagonal closure. —

**Theorem 6.5 (Two diagonal inputs suffice).** — *Let  $\mathcal{T} = \sum_\beta T_\beta(H^2, H^2)Q^{|\beta|}$ . Then*

$$(1 - \theta_Q^2 \mathcal{D}_0/20)\mathcal{T} = 2\theta_Q^2 \mathcal{D}_1 + 12\theta_Q \mathcal{D}_0,$$

and

$$\theta_Q^2 \mathcal{D}_2 = -6\theta_Q \mathcal{D}_1 - 12\mathcal{D}_0 + \frac{\theta_Q \mathcal{D}_1 \theta_Q^2 \mathcal{D}_0 - \theta_Q \mathcal{D}_0 \theta_Q^2 \mathcal{D}_1}{10} + \frac{3}{10} \mathcal{D}_0 \theta_Q^2 \mathcal{D}_0 - \frac{3}{5} (\theta_Q \mathcal{D}_0)^2, \quad (6.14)$$

$$\theta_Q^2 \mathcal{D}_3 = \frac{\mathcal{D}_0 \theta_Q^2 \mathcal{D}_1 - \mathcal{D}_1 \theta_Q^2 \mathcal{D}_0}{10} + \frac{3}{5} \mathcal{D}_0 \theta_Q \mathcal{D}_0 - 2\theta_Q \mathcal{D}_2. \quad (6.15)$$

These determine  $\mathcal{D}_2, \mathcal{D}_3$  uniquely from  $\mathcal{D}_0, \mathcal{D}_1$  as zero-constant formal power series.

*Proof.* — As in the proof of Theorem 6.2, the permutation symmetry of the extraction formula of Theorem [2, 17.1] gives  $\sum_{|\beta|=m} D_\beta = c_m H$ , and pairing with  $H^3/6$ , using  $\int_{F_4} H^4 = 120$ , gives  $c_m = [Q^m] \mathcal{D}_0/20$ . Put

$$\mathcal{J} = \sum_{\beta > 0} J_\beta(H; H^2)Q^{|\beta|}, \quad \mathcal{K} = \sum_{\beta > 0} K_\beta(H^2)Q^{|\beta|}, \quad V = 1 - \theta_Q^2 \mathcal{D}_0/20.$$

Aggregate (6.11) with  $E = E' = H$  and  $\nu = H^2$ , and (6.12) with  $\lambda = H$  and  $\nu = H^2$ , using  $A_\beta(H^2) = 2N_{1,\beta}$  and  $P_\beta(H^3) = 6N_{0,\beta}$ . The intermediate equations are  $V\mathcal{T} = 2\theta_Q^2 \mathcal{D}_1 + 12\theta_Q \mathcal{D}_0$ ,  $\mathcal{J} = -2\theta_Q \mathcal{D}_1 - 6\mathcal{D}_0 - (\theta_Q \mathcal{D}_0)\mathcal{T}/20$ , and  $\mathcal{K} = 2\mathcal{D}_1 + \mathcal{D}_0 \mathcal{T}/20$ . Equation (6.13) with  $E = \lambda = H$  gives  $\theta_Q^2 \mathcal{D}_2 = V\mathcal{J} - 4\theta_Q \mathcal{D}_1 - 6\mathcal{D}_0$  and  $\theta_Q^2 \mathcal{D}_3 = V\mathcal{K} - 2\theta_Q \mathcal{D}_2 - 2\mathcal{D}_1$ . Eliminate  $\mathcal{T}$ . Formal  $\theta_Q^{-2}$  with zero constants is unique.  $\square$

Thus the diagonal reconstruction closes on two inputs; no minimality statement is intended. It does not identify  $\mathcal{T}$  with a Yukawa coupling, nor show that a zero of  $V$  is a genuine singularity.

## 7. Endpoint values and physical threshold behavior

We use the primitives  $C_j$  and their normalization of Section [2, 18] of the companion paper [2], and put  $g_3 = 2\zeta(3)/3$ ,  $g_5 = 2\zeta(5)/5$ . Its endpoint theorem gives, for  $L = 3$ ,

$$(C_1, C_2, C_3, C_4)_\infty = (R_*, 0, g_3, g_3 R_*), \quad R_* = -7\zeta(3)/\pi^2, \quad Q_* = e^{R_*}.$$

Indeed the torus logarithmic formula gives  $R_* = -2m_{\text{Mah}}(1 + z_1 + z_2 + z_3)$ , where  $m_{\text{Mah}}(P) = \int_{[0,1]^n} \log |P(e^{2\pi i t_1}, \dots, e^{2\pi i t_n})| dt_1 \cdots dt_n$  denotes the logarithmic Mahler measure of the multivariate polynomial  $P$ . Here  $m_{\text{Mah}}(1 + z_1 + z_2 + z_3) = 7\zeta(3)/(2\pi^2)$  by [25, §6.2, equation (66)]. Consequently

$$(K_2, K_3, K_4)_\infty = (-R_*^2/2, g_3 + R_*^3/3, -R_*^4/8).$$

At  $L = 4$ , write  $R_* = C_1(\infty)$  and  $S_* = C_2(\infty)$ . The same theorem gives

$$(C_1, C_2, C_3, C_4, C_5)_\infty = (R_*, S_*, g_3, g_3 R_*, g_3 S_* + g_5).$$

Thus all descendant endpoint values follow by the finite formula  $K_k(\infty) = \sum_{j=0}^k (-R_*)^{k-j} C_j(\infty)/(k-j)!$ . These are exact reductions to two constants, not special-value evaluations of those constants. Explicit convergent subtracted Bessel integrals for them are supplied in Lemma [2, 18.2] of the companion paper [2].

By Proposition [2, 8.2] of the companion paper [2], and Remark [2, 8.4], the physical four-loop Feynman integral is holomorphic at the pseudothresholds  $s = 1, 9$ , since both lie in its Taylor disk  $|s| < 25$ . At the physical threshold, with  $\delta = 25 - s \downarrow 0$ ,

$$I_\ominus^{(4)}(25 - \delta) = T + \alpha\delta \log \delta + c\delta + \frac{\alpha}{20}\delta^2 \log \delta + d\delta^2 + O(\delta^3 |\log \delta|), \quad \alpha = \frac{\pi^2}{5\sqrt{5}}.$$

Here  $T, c, d$  are finite constants. To verify the normalization, the large- $t$  part of  $16tI_0(\sqrt{st})K_0(t)^5$  is  $2\pi^2 s^{-1/4} t^{-2} e^{-(5-\sqrt{s})t} (1 + O(t^{-1}))$ . Integration gives  $2\pi^2 s^{-1/4} \epsilon \log \epsilon$ , where  $\epsilon = 5 - \sqrt{s} = \delta/10 + O(\delta^2)$ , proving  $\alpha$ . The threshold indicial polynomial of the physical fourth-order equation is  $240000z(z-1)^2(z-2)$ . Its Frobenius recurrence gives only first powers of the logarithm and gives the next logarithmic coefficient  $\alpha/20$ ; the local Frobenius series is convergent at least for  $|s - 25| < 16$ , up to the next singular point  $s = 9$  of the differential equation; no claim that  $s = 9$  is a singularity of this particular physical solution is needed here. This is compatible with the continuous but non-differentiable threshold value established by the positive coefficient asymptotics in Proposition [2, 8.2] of the companion paper [2].

## 8. Exact three-loop period connection polynomials

There is a second arithmetic question determining the coefficients of the logarithmic periods at  $x = 0$ . This admits a separate analytic solution at three loops. Set

$$\mathcal{P}_3 = \theta_y^3 + y(20\theta_y^3 + 30\theta_y^2 + 18\theta_y + 4) + 64y^2(\theta_y + 1)^3. \quad (8.1)$$

The homogeneous ambient periods  $W_0, W_1, W_2$  solve  $\mathcal{P}_3 W = 0$ . They are fixed by their leading terms  $1, \log y, \log^2 y/2$ , with no additional constant terms at the maximally unipotent point.

**Theorem 8.1.** — Along the positive  $y$  axis, as  $x = 1/y \downarrow 0$ ,

$$W_j(y) = xP_{j,0}(\log x) + O(x^2(1 + |\log x|^2)), \quad 0 \leq j \leq 2,$$

where

$$P_{0,0}(t) = \frac{3}{16\pi^2}(t - 6\log 2)^2, \quad P_{1,0}(t) = \frac{t - 6\log 2}{8}, \quad P_{2,0}(t) = \frac{\pi^2}{24}. \quad (8.2)$$

In particular the constant coefficient of  $P_{0,0}$  is  $27\log^2(2)/(4\pi^2)$ .

*Proof.* — Let

$$z = \frac{108y^2}{(1+4y)^3}, \quad p = \frac{1}{1+4y}, \quad a = \frac{1}{6}, \quad b = \frac{1}{3}, \quad f(z) = {}_2F_1(a, b; 1; z).$$

Let  $g_0(z)$  be the second local Gauss solution with  $g_0 = f \log z +$  a power series with zero constant term, and put  $h_0 = g_0 - (\log 108)f$ . Direct substitution in (8.1) gives the local identifications

$$W_0 = pf^2, \quad W_1 = \frac{1}{2}pfh_0, \quad W_2 = \frac{1}{8}ph_0^2. \quad (8.3)$$

For completeness, the Gauss equation used in that substitution is

$$f'' + \frac{1-3z/2}{z(1-z)}f' - \frac{1}{18z(1-z)}f = 0.$$

The symmetric products of any two of its solutions, multiplied by  $p$ , solve (8.1); this is a rational differential identity. It can be checked by collecting the three coefficients of  $f^2, ff', f'^2$ . Because  $\log z - \log 108 = 2\log y + O(y)$  at zero, the leading logarithms in (8.3) match the ambient normalization, proving the identities.

It is essential to continue these expressions through  $y = 1/2$ , rather than substitute the principal branch of  $f$  on both sides. Put

$$F_1(z) = {}_2F_1(a, b; 1/2; 1-z), \quad \kappa = \frac{\Gamma(1/2)}{\Gamma(1/6)\Gamma(1/3)}, \quad c = \frac{2\pi}{\sqrt{3}}.$$

The zero-balanced logarithmic connection formula at  $z = 0$  [26, §15.8(ii)] is

$$F_1 = \kappa[-g_0 + H_0f], \quad H_0 = 2\psi(1) - \psi(1/6) - \psi(1/3) = c + \log 108.$$

Consequently  $h_0 = cf - F_1/\kappa$ . At  $z = 1$ , the Gauss connection formula [26, §15.10(ii)] writes

$$f = AF_1 + B\sqrt{1-z}F_2(z), \quad A = \frac{\Gamma(1/2)}{\Gamma(5/6)\Gamma(2/3)},$$

where  $F_2$  is analytic at 1 and its explicit coefficient  $B$  is not needed. Reflection of gamma functions gives  $2A\kappa = 1/c$ . The exact factorization

$$1 - z(y) = \frac{(1+16y)(1-2y)^2}{(1+4y)^3} \quad (8.4)$$

shows that continuation through  $y = 1/2$  changes the sign of the analytic square root relative to its nonnegative real value. The point  $y = 1/2$  is ordinary for (8.1), so this continuation is unambiguous. It acts on the local pair by

$$f \mapsto 2AF_1 - f = -h_0/c, \quad h_0 \mapsto -cf. \quad (8.5)$$

As  $y \rightarrow \infty$  the argument  $z$  returns to zero. In the local expressions on the right of (8.5),

$$f = 1 + O(x), \quad h_0 = t - 6 \log 2 + O(x(1 + |t|)), \quad p = x/4 + O(x^2), \quad t = \log x.$$

Substitution in (8.3) proves (8.2).  $\square$

The branch reflection explains why a direct evaluation of the principal hypergeometric function at the small positive value  $z(y)$  for large  $y$  would give an incorrect connection matrix. The logarithm at infinity is created by the continuation through the pullback point  $z = 1$ . Once (8.2) is fixed, the differential recurrence at  $x = 0$  determines all subsequent homogeneous Frobenius coefficients. This does not assert an arithmetic formula for every four-loop period connection.

**Corollary 8.2.** — *At three loops each of  $K_2, K_3, K_4$  has a one-sided  $C^1$  extension at  $Q_*$  and an unbounded second derivative. Its Taylor radius as a power series in  $Q$  lies between  $1/12$  and  $Q_*$ . No assertion of Taylor convergence at  $Q_*$  is made.*

*Proof.* — The connection polynomials above give  $R_* - R \sim 3x \log^2 x / (16\pi^2)$ . Moreover  $\partial_R K_2 = W_1/A - R$ , whose ratio has a nonzero  $1/\log x$  term, and  $\partial_R K_3 = (W_2 - RW_1)/A - K_2$ , whose corresponding coefficient is nonzero because  $R_* \neq 0$ . Their derivatives are unbounded after conversion from  $x$  to  $R$ . Differentiation is legitimate in the convergent Frobenius series, not merely in an unspecified asymptotic remainder. First derivatives have finite limits. The relation  $K_4 = -K_2^2/2$  and  $K_2(Q_*) \neq 0$  give the same assertions for  $K_4$ . The lower bound follows from the principal mirror disk, Corollary [2, 10.5] of the companion paper [2]; a radius greater than  $Q_*$  would contradict this nonanalyticity.  $\square$

## 9. The four-loop connection quantities to certify

Set  $x = 1/y$ ,  $t = \log x$ ,  $\theta_y = y\partial_y$ . The ambient functions obey

$$P_4 W_j = \delta_{j4}, \quad 0 \leq j \leq 4, \quad W_j = (\log y)^j / j! + O(y \log^j y),$$

where

$$\begin{aligned} P_4 &= \theta_y^4 + yQ_1(\theta_y) + y^2Q_2(\theta_y) + y^3Q_3(\theta_y), \\ Q_1(z) &= 35z^4 + 70z^3 + 63z^2 + 28z + 5, \\ Q_2(z) &= 259z^4 + 1036z^3 + 1580z^2 + 1088z + 285, \\ Q_3(z) &= 225(z+1)^2(z+2)^2. \end{aligned}$$

The normalization specifies all degree-zero logarithmic coefficients, with no extra constant terms. Define  $C_0 = 1$ ,  $\theta_y C_{j+1} = W_j$ ,  $C_{j+1} \sim (\log y)^{j+1}/(j+1)!$ , and

$$R = C_1, \quad A = W_0, \quad Q = e^R, \quad K_k = \sum_{j=0}^k \frac{(-R)^{k-j}}{(k-j)!} C_j.$$

The diagonal descendant potentials are  $-20K_2, -60K_3, -120K_4, -120K_5$ .

The endpoint expansion has the form

$$W_j = x(a_j + b_j \log x) + O(x^2 |\log x|), \quad b_0 < 0.$$

Write  $R_\infty = \lim R$ ,  $Q_\infty = e^{R_\infty}$ , and

$$d_j = \frac{a_j b_0 - a_0 b_j}{b_0^2}, \quad E_k = \sum_{h=1}^{k-1} \frac{(-R_\infty)^{k-1-h}}{(k-1-h)!} d_h, \quad 2 \leq k \leq 5. \quad (9.1)$$

All five rows, including the inhomogeneous row  $j = 4$ , must be included.

**Theorem 9.1 (Certified individual singularities).** — *The quantities in (9.1) have signs*

$$E_2 > 0, \quad E_3 < 0, \quad E_4 > 0, \quad E_5 > 0.$$

*Each of  $K_2, K_3, K_4, K_5$  has a one-sided  $C^1$  extension to  $Q_\infty$ , but has an unbounded second derivative there. More precisely, if  $\Delta = R_\infty - R \downarrow 0$ ,*

$$\partial_Q^2 K_k \sim \frac{E_k}{Q_\infty^2 \Delta |\log \Delta|^2}. \quad (9.2)$$

*Consequently their individual  $Q$ -Taylor radii  $\mathcal{R}_k$  satisfy*

$$\frac{1}{20} \leq \mathcal{R}_k \leq Q_\infty, \quad k = 2, 3, 4, 5, \quad (9.3)$$

*where the endpoint is enclosed by*

$$0.336611738699193750820017 \leq Q_\infty \leq 0.336611738699193750820018.$$

The sign assertion is a computer-assisted statement. Its computational part consists of finite ball-arithmetic operations with the analytic tail bounds proved below; it is not inferred from agreement of decimal runs. The rational intervals in Table 2 are intentionally rounded outward and are sufficient to prove every sign. The proof uses only these rational enclosures, whose endpoints are exact decimals with denominator  $10^{24}$ ; truncated decimals elsewhere in the paper, such as  $Q_\infty = 0.3366117386\dots$ , are approximations printed for readability and play no role in any argument.

*Analytic part of Theorem 9.1.* — The convergent endpoint expansion and primitive tail below imply that  $C_j, K_k$  have finite limits. Since  $b_0 < 0$ , all ratios  $W_j/A$  have finite limits, so the first  $R$ - and  $Q$ -derivatives of  $K_k$  have finite limits. This gives the one-sided  $C^1$  extension. Furthermore,

$$\Delta \sim (-b_0)x|t|, \quad \frac{W_j}{A} = \frac{b_j}{b_0} + \frac{d_j}{t} + O(t^{-2}) + O(x).$$

|       | Lower endpoint              | Upper endpoint              |
|-------|-----------------------------|-----------------------------|
| $E_2$ | 1.959219916247366530180947  | 1.959219916247366530180948  |
| $E_3$ | -0.535347537401972845058210 | -0.535347537401972845058209 |
| $E_4$ | 0.420379682614378171767897  | 0.420379682614378171767898  |
| $E_5$ | 0.748770280780437029430306  | 0.748770280780437029430307  |

TABLE 2. Certified rational enclosures, with common denominator  $10^{24}$ .

Differentiation of the normalization gives

$$\partial_R K_k = \sum_{h=0}^{k-1} \frac{(-R)^{k-1-h}}{(k-1-h)!} \frac{W_h}{A} - K_{k-1}, \quad K_1 = 0.$$

Its inverse-logarithm coefficient is exactly  $E_k$ : variation of  $R$  and  $K_{k-1}$  from their endpoint limits contributes  $O(\Delta)$ . All expansions here arise from convergent logarithmic series and may be differentiated; in particular this step does not differentiate an uncontrolled big-O estimate. Using  $dR/dt = -A$  and  $\partial_Q^2 = Q^{-2}(\partial_R^2 - \partial_R)$  gives (9.2). The certified nonzero signs therefore prove unboundedness. A Taylor radius larger than  $Q_\infty$  would supply a holomorphic extension across that point, contradicting this conclusion. Corollary [2, 10.5] of the companion paper [2] gives the lower bounds.  $\square$

## 10. Certificate I: exact normalization and initial tails

The implementation uses the truncated parameter ring  $\mathbb{Q}[\varepsilon]/(\varepsilon^5)$ , with

$$u_0 = 1, \quad u_D(\varepsilon) = -(D + \varepsilon)^{-4} \sum_{l=1}^{\min(3,D)} Q_l(D - l + \varepsilon) u_{D-l}(\varepsilon). \quad (10.1)$$

Then

$$W_j(y) = [\varepsilon^j] e^{\varepsilon \log y} \sum_{D \geq 0} u_D(\varepsilon) y^D.$$

The degree-zero term gives  $\theta_y^4 W_4 = 1$  and zero source for the lower four rows. The positive-degree recursion gives the stated equations and fixes the normalized solutions uniquely. As an independent normalization check, the code compares the 50 rational coefficients in total degrees zero through nine (five parameter coefficients per degree) with the square-free ambient formula from the square-free ambient seed; all agree exactly.

Use the coefficient  $\ell^1$  norm in the truncated parameter ring. For  $D \geq 10$ ,

$$\|(D + \varepsilon)^{-4}\| \leq (D - 1)^{-4}, \quad \|Q_l(D - l + \varepsilon)\| \leq Q_l(D).$$

Each  $Q_l(D)/(D-1)^4$  decreases with  $D \geq 10$  because its coefficients are positive and its degree is at most four. The finite rational check

$$\sum_{l=1}^3 \frac{Q_l(10)}{9^4 100^l} = \frac{15497623}{21870000} < 1$$

and the ten base cases give

$$\|u_D\| \leq 2 \cdot 100^D \quad (D \geq 0). \quad (10.2)$$

This is a deliberately coarse bound, sufficient at  $y_0 = 1/512$ . Let  $q_0 = 100/512$ , and truncate after degree  $N_g$ . A common bound for each omitted logarithmic jet  $\theta_y^a W_j$ ,  $0 \leq a \leq 3$ , and each primitive  $C_{j+1}$  is

$$e_g = 1024 q_0^{N_g+1} (N_g + 2)^3 \frac{1 + 4q_0 + q_0^2}{(1 - q_0)^4}. \quad (10.3)$$

Indeed, evaluation of the logarithmic polynomial costs at most  $e^{|\log y_0|} = 512$ , and each logarithmic differentiation costs at most  $D + 1$  in the exponential-factorial basis. The sum is bounded using  $D = N_g + 1 + m$  and  $\sum_{m \geq 0} (m+1)^3 q^m = (1 + 4q + q^2)/(1 - q)^4$ . For primitives, the inverse of  $D + \partial_{\log y}$  has norm at most  $1/(D-1)$  for  $D > 1$ , so the same bound suffices for the omitted tail. The degree-zero primitive is integrated exactly. Conversion to ordinary  $y$ -derivatives is done only after these bounds have been added to the balls.

## 11. Certificate II: finite continuation with Cauchy bounds

At a positive center  $c$ , put  $z = (y - c)/c$ . The augmented state is

$$U = (W, \theta_y W, \theta_y^2 W, \theta_y^3 W, C, 1).$$

It satisfies a homogeneous first-order system in  $z$ ; the last component encodes the possible source 1. Write  $P_4 = \sum_{a=0}^4 p_a(y) \theta_y^a$ , so  $p_4(y) = (1+y)(1+9y)(1+25y)$ . Fix  $r = 1/4$ . On  $|z| \leq r$ , an induced maximum-row-sum bound for that system is

$$M(c) = \frac{1}{1-r} \max \left\{ 1, \frac{1 + \sum_{a=0}^3 p_a(c(1+r))}{(1 + c(1-r))(1 + 9c(1-r))(1 + 25c(1-r))} \right\}. \quad (11.1)$$

The numerator estimate uses positive polynomial coefficients; the denominator uses  $|1 + ac(1+z)| \geq 1 + ac(1-r)$  for  $a = 1, 9, 25$ . Integration along any radial segment gives

$$\sup_{|z| \leq r} \|U(z)\|_\infty \leq B := \|U(0)\|_\infty e^{M(c)r}.$$

This bound applies to every exact initial vector contained in the current balls, because  $\|U(0)\|_\infty$  is rounded upward.

Each step has  $0 < h \leq c/16$ , so  $\eta = h/(rc) \leq 1/4$ . Taylor coefficients through degree  $N$  are computed from the scalar ordinary differential equation by ball arithmetic. With  $S_3(v) = (1 + 4v + v^2)/(1 - v)^4$ , a common base remainder is

$$e = B4^{-(N+1)}S_3(1/4).$$

The omitted remainder for the ordinary derivative  $W^{(a)}$  at  $c + h$ ,  $0 \leq a \leq 3$ , is bounded by

$$e \left( \frac{4(N+1)}{rc} \right)^a. \quad (11.2)$$

To see this, Cauchy's inequality bounds the  $z$ -Taylor coefficient by  $Br^{-m}$ , and the derivative tail is bounded by the geometric sum with  $(N+1+j)^a \leq (N+1)^a(j+1)^a$ . The bound for  $C$ , computed through degree  $N$  using  $C' = W/y$ , is simply  $e$ . The code adds these bounds after every step; subsequent operations propagate the full accumulated balls. Roundoff and truncation are therefore both included.

## 12. Certificate III: convergent endpoint matching

For  $W = xF$  and  $D = x\partial_x$ , the equation becomes

$$[225D^2(D-1)^2 + xD^2(259D^2 + 26) + x^2Q_1(D) + x^3(D+1)^4]F = bx^2.$$

Write  $F = \sum_{k \geq 0} x^k P_k(\log x)$ , with  $P_k$  linear. The four coefficients of  $P_0, P_1$  are free, and for  $k \geq 2$ ,

$$\begin{aligned} 225(k+\partial)^2(k-1+\partial)^2P_k = & -(k-1+\partial)^2[259(k-1+\partial)^2 + 26]P_{k-1} \\ & - Q_1(k-2+\partial)P_{k-2} - (k-2+\partial)^4P_{k-3} + b\delta_{k2}. \end{aligned}$$

Here  $\partial = \partial_{\log x}$ ,  $P_{-1} = 0$ . The inverse on linear polynomials is

$$\frac{1}{225k^2(k-1)^2} \left( 1 - \left( \frac{2}{k} + \frac{2}{k-1} \right) \partial \right).$$

We use four homogeneous basis vectors with unit initial data and the particular solution with  $P_0 = P_1 = 0$ ,  $b = 1$ .

For the ordinary coefficient norm of linear polynomials,  $\|\partial\| \leq 1$ . For  $k \geq 4$  the displayed recurrence has the bounds

$$\|P_k\| \leq 5\|P_{k-1}\| + 2\|P_{k-2}\| + \frac{1}{100}\|P_{k-3}\|.$$

Explicit dominating coefficients are  $1807/405 < 5$ ,  $871/450 < 2$ , and  $13/1350 < 1/100$ . The first four coefficients of each of the five basis solutions are checked, and induction then gives

$$\|P_k\| \leq 2 \times 8^k, \quad (12.1)$$

since  $5/8 + 2/8^2 + 1/(100 \cdot 8^3) < 1$ .

At a matching point  $x = 1/c \leq 1/32$ , put  $q = 8x$  and  $H = \max(1, |\log x|)$ . After degree  $N_f$ , the omitted tail of  $D^a W$ ,  $0 \leq a \leq 3$ , is bounded by

$$2xHq^{N_f+1}(N_f+3)^a S_3(q). \quad (12.2)$$

The primitive tail has the exact expansion

$$C(\infty) - C(y) = \sum_{k \geq 0} \frac{x^{k+1}}{k+1} \left( P_k(\log x) - \frac{P'_k(\log x)}{k+1} \right),$$

and its omitted part is bounded by  $4xHq^{N_f+1}/(1-q)$ . These estimates follow directly from (12.1); they bound the actual infinite series, not just its last computed term.

The four transported jets are matched to the four enclosed basis jets by a validated real-ball linear solve. For the inhomogeneous row, the enclosed particular solution is subtracted first. The solution balls enclose  $(a_j, b_j, c_j, e_j)$ , where  $P_{j,1}(t) = c_j + e_j t$ . The primitive-tail formula then encloses every  $C_j(\infty)$ , including  $R_\infty$ . Finally ordinary ball operations evaluate (9.1); the ball for  $b_0$  lies strictly below zero, so the division is valid.

### 13. Execution and interpretation of the certificate

The first run uses 1400-bit midpoint precision,  $N_g = 450$ , Taylor degree  $N = 400$ , endpoint degree  $N_f = 250$ , and matching at  $y = 32$ . It takes 161 positive steps. The independent matching check uses 1500 bits,  $N = 420$ , and  $y = 48$ , with the same germ and endpoint degrees. Both runs independently certify all four signs and return the rational enclosures in Table 2.

For scale, the initial bound (10.3) is less than  $5.4 \times 10^{-309}$ . The largest step base remainder  $e$  is below  $1.25 \times 10^{-218}$  in the first run. These small local bounds alone are not the certificate since their propagated effect is retained throughout. After endpoint matching, the internal radii of the four  $E_k$  balls are below  $6 \times 10^{-139}$  in the first run. The exported decimal ball strings are deliberately shorter, and the separate rational intervals have width  $10^{-24}$ ; no ambiguity about rounded decimal endpoints is used to infer a sign.

The implementation uses the real-ball and validated matrix-solve facilities of Arb through python-flint [27, 28]. It does not use an approximate matrix-solve mode. The finite verification also checks the ten base cases of (10.2), the rational induction constants, the 50 exact square-free germ coefficients, and every exported rational sign. This is a reproducible computer-assisted proof with the stated analytic majorants and the standard correctness assumptions for Python/FLINT/Arb; it is not a formal proof-assistant verification. Agreement between the two matching runs is a cross-check, not a substitute for the error bounds.

## Appendix A. Reproducible interval certificate

The following two listings are complete programs. These programs are available from the Github

<https://github.com/pierrevanhove/Sunset-Gromov-Witten> repository.

They require Python 3.10 or later and `python-flint==0.9.0`. Run without Python's optimization flag, so assertions remain enabled:

```
python3 certify.py
```

```
python3 certify.py --bits 1500 --order 420 --match 48
```

```
python3 validate.py
```

All numerical data are generated from the rational differential equation.

```

1 """Rigorous real-ball continuation for four-loop endpoint connections.
2 Every truncation has an explicit majorant; no floating-point input data.
3 """
4 from flint import arb, arb_mat, ctx
5 from fractions import Fraction as F
6 from math import factorial, comb
7 from pathlib import Path
8 import json, time, argparse
9 Q=[[0,0,0,0,1],[5,28,63,70,35],[285,1088,1580,1036,259],[900,2700,2925,1350,225]]
10 S=[[int(k==r==0) for r in range(5)] for k in range(5)]
11 for k in range(1,5):
12     for r in range(1,k+1):S[k][r]=S[k-1][r-1]+r*S[k-1][r]
13 ODE=[[sum(Q[j][k]*S[k][r] for k in range(r,5) for j in range(4) if d==j+r) for d
14         in range(r+4)] for r in range(5)]
15 def A(v):return arb(v.numerator)/v.denominator if isinstance(v,F) else arb(v)
16 def zero(n):return [arb(0) for _ in range(n)]
17 def add(a,b):return [x+y for x,y in zip(a,b)]
18 def scale(a,c):return [x*c for x in a]
19 def deriv(p):return [(k+1)*p[k+1] for k in range(len(p)-1)]+[arb(0)]
20 def op(p,k,r):
21     for _ in range(r):p=add(scale(p,k),deriv(p))
22     return p
23 def pv(p,x):
24     v=arb(0)
25     for a in reversed(p):v=v*x+a
26     return v
27
28 def mul(a,b):return [sum((a[j]*b[k-j] for j in range(k+1)),arb(0)) for k in range
29     (5)]
30 def qshift(q,d):return [sum(q[a]*comb(a,k)*d**(a-k) for a in range(k,5)) for k in
31     range(5)]
32 def norm(p):return sum((x.abs_upper() for x in p),arb(0)).upper()
33 def maxabs(p):return max(x.abs_upper() for x in p)
34 def widen(x,e):return x+arb(0,e.abs_upper())
35
36 def germ_coeff(N):

```

```

35     us=[[arb(1)]+zero(4)]
36     for d in range(1,N+1):
37         rhs=zero(5)
38         for l in range(1,min(3,d)+1):rhs=add(rhs,mul(list(map(arb,qshift(Q[l],d-1)
39         )),us[d-1]))
39         inv=[arb((-1)**r*comb(r+3,3))/arb(d)**(4+r) for r in range(5)]
40         us.append(scale(mul(inv,rhs),-1))
41         # Base cases of the coefficient majorant. Strict bound 2*100^d allows rounding
42         .
42         for d in range(10):assert norm(us[d]) < 2*arb(100)**d
43         assert sum(F(sum(Q[l][a]*10**a for a in range(5)),9**4*100**1) for l in range
44         (1,4))<1
44         return us
45
46 def initial(N):
47     y=arb(1)/512;t=y.log();us=germ_coeff(N);W=[];C=[]
48     q=arb(100)/512
49     # Uniform tail bound for theta jets of order <=3 and primitive.
50     e=1024*q**(N+1)*(N+2)**3*(1+4*q+q*q)/(1-q)**4
51     for j in range(5):
52         jets=zero(4);c=arb(0)
53         for d,row in enumerate(us):
54             p=[row[j-r]/factorial(r) for r in range(j+1)]
55             for r in range(4):jets[r]+=y**d*pv(op(p,d,r),t)
56             if d:
57                 prim=zero(len(p));dp=p
58                 for r in range(len(p)):
59                     prim=add(prim,scale(dp,arb((-1)**r)/arb(d)**(r+1)));dp=deriv(
60                     dp)
61                     c+=y**d*pv(prim,t)
62                 else:c+=t**(j+1)/factorial(j+1)
62             jets=[widen(v,e) for v in jets];c=widen(c,e)
63             W.append([jets[0],jets[1]/y,(jets[2]-jets[1])/y**2,(jets[3]-3*jets[2]+2*
64             jets[1])/y**3]);C.append(c)
64         return y,W,C,e
65
66 def shifted(poly,y):return [sum((arb(poly[j])*comb(j,k)*y**(j-k) for j in range(k,
67     len(poly))),arb(0)) for k in range(len(poly))]
68
69 def step(y,W,C,h,N):
70     aa=[shifted(p,y) for p in ODE];r=arb(1)/4;q=arb(1)/4
71     # Supremum of augmented logarithmic-jet first-order system on |z|<=r.
72     ymax=y*(1+r);den=(1+y*(1-r))*(1+9*y*(1-r))*(1+25*y*(1-r))
73     numer=arb(1)+sum((sum(Q[l][:4])*ymax**l for l in range(1,4)),arb(0))
74     M=max(arb(1),(numer/den).upper()/(1-r))
75     nw=[];nc=[];largest=arb(0)
76     for j,jet in enumerate(W):
77         theta=[jet[0],y*jet[1],y*y*jet[2]+y*jet[1],y**3*jet[3]+3*y*y*jet[2]+y*jet
78         [1]]
79         B=maxabs(theta+[C[j],arb(1)]*(M**r).exp())
80         e=B*q**(N+1)*(1+4*q+q*q)/(1-q)**4
81         largest=max(largest,e.upper())
82         f=[jet[k]/factorial(k) for k in range(4)]
83         for n in range(N-3):

```

```

81         rhs=arb(int(j==4 and n==0))
82         for rr in range(5):
83             for k,a in enumerate(aa[rr]):
84                 if k>n or rr==4 and k==0: continue
85                 ix=n-k+rr
86                 factor=1
87                 for v in range(n-k+1,ix+1): factor*=v
88                 rhs-=a*f[ix]*factor
89                 f.append(rhs/(aa[4][0]*(n+1)*(n+2)*(n+3)*(n+4)))
90         vals=[];df=f
91         for rr in range(4):
92             vals.append(widen(pv(df,h),e*(4*(N+1)/(r*y)**rr));df=deriv(df)
93         cf=[C[j]];prev=arb(0)
94         for n in range(N):
95             qq=(f[n]-prev)/y;cf.append(qq/(n+1));prev=qq
96             nw.append(vals);nc.append(widen(pv(cf,h),e))
97         return nw,nc,largest
98
99 def frob(v,source,N):
100     ps=[list(map(arb,v[:2])),list(map(arb,v[2:]))]
101     for k in range(2,N+1):
102         p=ps[k-1];rhs=add(scale(op(p,k-1,4),259),scale(op(p,k-1,2),26))
103         for a,c in enumerate(Q[1]): rhs=add(rhs,scale(op(ps[k-2],k-2,a),c))
104         if k>=3: rhs=add(rhs,op(ps[k-3],k-2,4))
105         if k==2: rhs[0]-=source
106         rhs=add(rhs,scale(deriv(rhs),-arb(2)/k-arb(2)/(k-1)))
107         ps.append(scale(rhs,-arb(1)/(225*k*k*(k-1)**2)))
108     for k in range(4): assert norm(ps[k]) < 2*arb(8)**k
109     return ps
110
111 def eval_frob(ps,x):
112     N=len(ps)-1;t=x.log();q=8*x;jets=zero(4);tail=arb(0)
113     for k,p in enumerate(ps):
114         for r in range(4): jets[r]+=x**(k+1)*pv(op(p,k+1,r),t)
115         tail+=x**(k+1)*pv(add(p,scale(deriv(p),-arb(1)/(k+1))),t)/(k+1)
116     logbound=max(arb(1),t.abs_upper())
117     base=2*x*logbound*q**(N+1)*(1+4*q+q*q)/(1-q)**4
118     jets=[widen(v,base*(N+3)**r) for r,v in enumerate(jets)]
119     tail=widen(tail,4*x*logbound*q**(N+1)/(1-q))
120     return jets,tail
121
122 def box(v):
123     den=10**24
124     lo=int((v.lower()*den).lower().floor().unique_fmpz())
125     hi=int((v.upper()*den).upper().ceil().unique_fmpz())
126     assert v >= arb(lo)/den and v <= arb(hi)/den
127     return {'ball':v.str(45),'outer_lower_numerator':str(lo),'
128             'outer_upper_numerator':str(hi),'outer_denominator':str(den),'radius':v.rad().
129             str(8)}
130
131 def run(bits=1400,N=400,Ng=450,Nf=250,match=32):
132     assert match>=32 and N>=4 and Ng>=10 and Nf>=4
133     ctx.prec=bits;start=time.time();y,W,C,initial_tail=initial(Ng);steps=0;maxrem=
134     arb(0)

```

```

131 while y<match:
132     h=min(y/16,arb(match)-y);W,C,e=step(y,W,C,h,N);y+=h;steps+=1;maxrem=max(
        maxrem,e)
133     if steps%40==0:print('step',steps,'y',y.str(8),'radius',maxabs([a.rad()
        for w in W for a in w]).str(5),flush=True)
134     x=1/y;bases=[frob([int(a==j) for a in range(4)],0,Nf) for j in range(4)];part=
        frob([0]*4,1,Nf)
135     be=[eval_frob(b,x) for b in bases];pe=eval_frob(part,x)
136     matrix=arb_mat([[be[j][0][r] for j in range(4)] for r in range(4)])
137     connections=[];limits=[]
138     for j,w in enumerate(W):
139         target=[w[0],-y*w[1],y*w[1]+y*y*w[2],-y*w[1]-3*y*y*w[2]-y**3*w[3]]
140         if j==4:target=[a-b for a,b in zip(target,pe[0])]
141         sol=matrix.solve(arb_mat([[v] for v in target]));weights=[sol[k,0] for k
        in range(4)]
142         connections.append(weights);limits.append(C[j]+sum((weights[k]*be[k][1]
        for k in range(4)),arb(0))+pe[1] if j==4 else 0))
143     R=limits[0];a0,b0=connections[0][:2]
144     assert b0<0
145     ds=[(v[0]*b0-a0*v[1])/b0**2 for v in connections]
146     Es=[sum(((R)**(k-1-h)/factorial(k-1-h)*ds[h] for h in range(1,k)),arb(0)) for
        k in range(2,6)]
147     for e,sign in zip(Es,[1,-1,1,1]):assert sign*e>0
148     out={'bits':bits,'taylor_order':N,'germ_order':Ng,'frobenius_order':Nf,'
        matching_y':match,'steps':steps,'initial_tail_bound':box(initial_tail),'
        largest_step_tail_W_bound':box(maxrem),'R_infinity':box(R),'Q_infinity':box(R.
        exp()),'connections':[[box(v) for v in row] for row in connections],'C_limits'
        :[box(v) for v in limits],'E':[box(v) for v in Es],'seconds':time.time()-start
        , 'certificate_passed':True}
149     Path('certificate_'+str(match)+'.json').write_text(json.dumps(out,indent=2)+'\
        n')
150     print('R',R.str(40),'E',[v.str(30) for v in Es],'seconds',out['seconds'],flush
        =True)
151     return out
152 if __name__=='__main__':
153     p=argparse.ArgumentParser();p.add_argument('--bits',type=int,default=1400);p.
        add_argument('--order',type=int,default=400);p.add_argument('--germ',type=int,
        default=450);p.add_argument('--frob',type=int,default=250);p.add_argument('--
        match',type=int,default=32);a=p.parse_args();run(a.bits,a.order,a.germ,a.frob,
        a.match)

```

```

1 """Exact normalization checks and rational checks on exported enclosures."""
2 from fractions import Fraction as F
3 from math import factorial,comb
4 from pathlib import Path
5 import json
6 from certify import Q
7
8 def conv(a,b,N):
9     out=[F(0)]*(N+1)
10    for i,x in enumerate(a):

```

```

11         for j,y in enumerate(b[:N+1-i]): out[i+j]+=x*y
12     return out
13
14 def exact_recurrence(N):
15     out=[F(1)]+[F(0)]*4
16     for d in range(1,N+1):
17         rhs=[F(0)]*5
18         for l in range(1,min(d,3)+1):
19             p=[F(sum(Q[l][a]*comb(a,k)*(d-l)**(a-k) for a in range(k,5))) for k in
20                 range(5)]
21             v=conv(p,out[d-1],4); rhs=[a+b for a,b in zip(rhs,v)]
22             inv=[F((-1)**k*comb(k+3,3),d**(k+4)) for k in range(5)]
23             out.append([-v for v in conv(rhs,inv,4)])
24     return out
25
26 def squarefree(N):
27     f=[F(1,factorial(d)**2) for d in range(N+1)]; g=[]; h=F(0)
28     for d in range(N+1):
29         if d:h+=F(1,d)
30         g.append(-2*h*f[d])
31     ps=[]
32     for k in range(5):
33         a=[F(1)]+[F(0)]*N
34         for i in range(5): a=conv(a,g if i<k else f,N)
35         ps.append(a)
36     out=[]; num=[F(1)]+[F(0)]*4
37     for d in range(N+1):
38         if d:
39             for _ in range(2): num=[num[k]+(num[k-1]/d if k else 0) for k in range
40                 (5)]
41             out.append([( (-1)**d*factorial(d)**2*sum((num[a]*ps[k-a][d]/factorial(k-a)
42                 for a in range(k+1)),F(0)) for k in range(5))])
43     return out
44
45 def interval(b): return F(int(b['outer_lower_numerator']),int(b['outer_denominator',
46     ])),F(int(b['outer_upper_numerator']),int(b['outer_denominator']))
47
48 if __name__=='__main__':
49     a=exact_recurrence(9); b=squarefree(9); assert a==b
50     base=[sum(map(abs,row)) <=2*100**d for d,row in enumerate(a)]; assert all(base)
51     germ_factor=sum(F(sum(Q[l][k]*10**k for k in range(5)),9**4*100**1) for l in
52         range(1,4)); assert germ_factor<1
53     H1=(F(259)+F(26,16))*F(16,9)*F(13,6)/225
54     H2=F(201,225)*F(13,6); H3=F(1,225)*F(13,6)
55     assert H1<5 and H2<2 and H3<F(1,100)
56     certs=[json.loads(Path('certificate_'+str(m)+'.json').read_text()) for m in
57         [32,48]]
58     Es=[]
59     for c in certs:
60         assert c['certificate_passed']
61         bounds=[interval(v) for v in c['E']]
62         assert bounds[0][0]>0 and bounds[1][1]<0 and bounds[2][0]>0 and bounds
63             [3][0]>0

```

```

57     Es.append(bounds)
58     for left,right in zip(Es[0],Es[1]):assert max(left[0],right[0])<=min(left[1],
    right[1])
59     out={'exact_squarefree_normalization_coefficients':50,'germ_base_cases':10,'
    germ_induction_factor':str(germ_factor),'frobenius_H1_bound':str(H1),'
    frobenius_H2_bound':str(H2),'frobenius_H3_bound':str(H3),'
    two_certificates_pass':True,'exported_rational_E_intervals':[[str(a),str(b)]
    for a,b in Es[0]],'status':'All assertions passed. The analytic tail proofs in
    the report are part of the certificate.'}
60     Path('validation.json').write_text(json.dumps(out,indent=2)+'\n')
61     Path('exact_germ_check.json').write_text(json.dumps([[str(v) for v in row] for
    row in a],indent=2)+'\n')
62     print(json.dumps(out,indent=2))

```

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